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(54) **MULTI-LAYER EXTRUDED UPPERS FOR ARTICLES OF FOOTWEAR AND OTHER FOOT-RECEIVING DEVICES**

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(73) Assignee: **NIKE, Inc.**, Beaverton, OR (US)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 98 days.

This patent is subject to a terminal disclaimer.

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(65) **Prior Publication Data**

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Related U.S. Application Data

(63) Continuation of application No. 16/380,219, filed on Apr. 10, 2019, now Pat. No. 11,583,034.
(Continued)

(51) **Int. Cl.**
A43B 23/02 (2006.01)
A43B 7/12 (2006.01)
(Continued)

(52) **U.S. Cl.**
CPC **A43B 23/024** (2013.01); **A43B 7/12** (2013.01); **A43B 23/0205** (2013.01);
(Continued)

(58) **Field of Classification Search**
CPC B29D 35/00; B29C 64/00; B32B 5/12; B32B 5/26; B32B 2437/02
See application file for complete search history.

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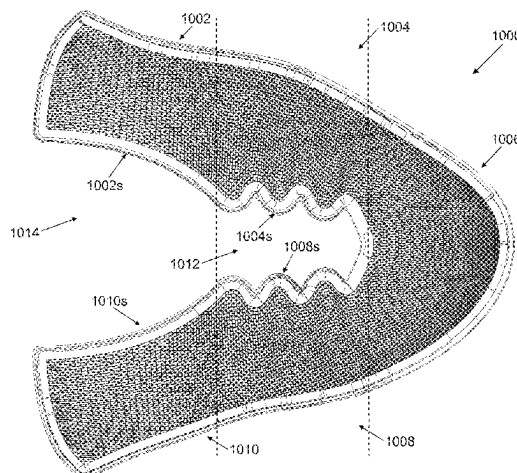
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(57) **ABSTRACT**

Upper components for footwear include: (a) a first layer including a first filament including first plural, non-intersecting, spaced apart path segments (wherein the first filament may have a width dimension of less than 3 mm wide); and (b) a second layer including a second filament including second plural, non-intersecting, spaced apart path segments (wherein the second filament may have a width dimension of less than 3 mm), wherein the second layer is fused to the first layer at locations where the layers contact one another. Additional layers of material, including additional layers with filament, may be included in the upper. The filament material in the different layers may be the same or different from one another (e.g., a thermoplastic material, a thermoplastic polyurethane material, a hydrophobic material, a water-repelling material, a non-water absorbing material, etc.). One or more layers each may be formed as a continuous extruded path of filament.

18 Claims, 40 Drawing Sheets



Related U.S. Application Data

- (60) Provisional application No. 62/655,519, filed on Apr. 10, 2018, provisional application No. 62/655,539, filed on Apr. 10, 2018.

(51) **Int. Cl.**

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B32B 5/12 (2006.01)
B32B 5/26 (2006.01)
B33Y 80/00 (2015.01)
B29L 31/50 (2006.01)

(52) **U.S. Cl.**

CPC **A43B 23/0235** (2013.01); **A43B 23/0245**
 (2013.01); **A43B 23/0255** (2013.01); **B29D**
35/126 (2013.01); **B32B 5/26** (2013.01); **B33Y**
80/00 (2014.12); **B29L 2031/505** (2013.01);
B32B 5/12 (2013.01); **B32B 2307/40**
 (2013.01); **B32B 2307/73** (2013.01); **B32B**
2437/00 (2013.01); **B32B 2437/02** (2013.01)

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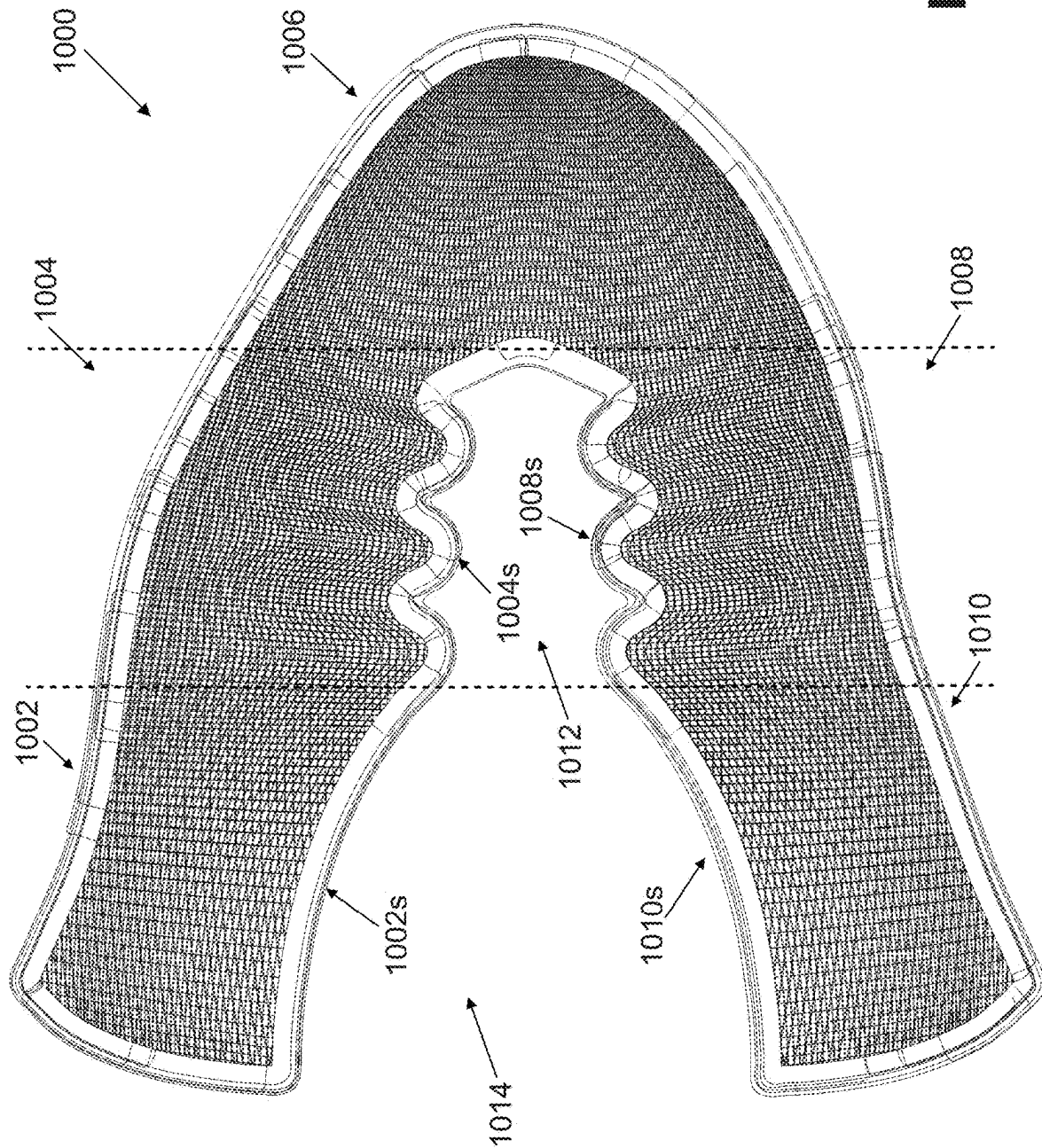
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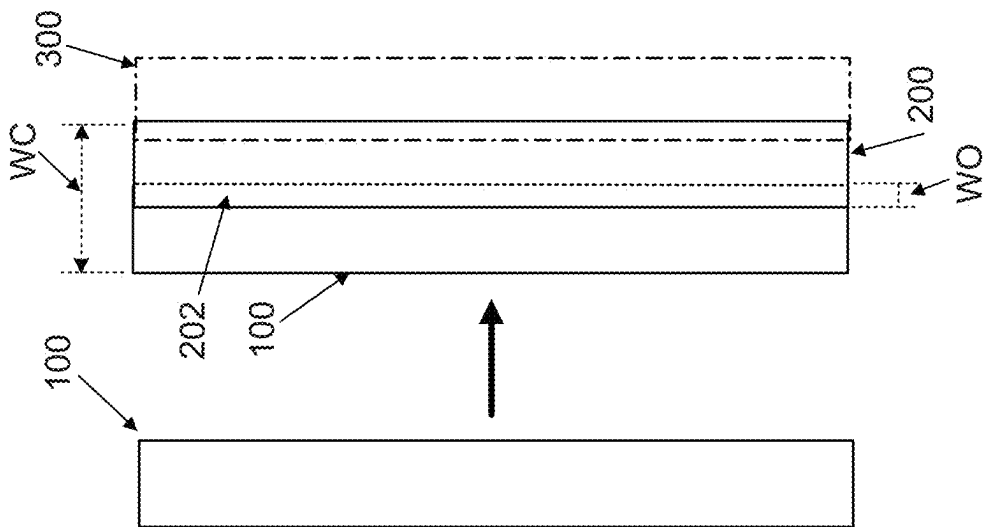


FIG. 2C

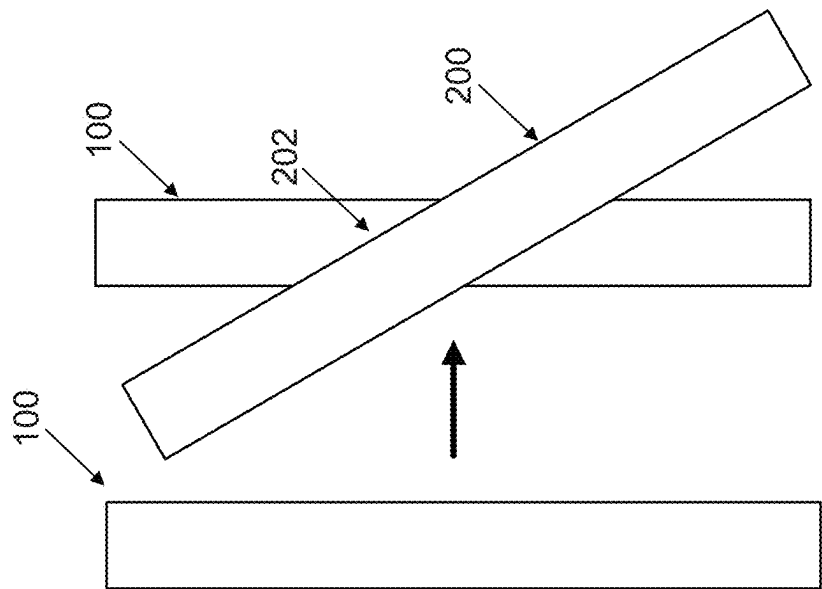


FIG. 2B

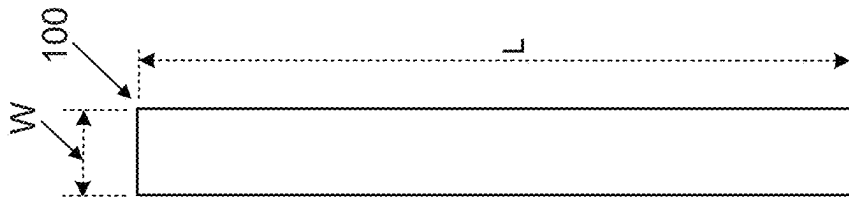
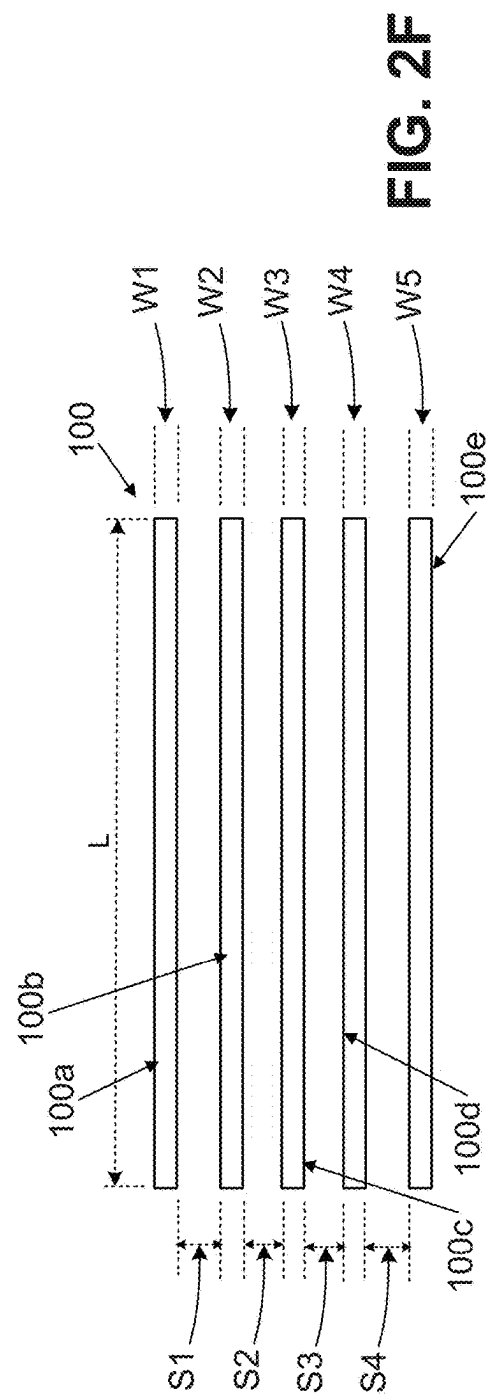
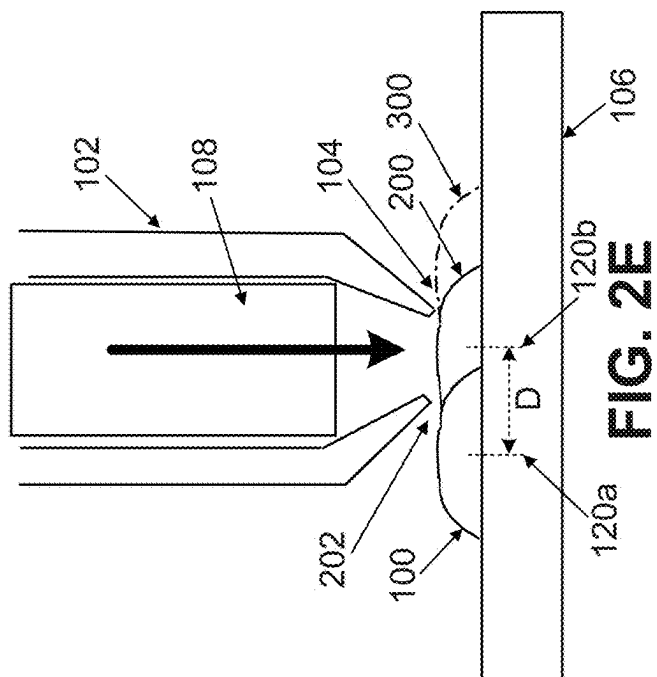
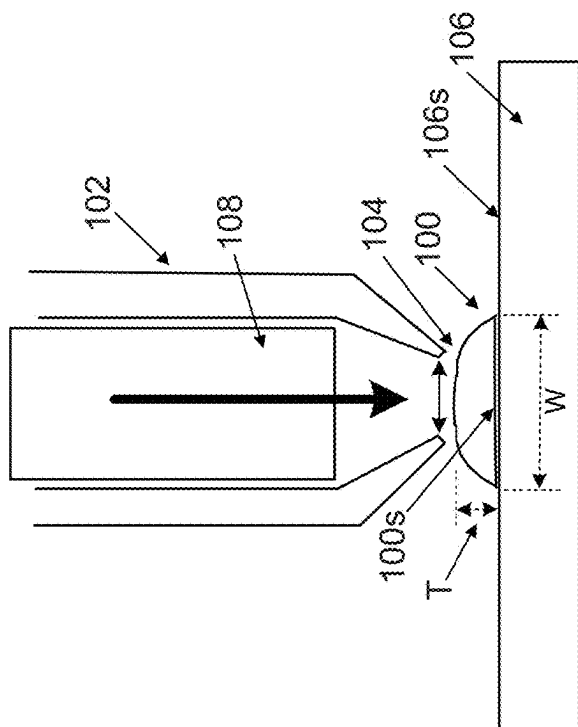


FIG. 2A



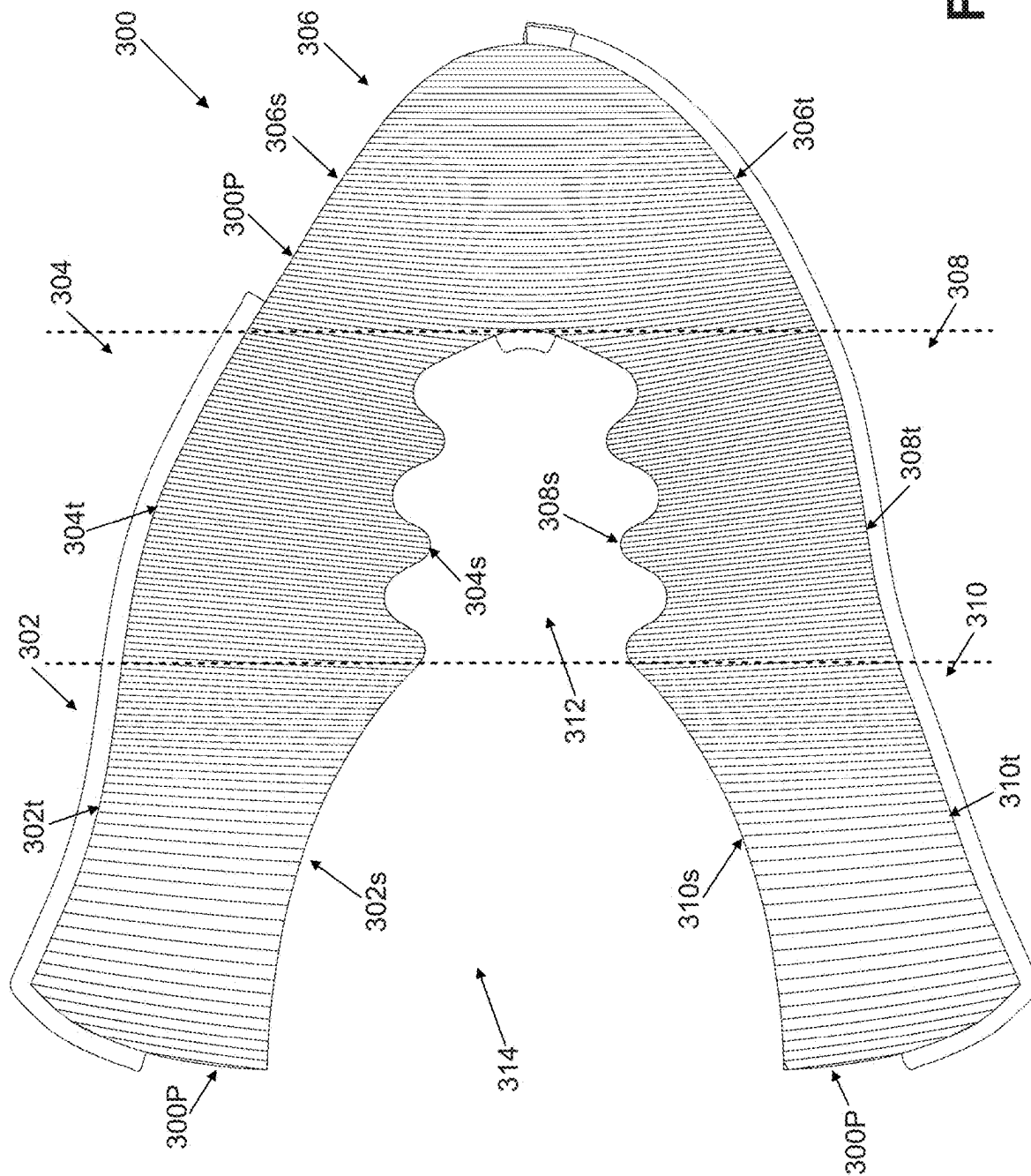


FIG. 3A

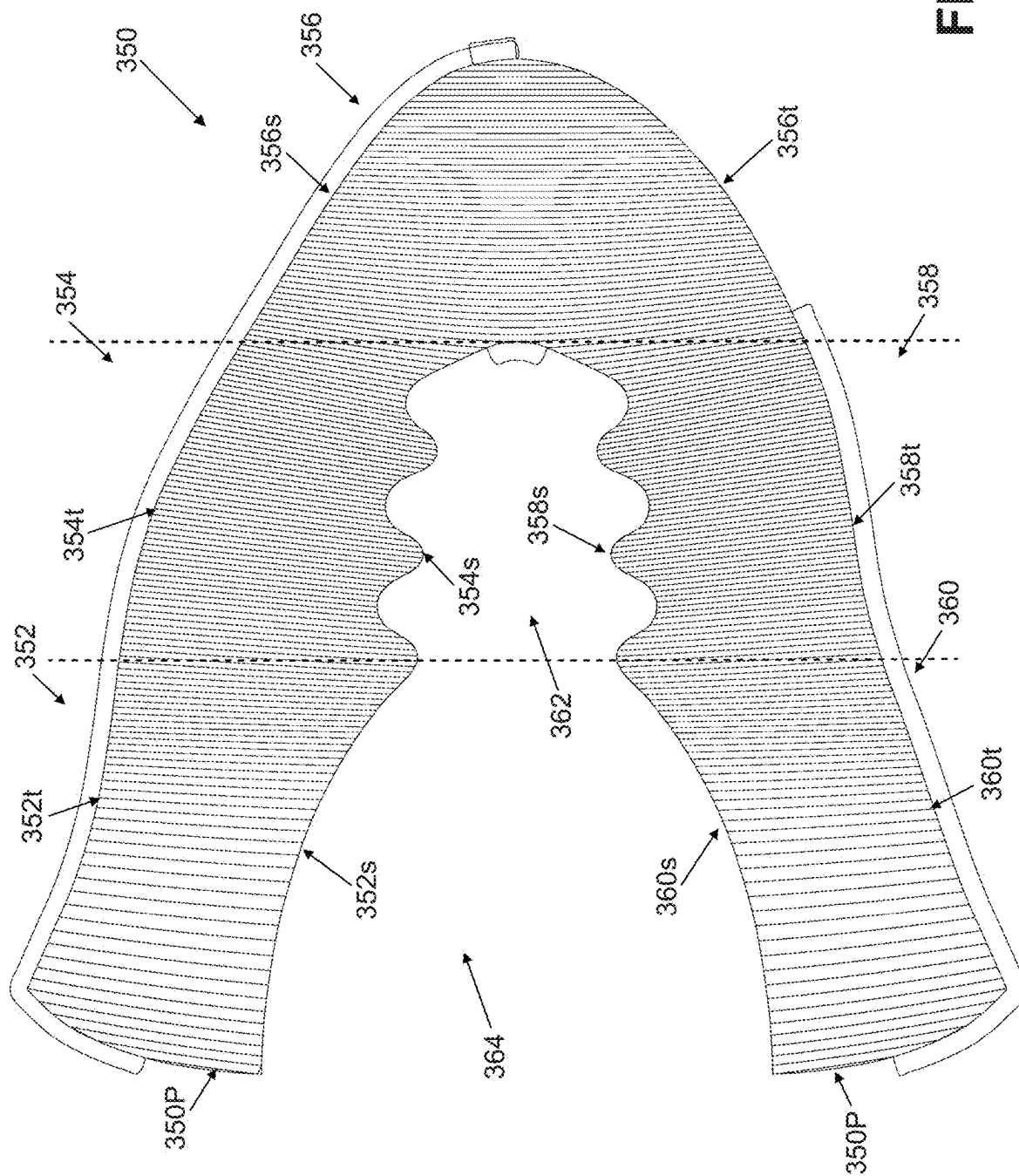
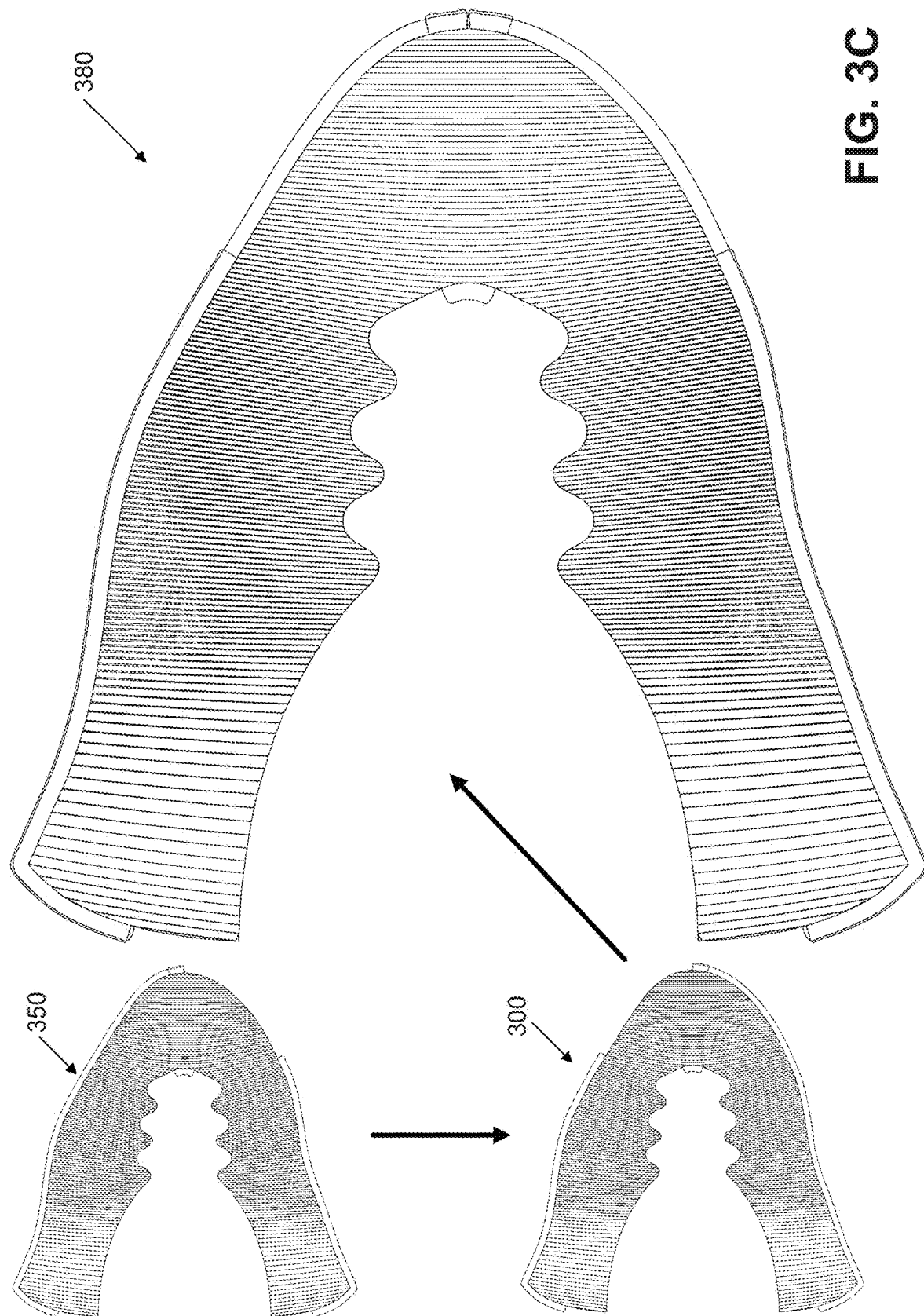
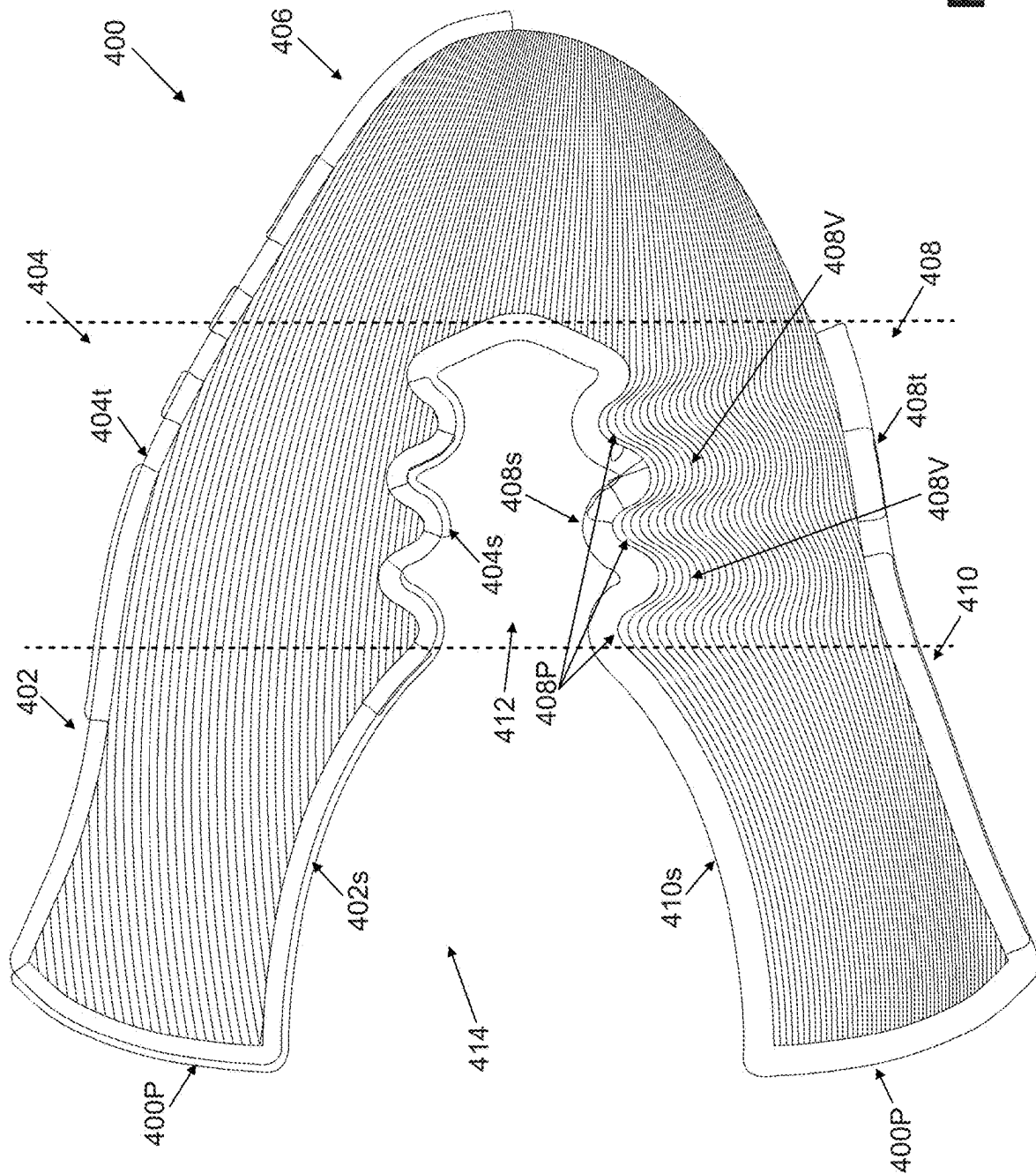
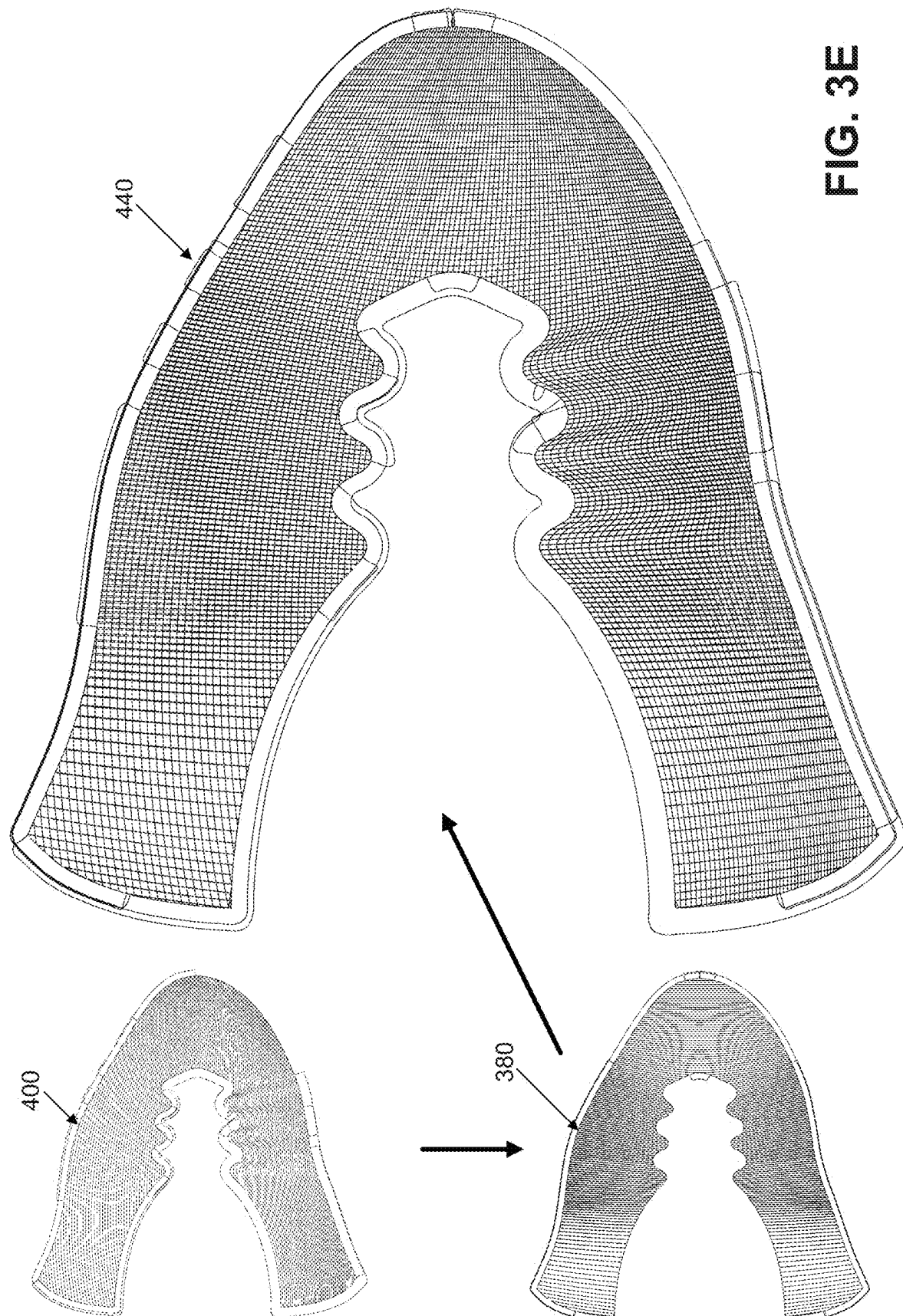


FIG. 3B



3E





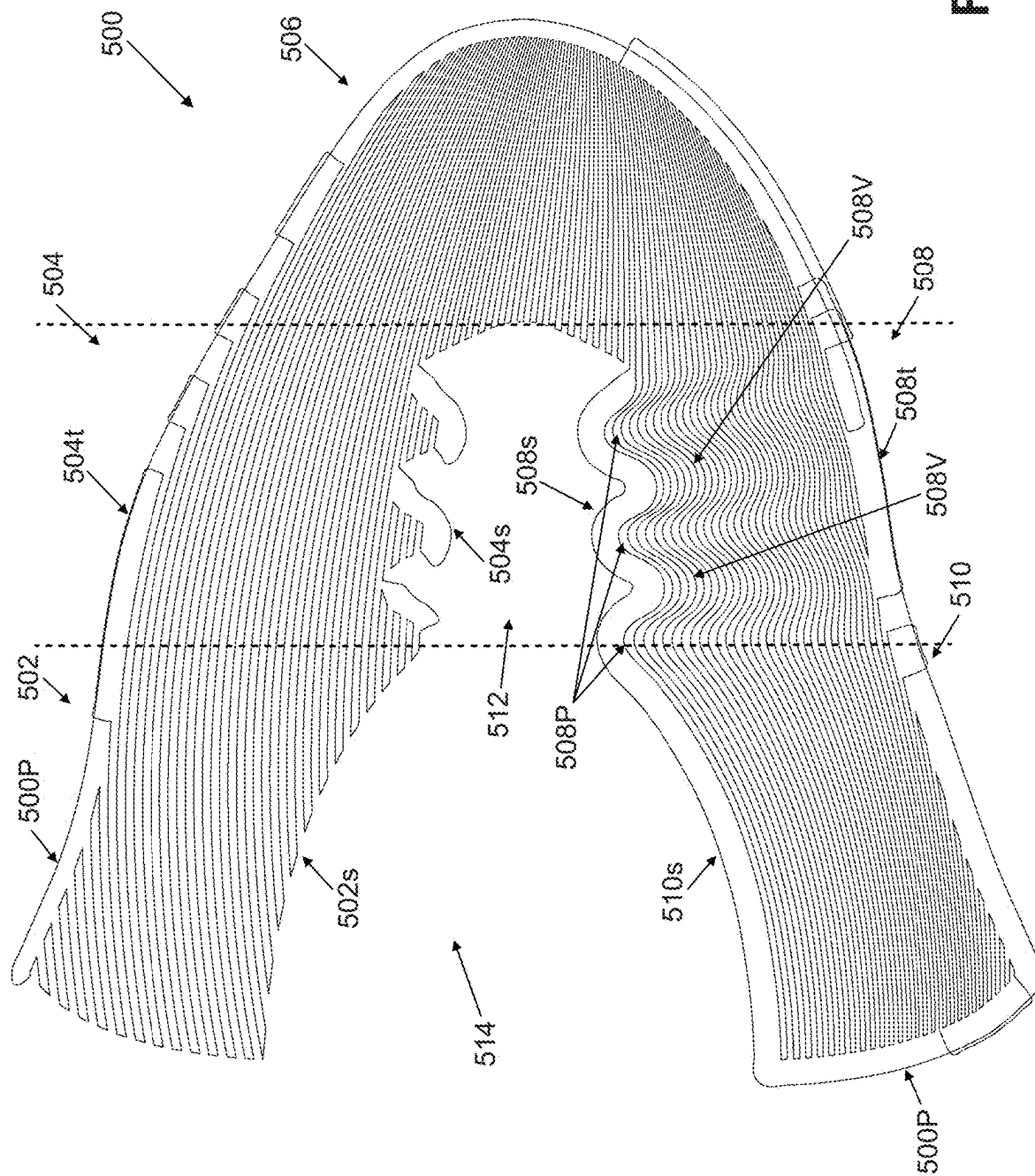
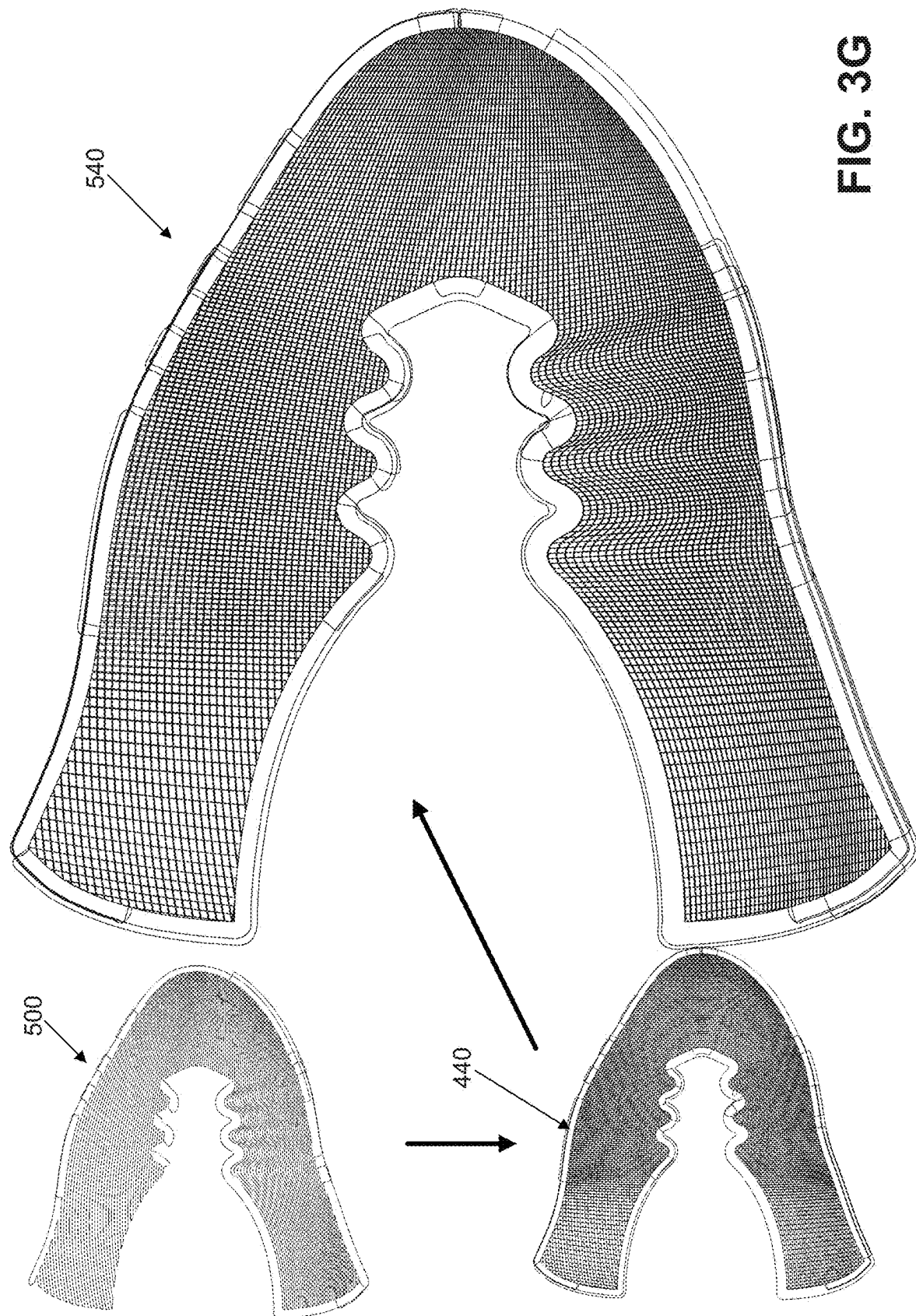


FIG. 3F



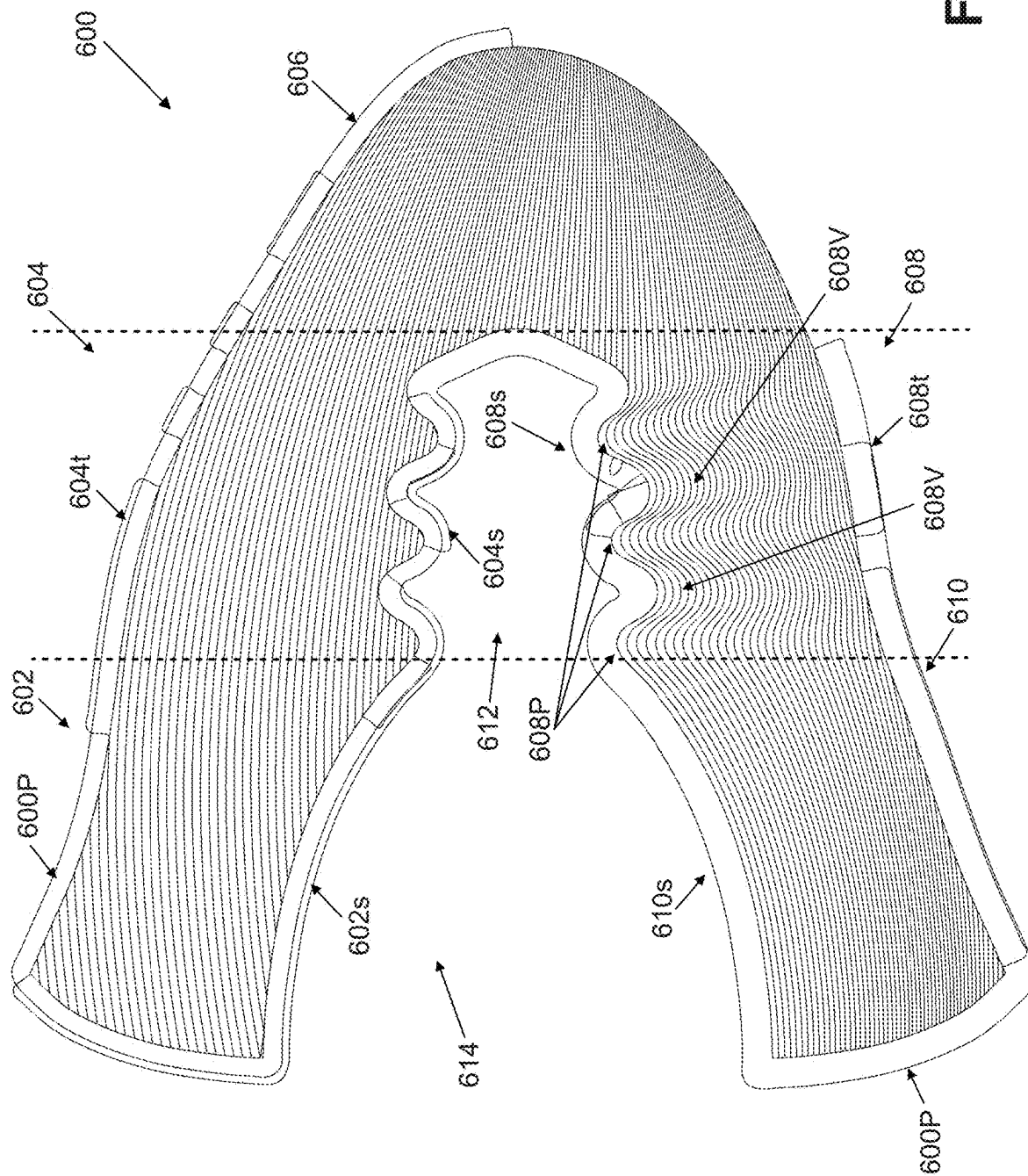
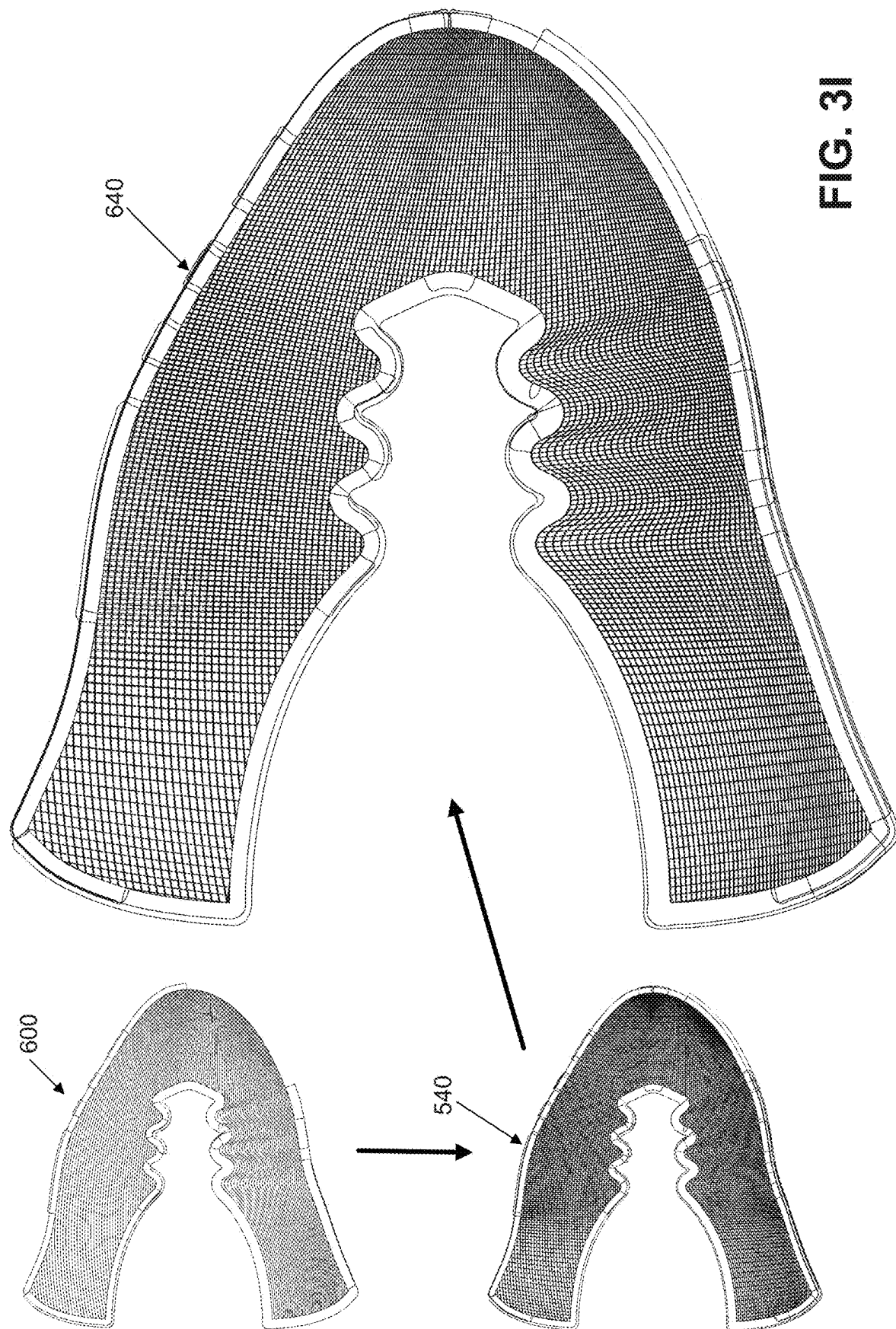


FIG. 3H



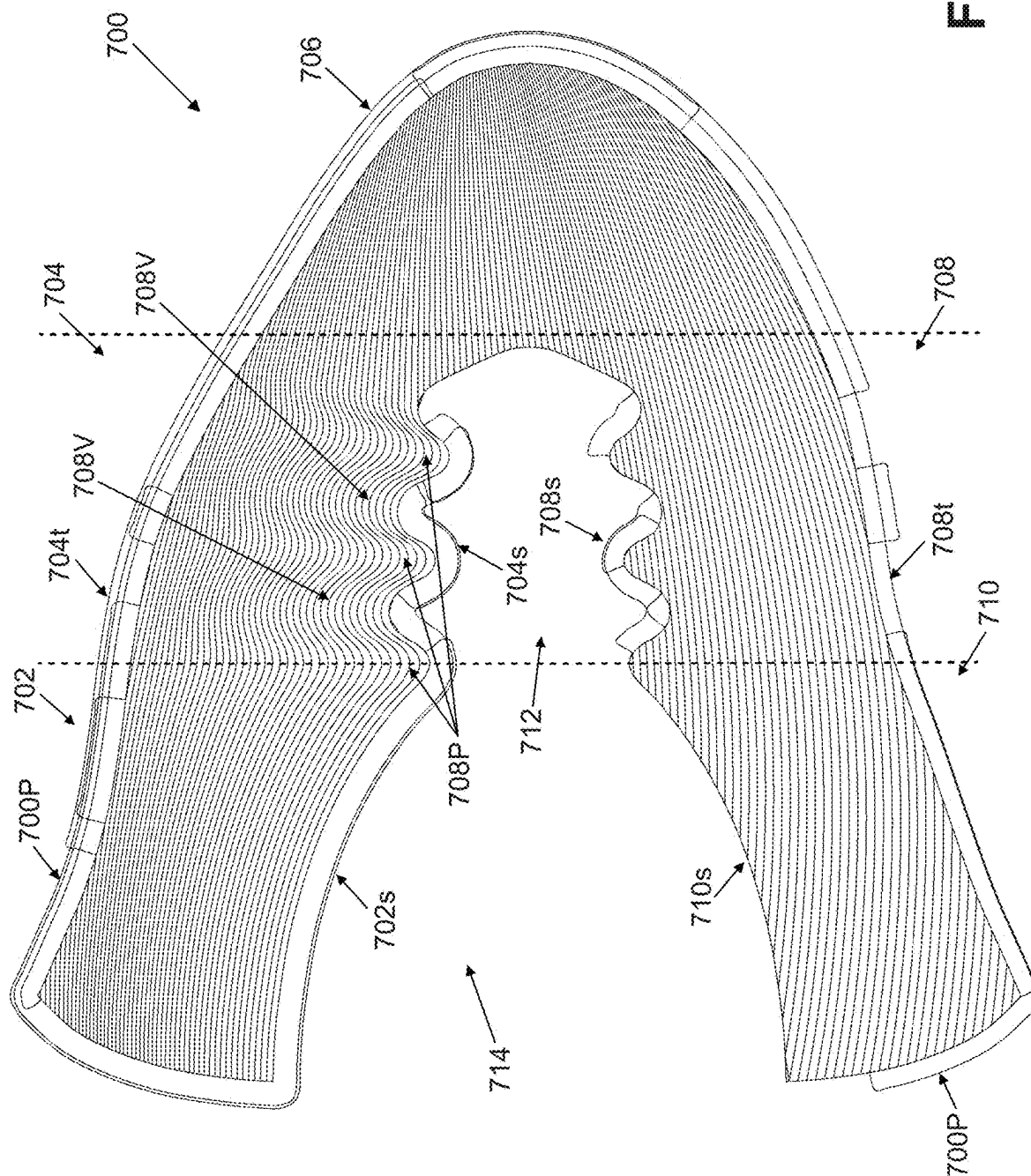
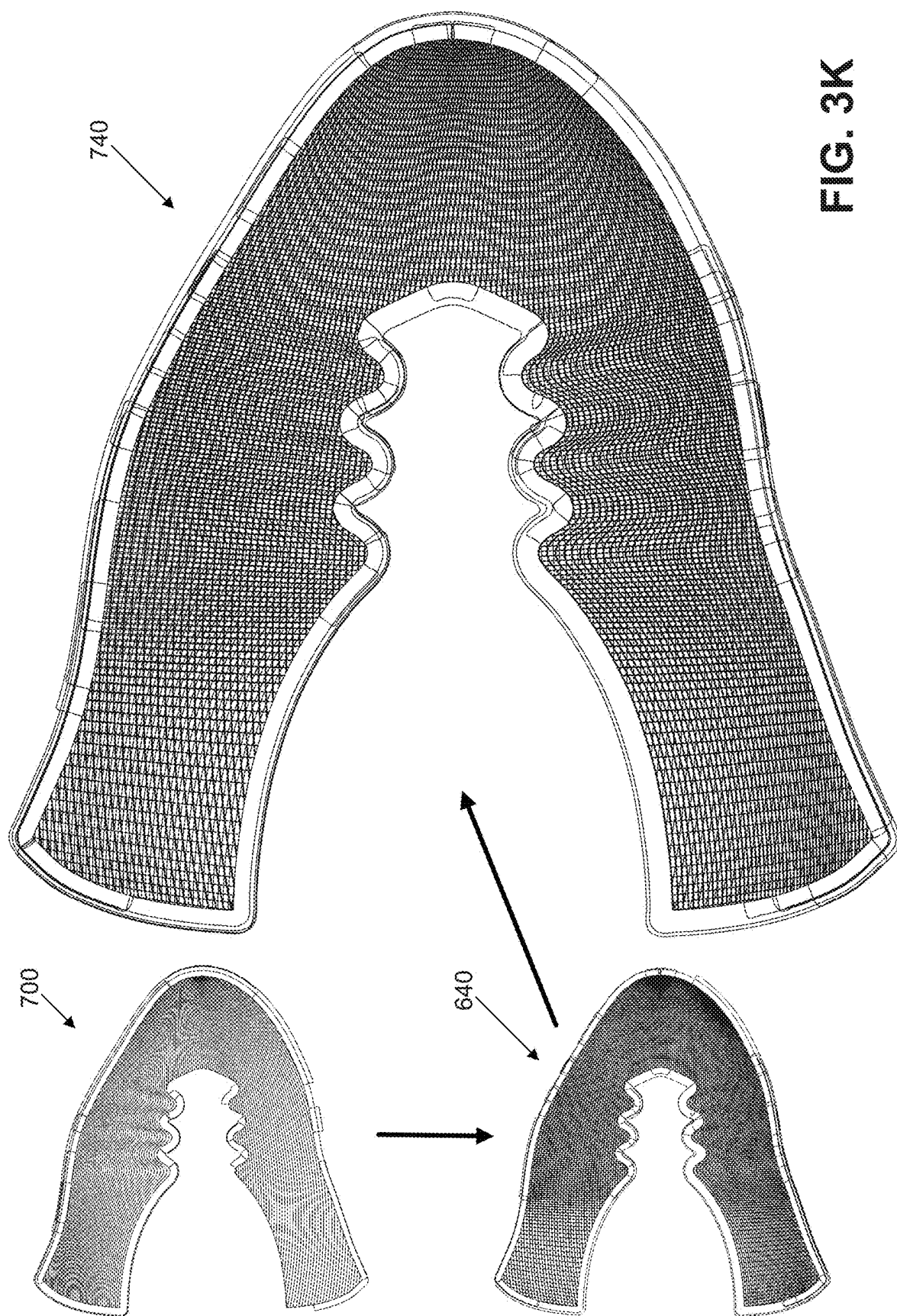


FIG. 3J



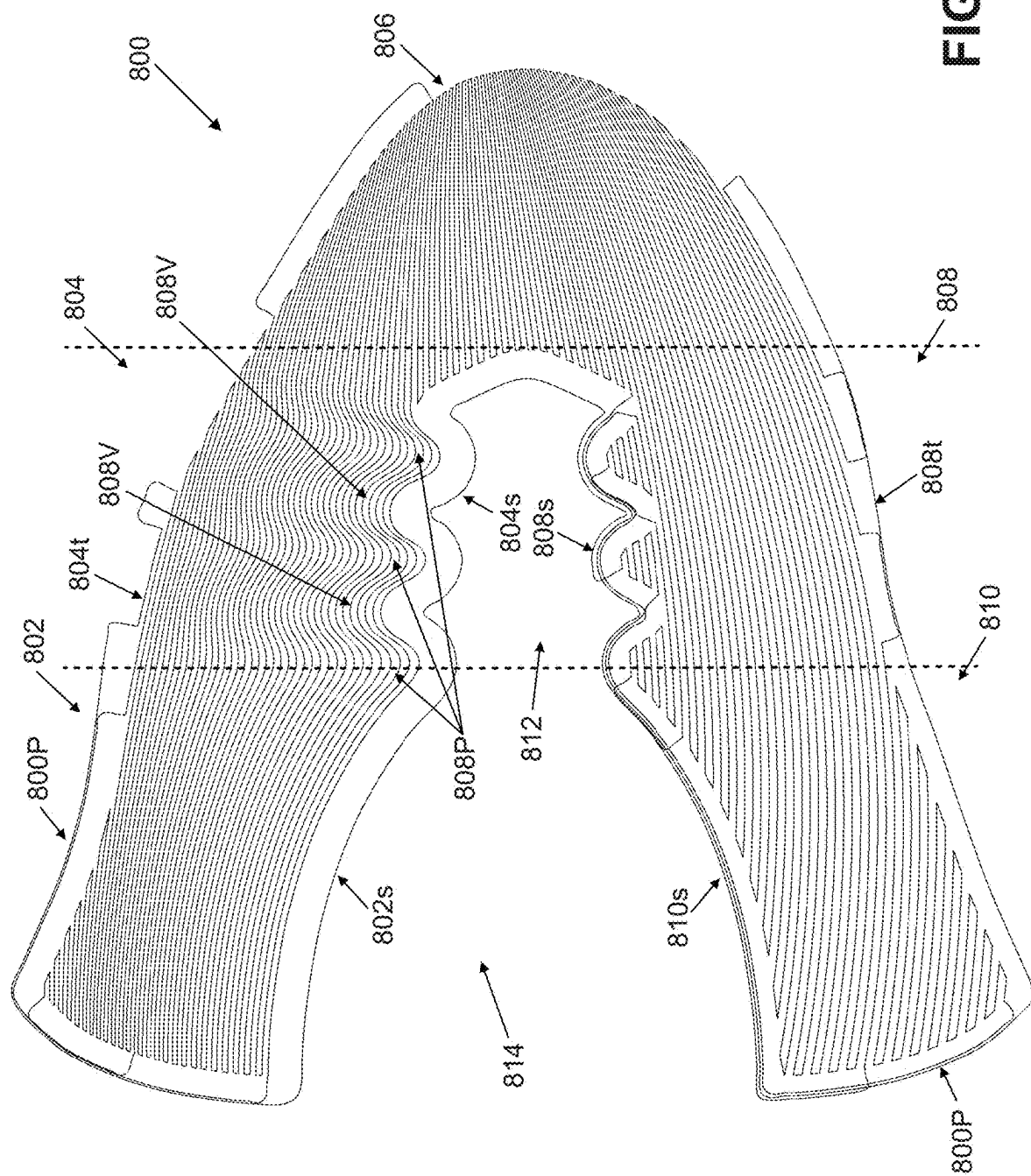
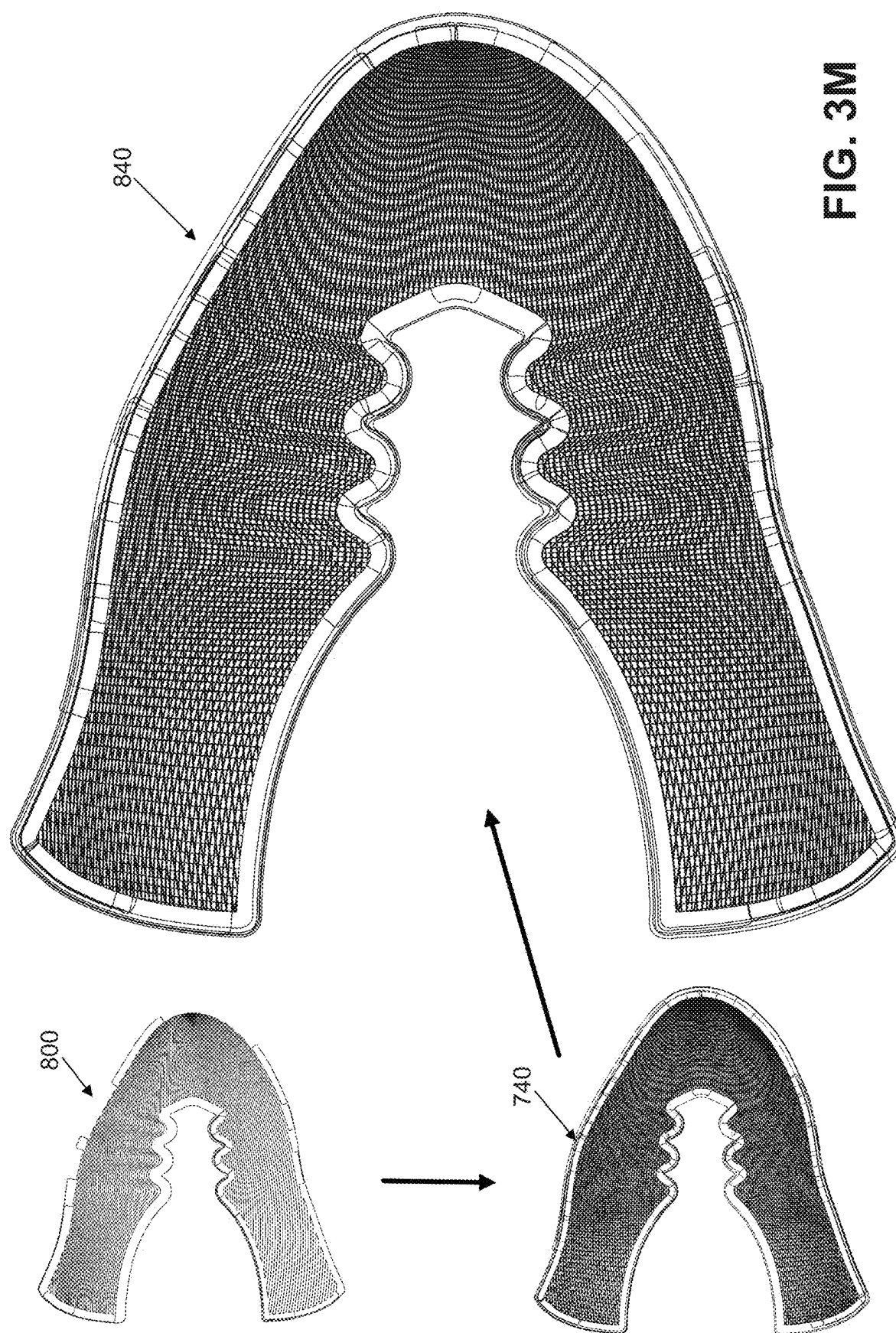


FIG. 3L



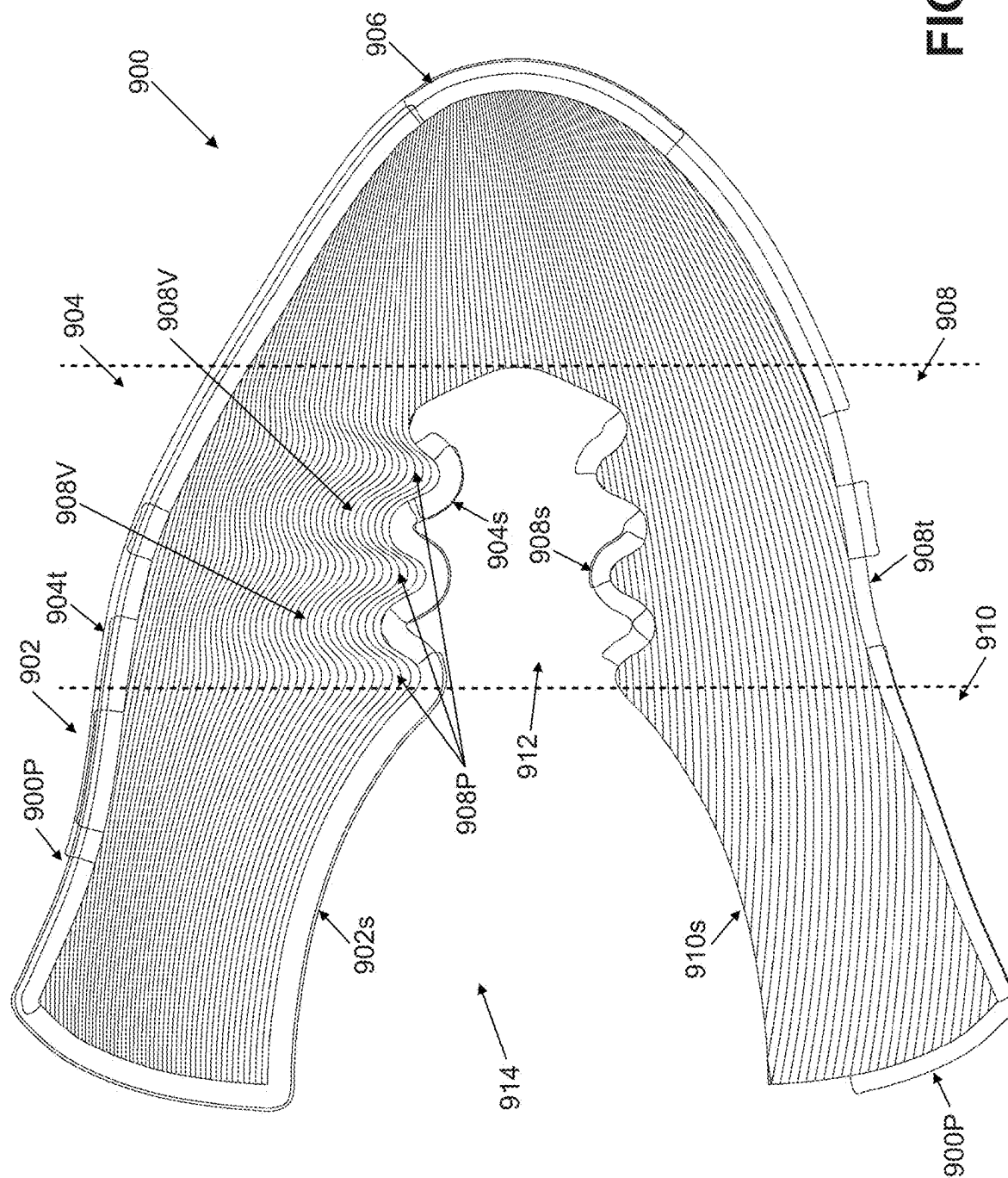
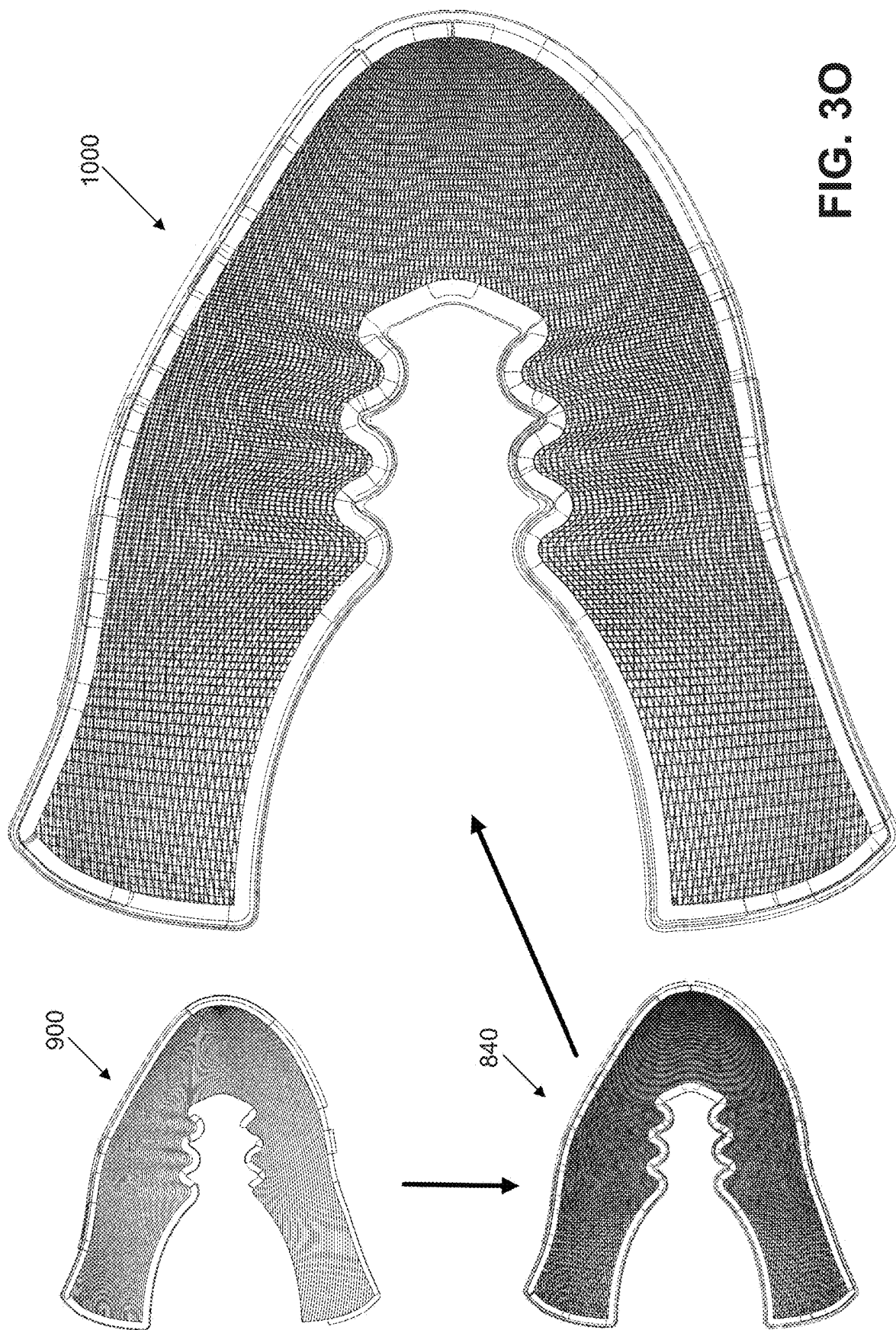


FIG. 3N



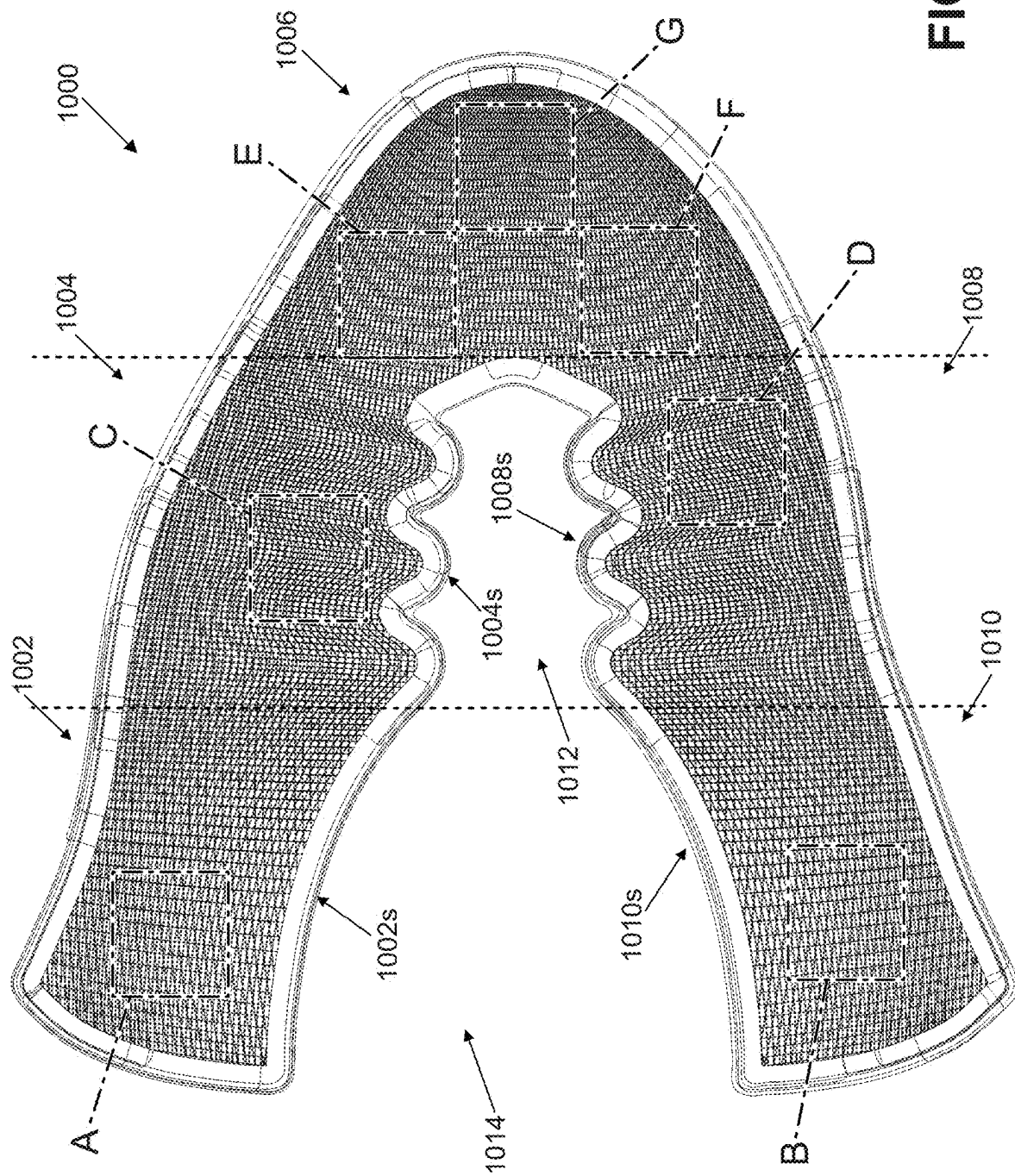


Fig. 3

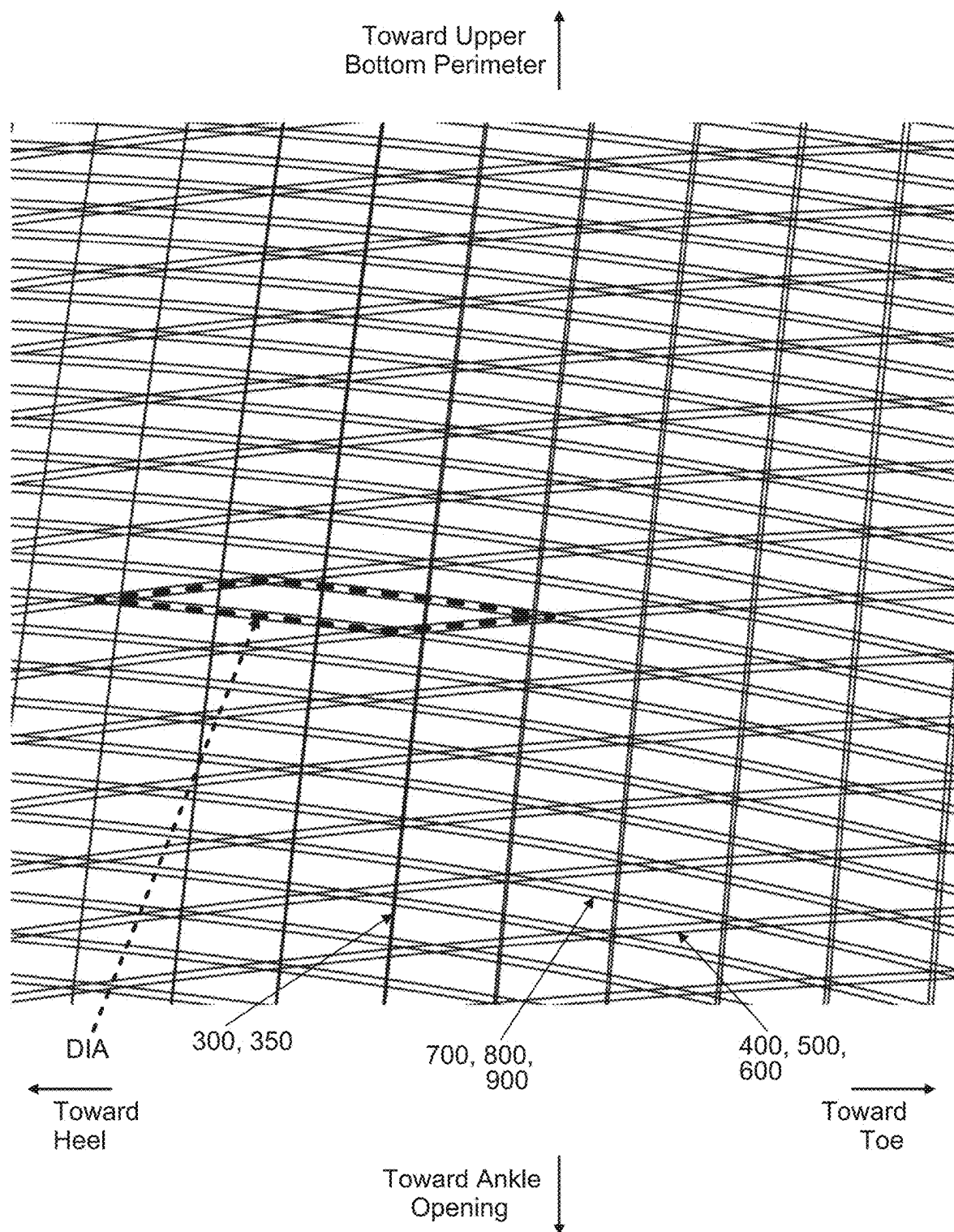


FIG. 3Q
Area A

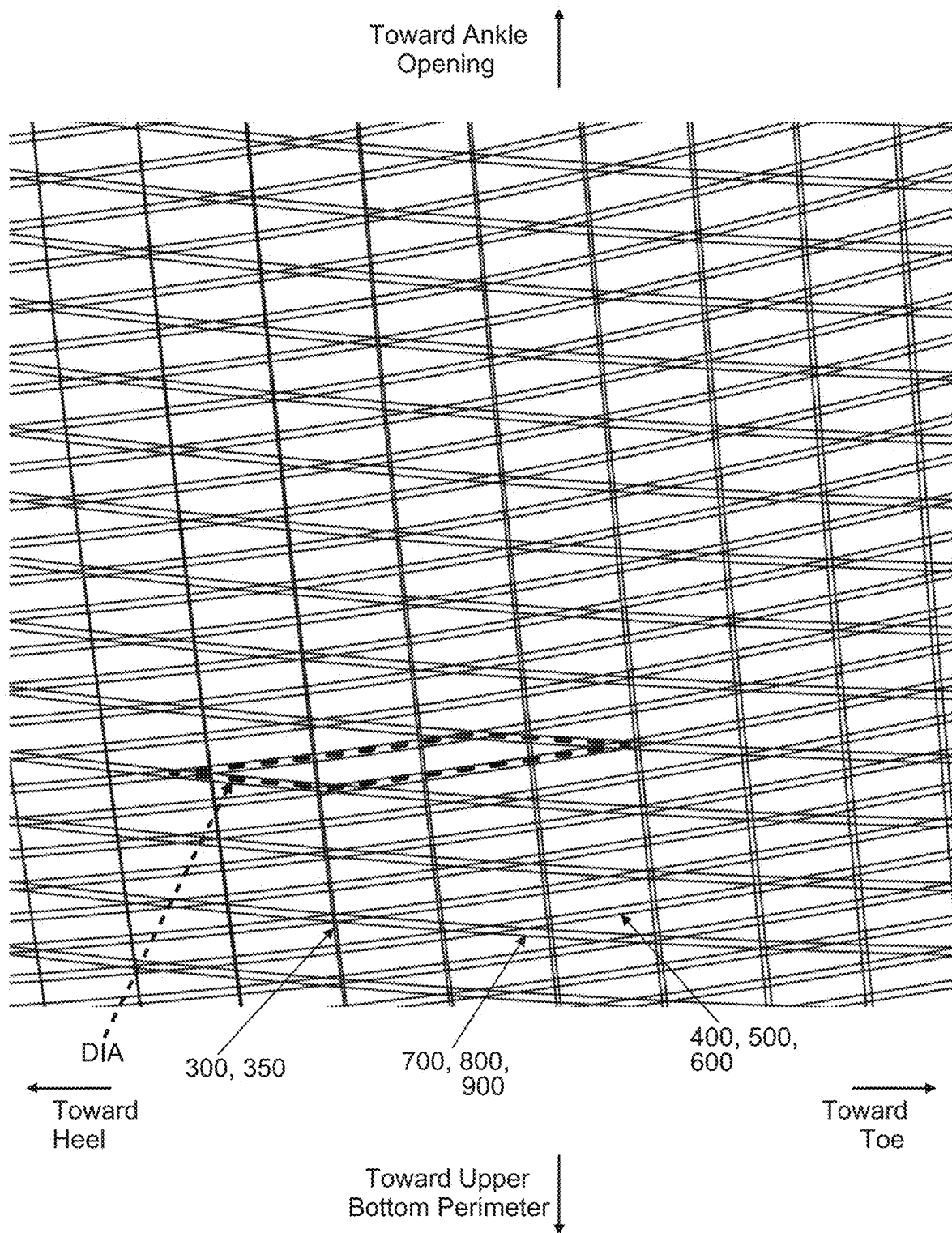


FIG. 3R
Area B

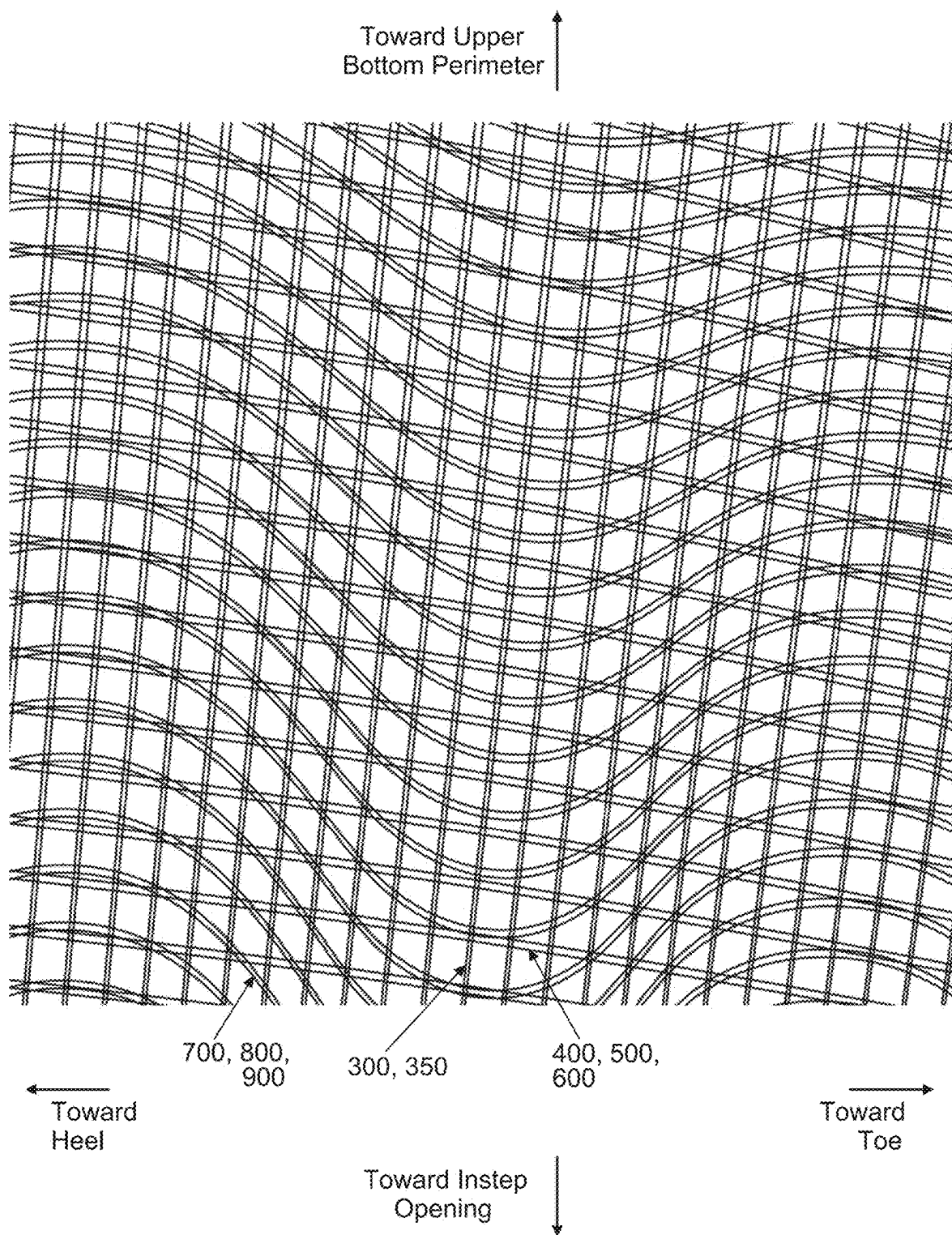


FIG. 3S
Area C

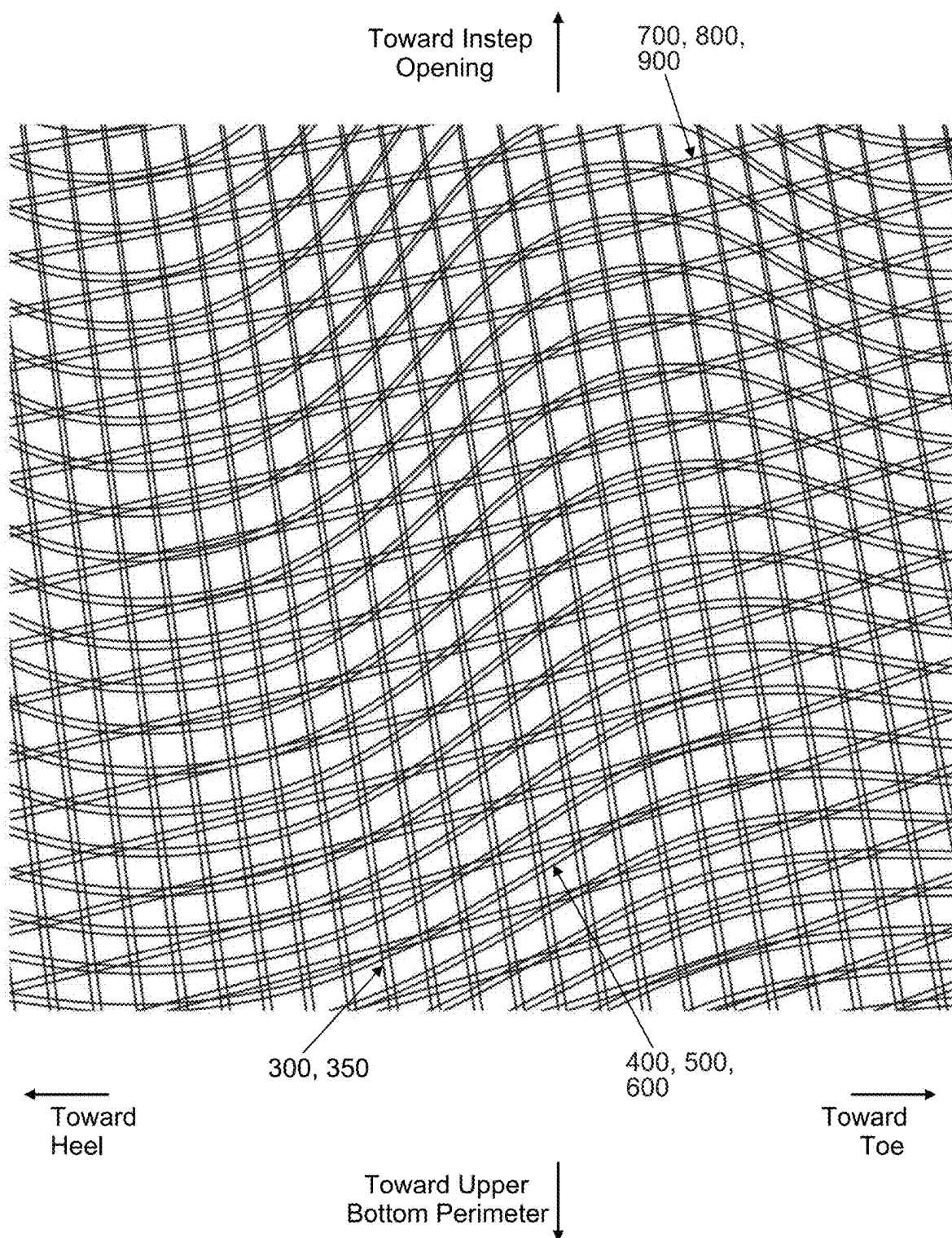


FIG. 3T
Area D

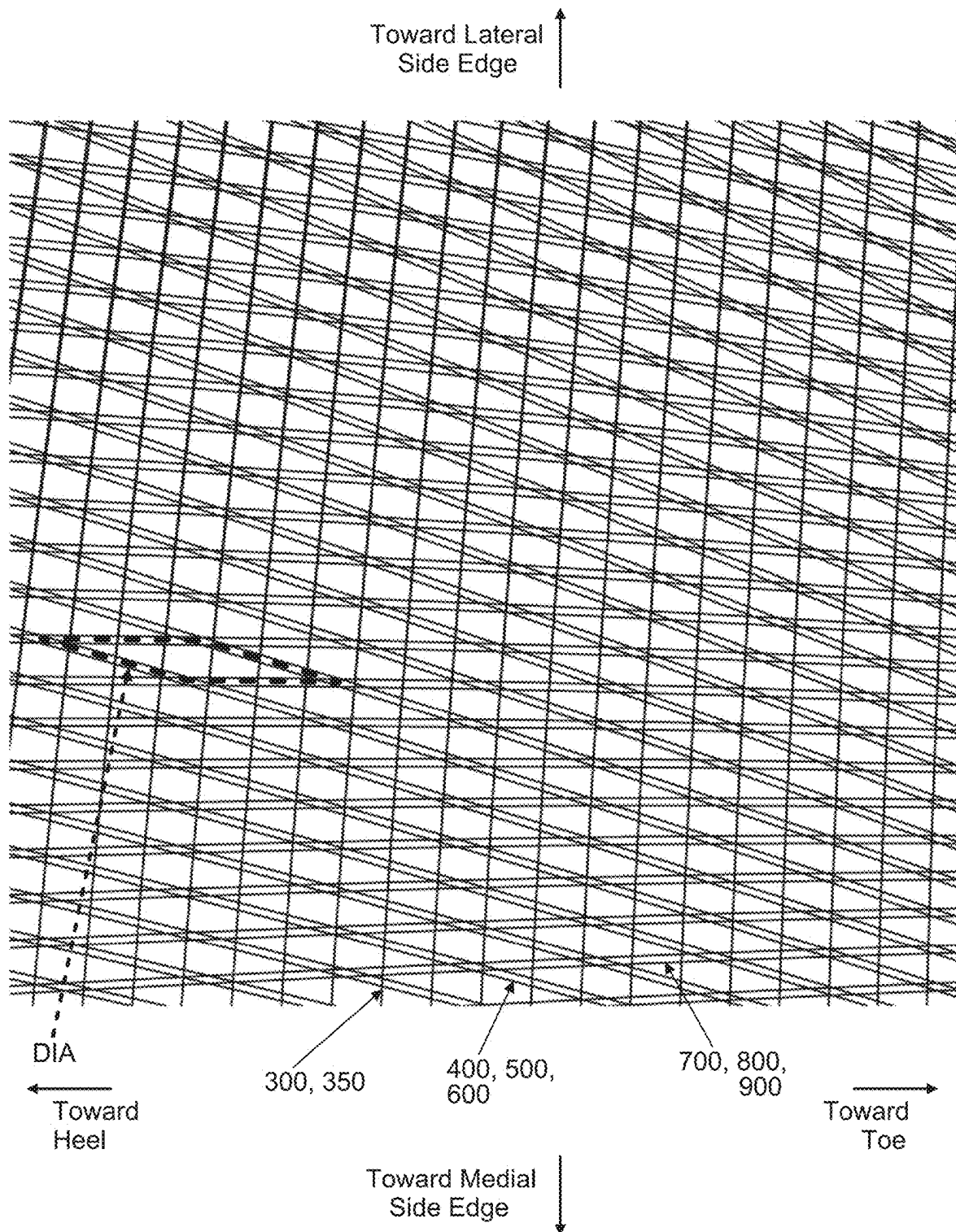


FIG. 3U
Area E

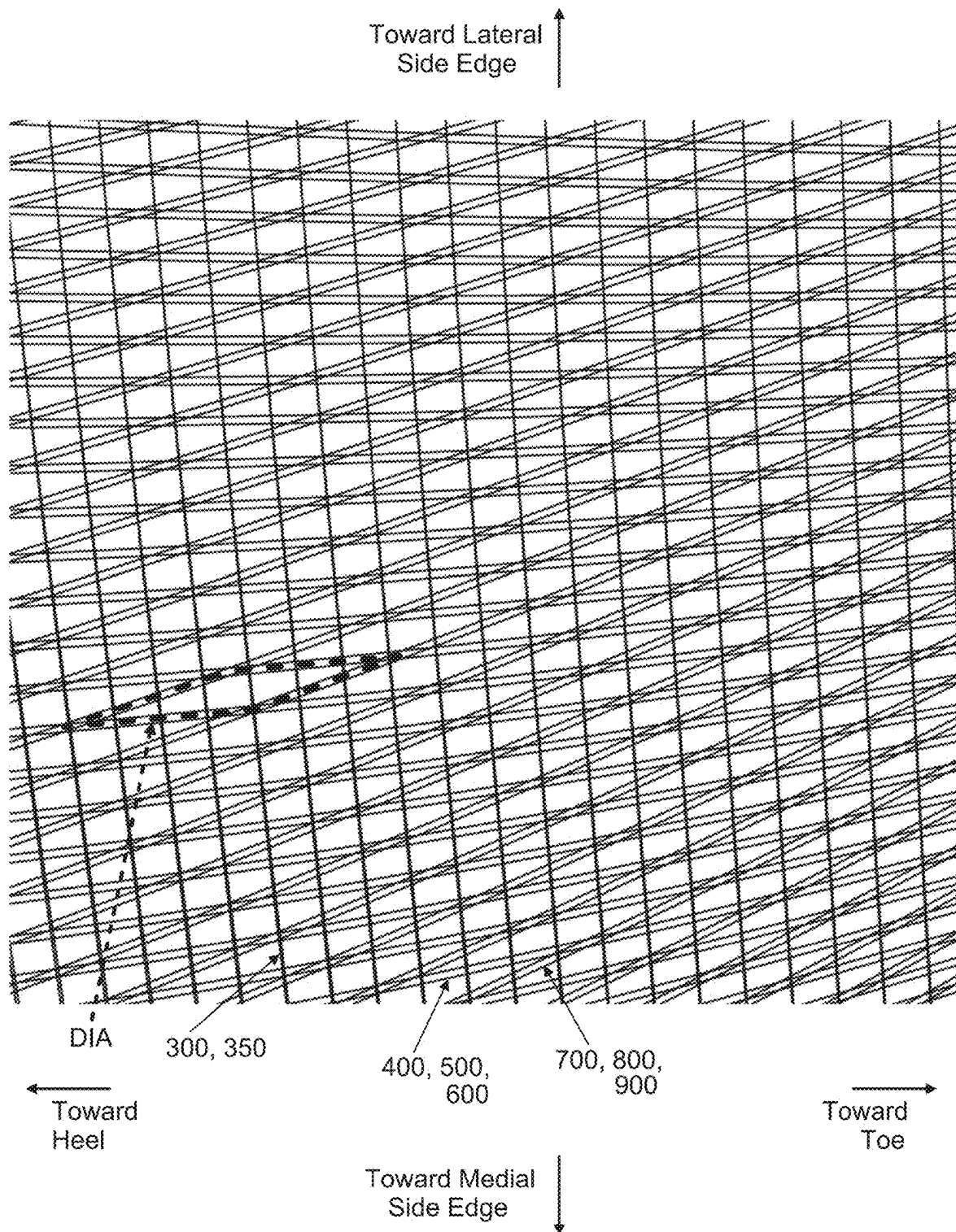


FIG. 3V
Area F

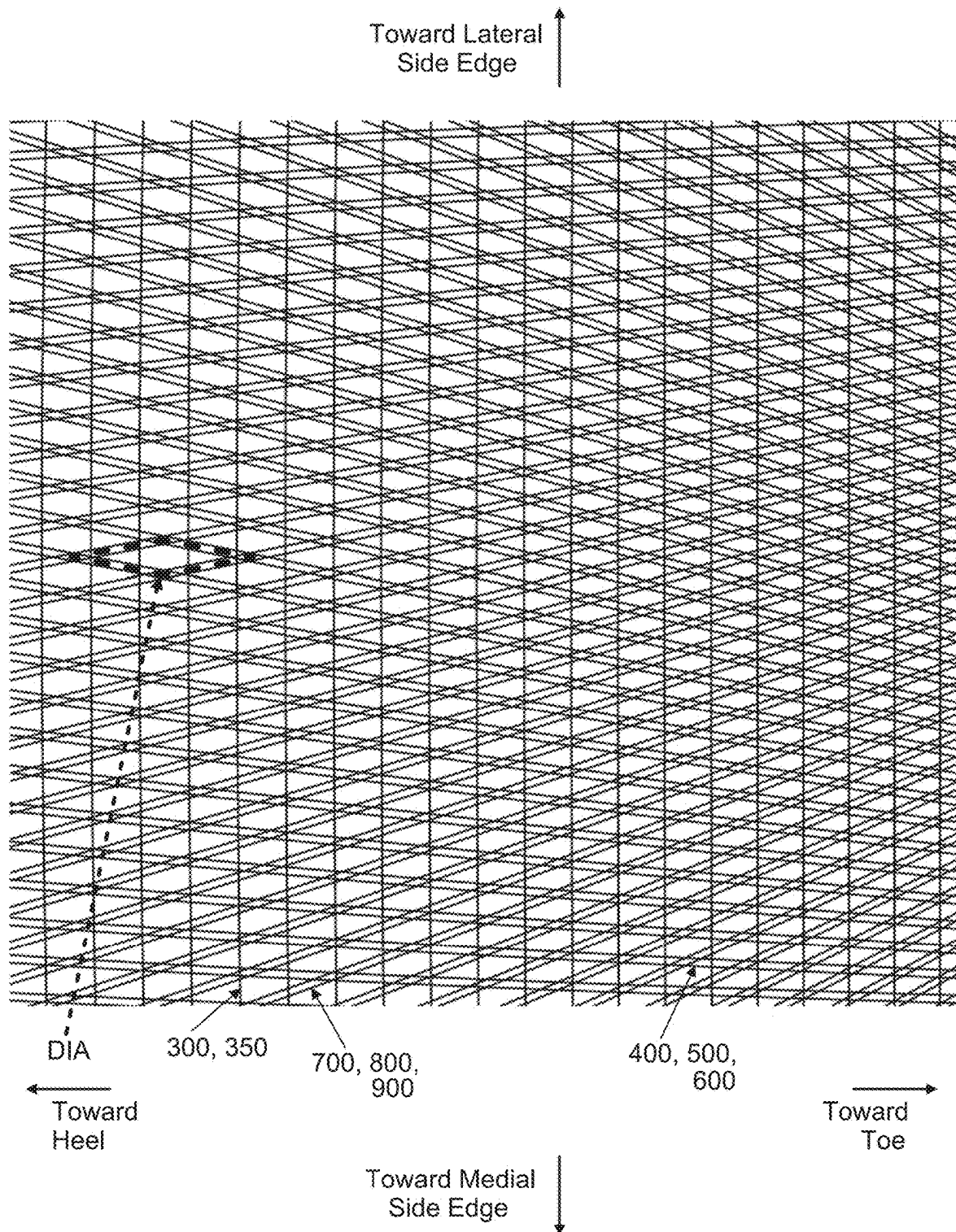


FIG. 3W
Area G

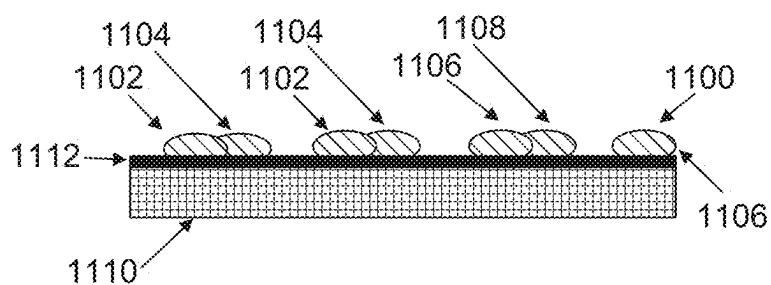


FIG. 4A

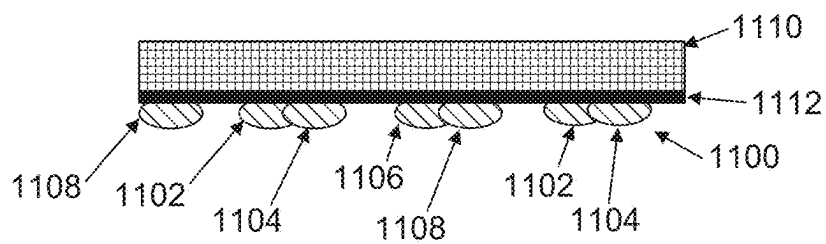


FIG. 4B

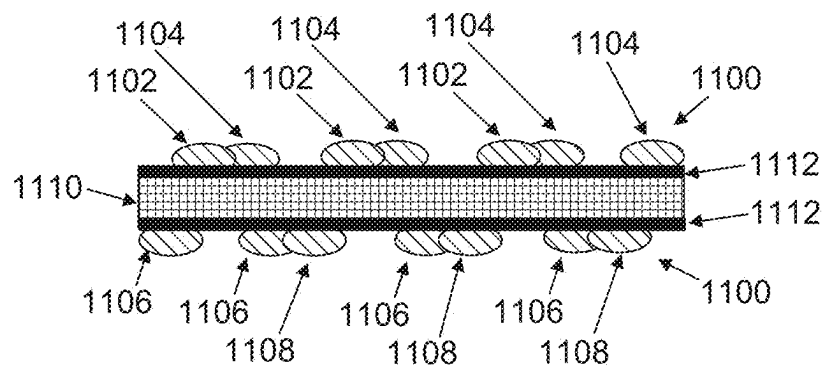
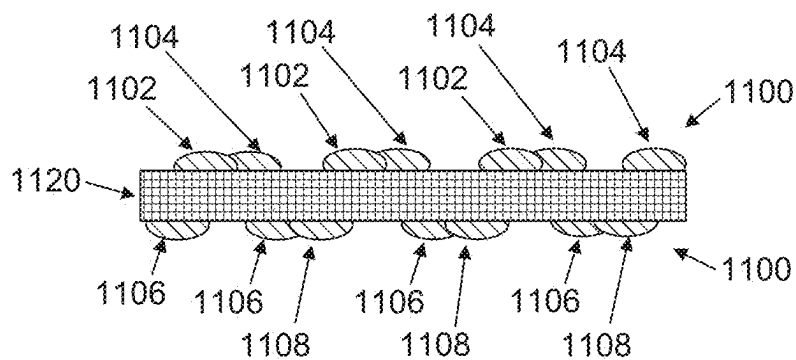
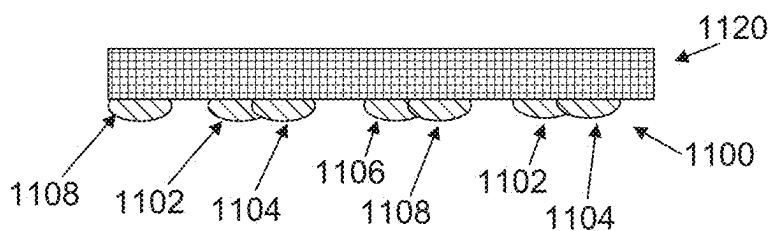
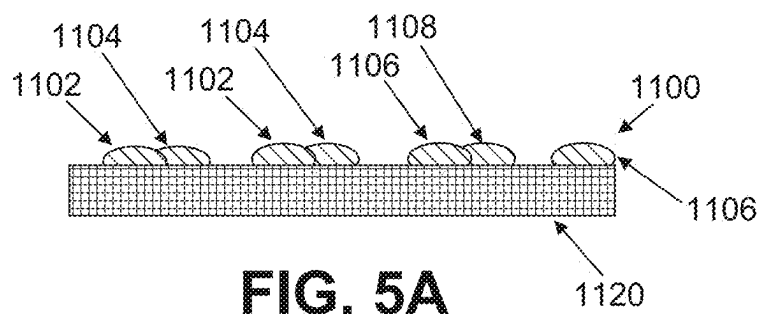


FIG. 4C



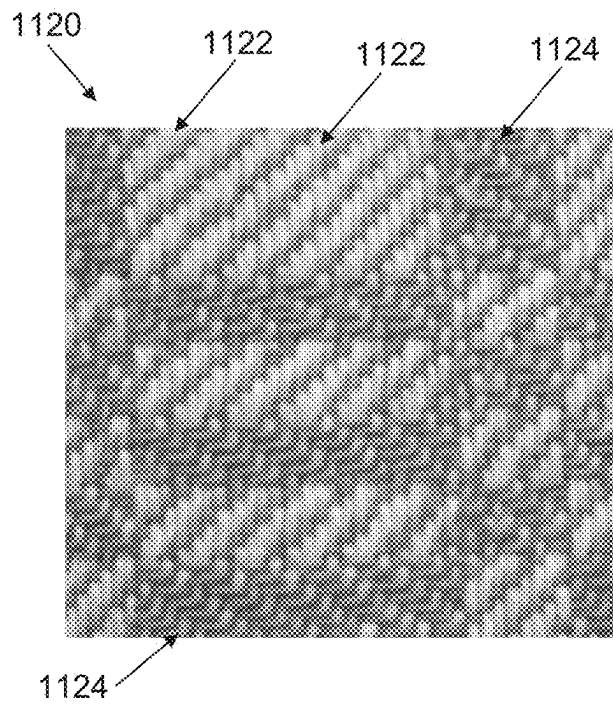


FIG. 5D

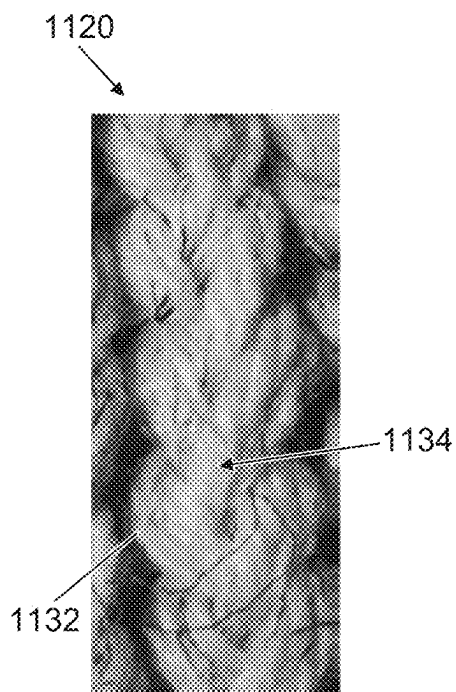


FIG. 5E

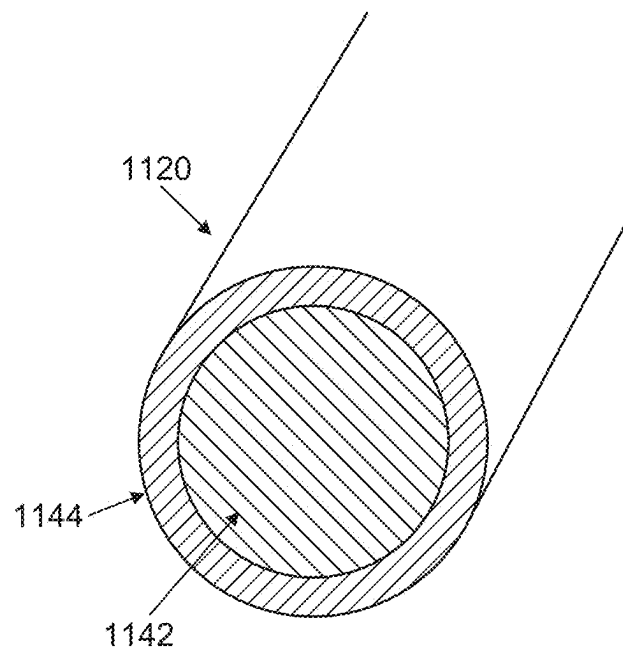


FIG. 5F

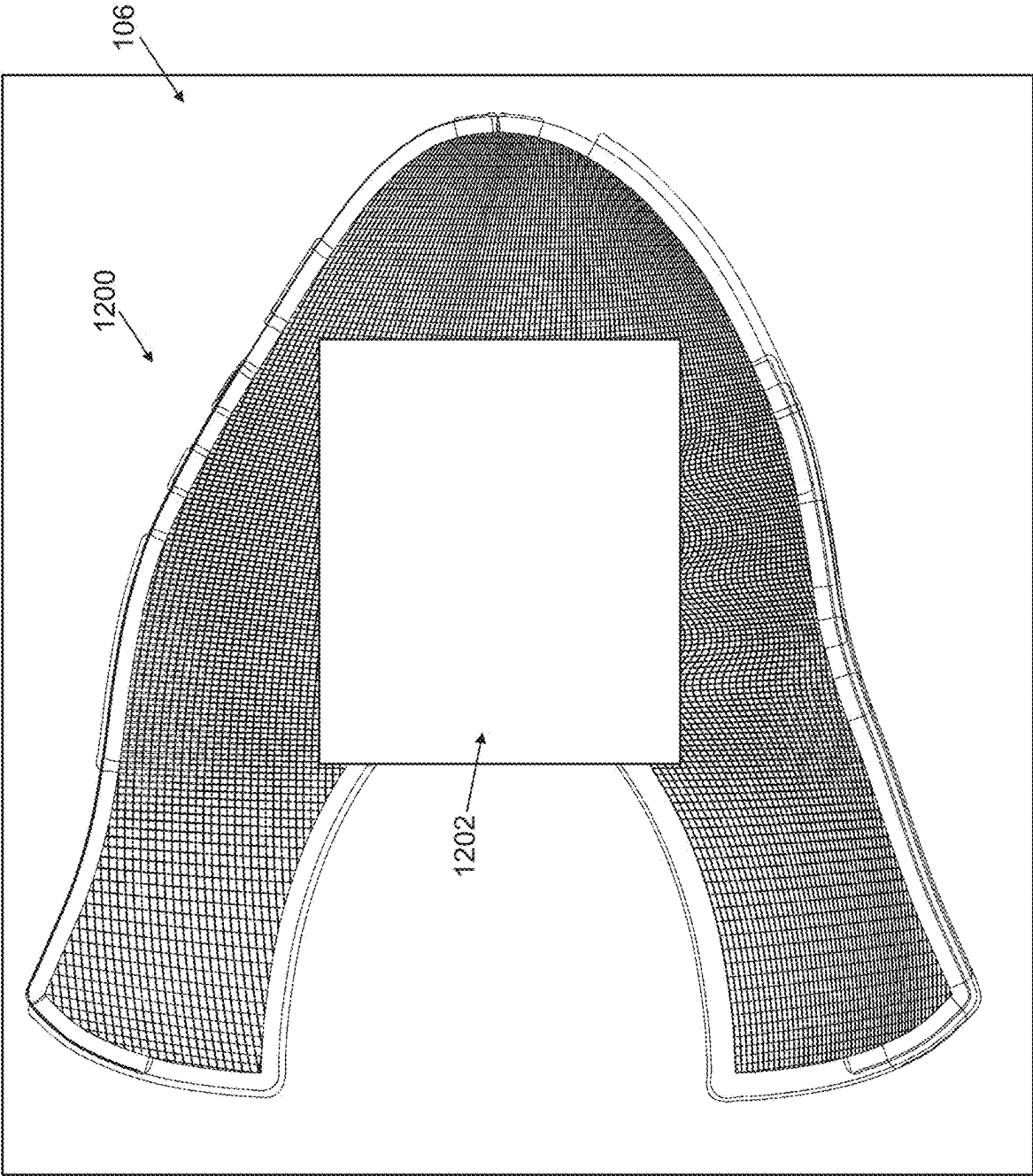
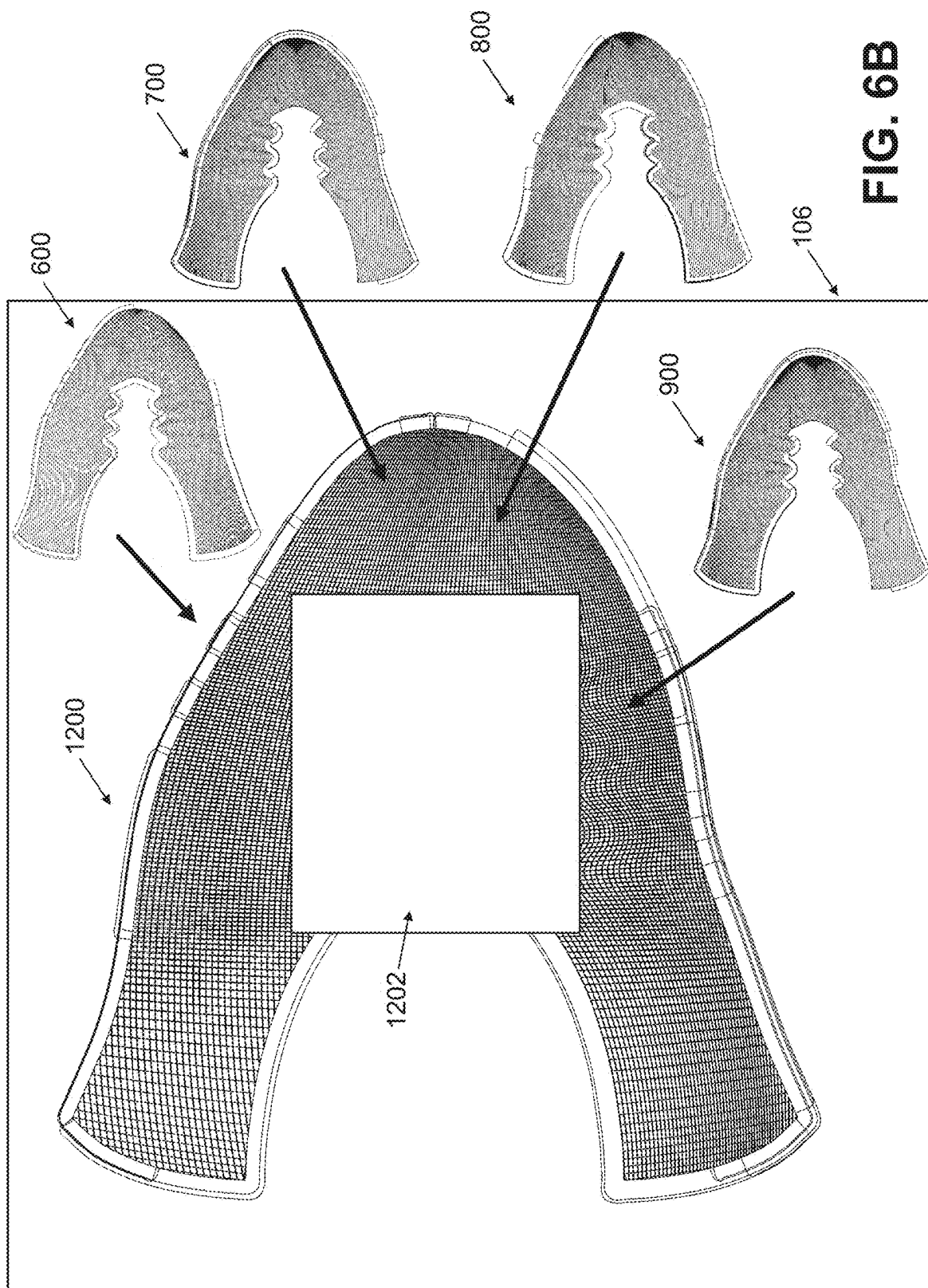


FIG. 6A



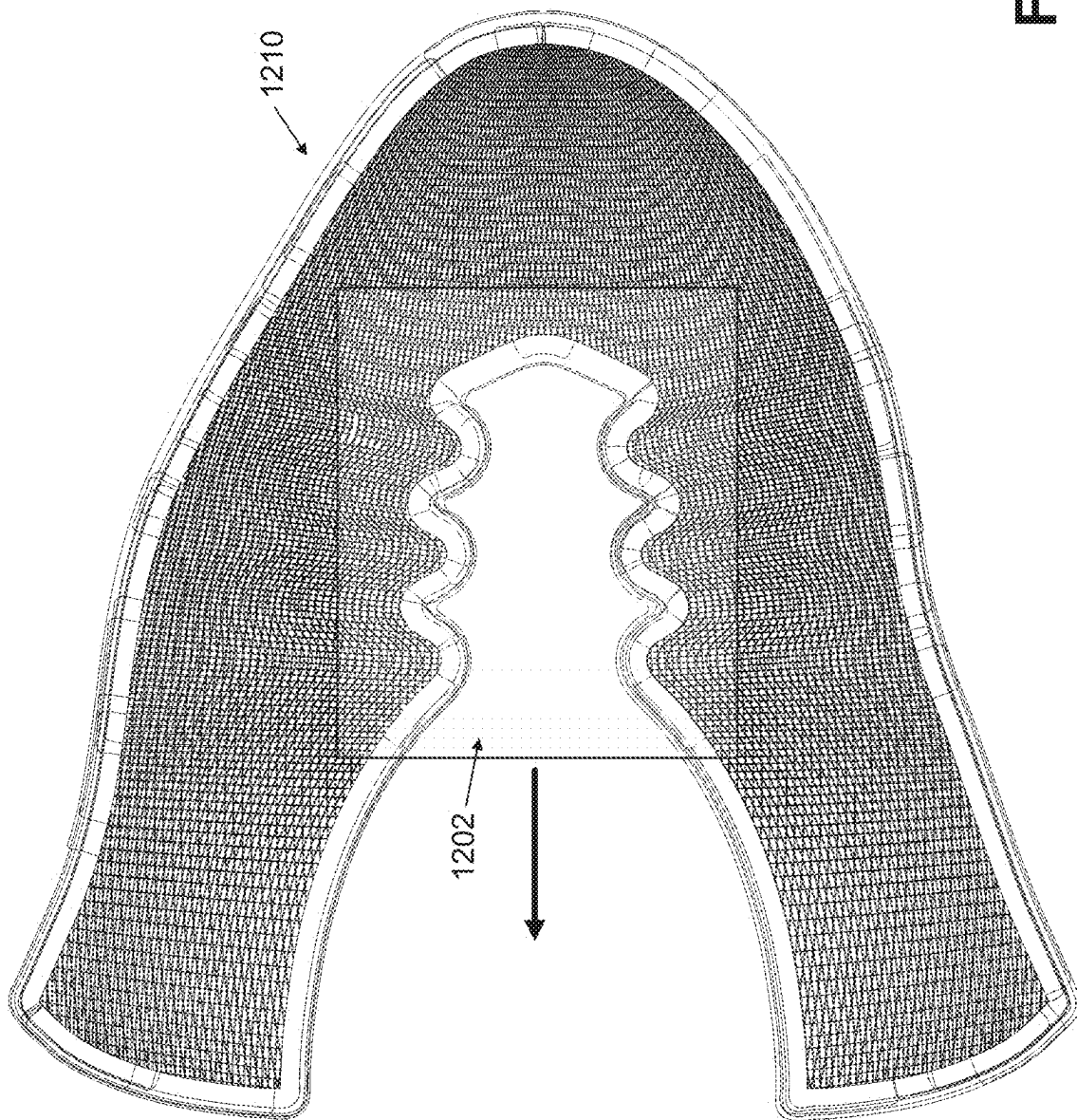


FIG. 6C

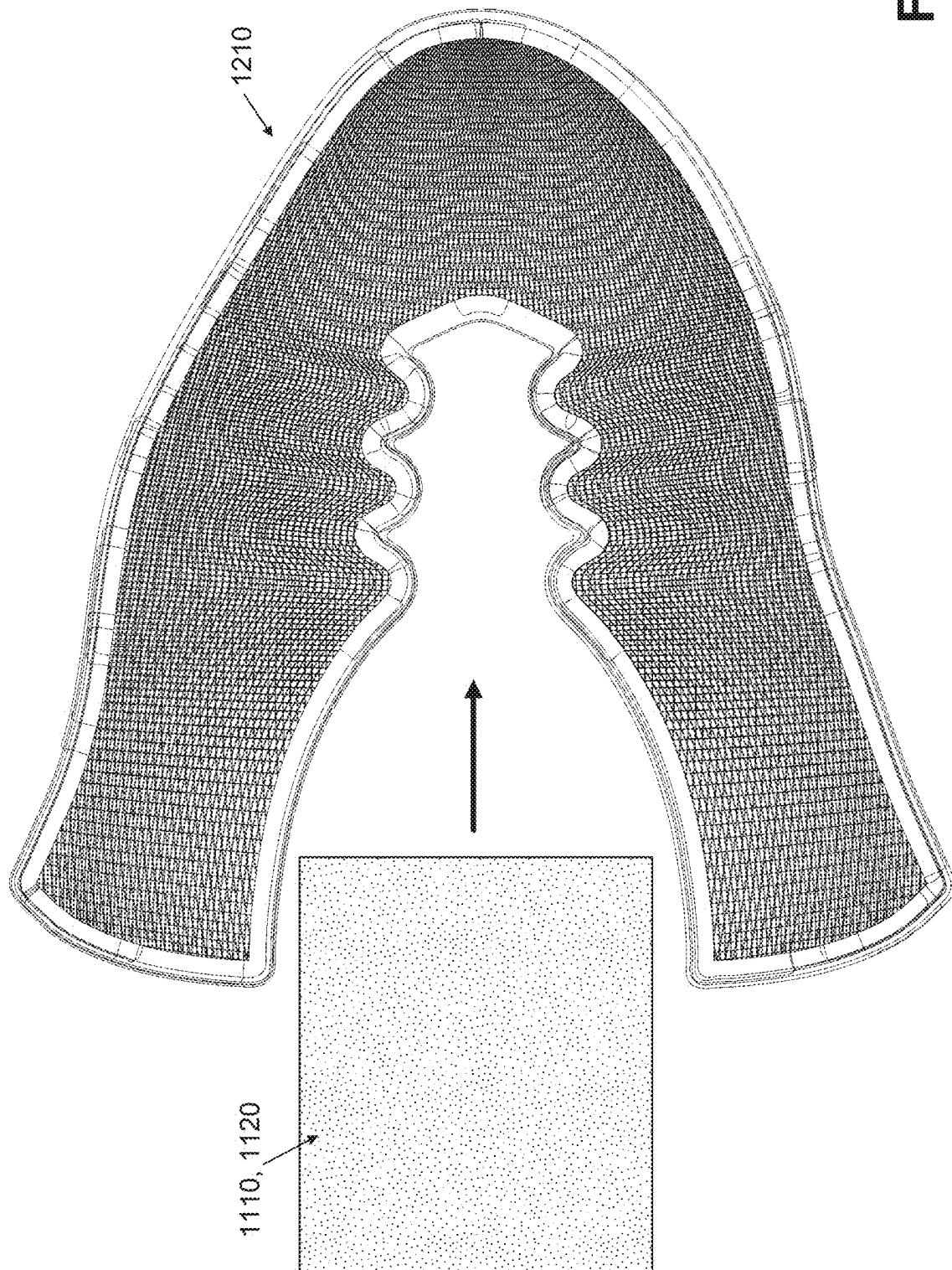


FIG. 6D

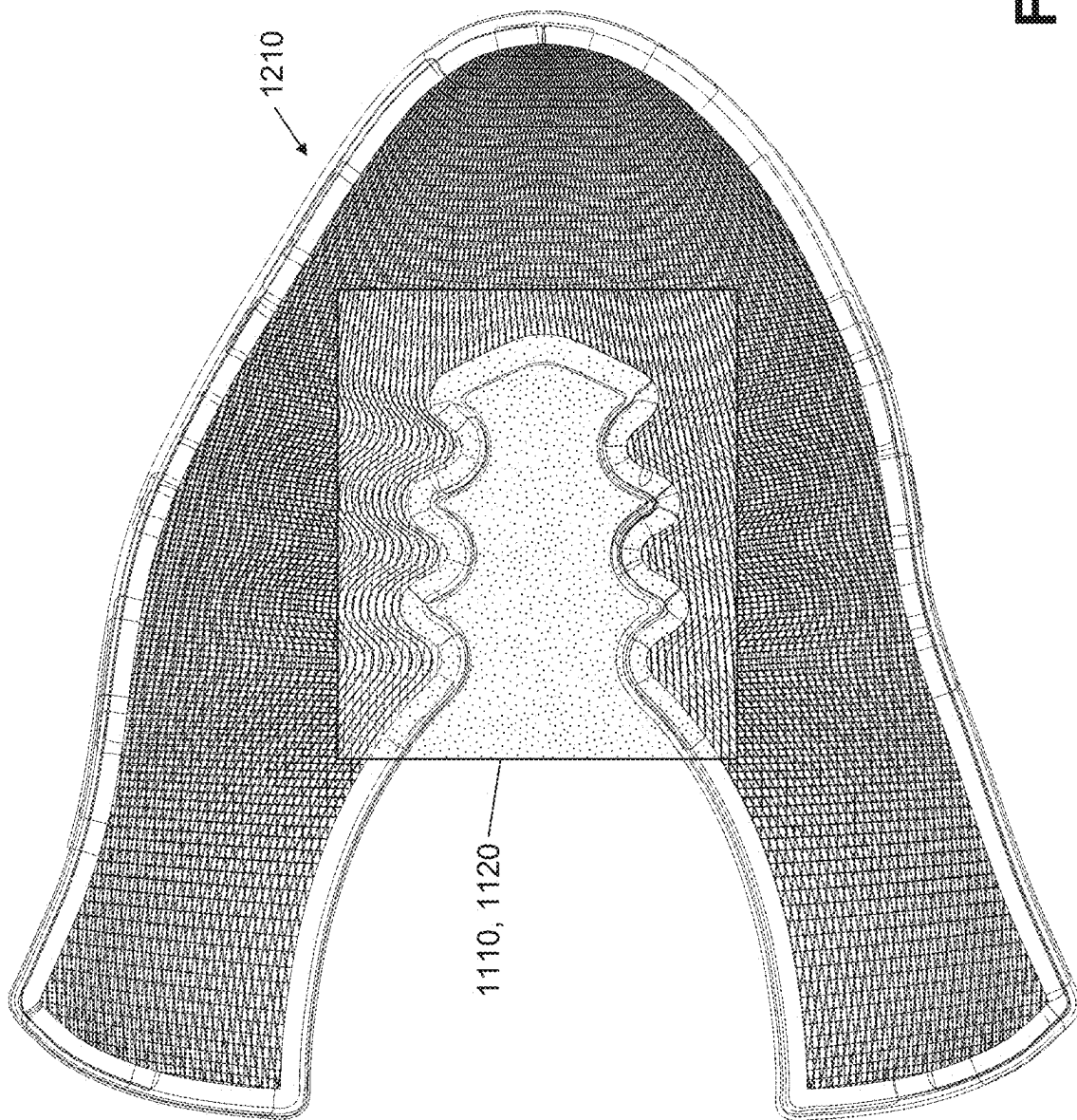


FIG. 6E

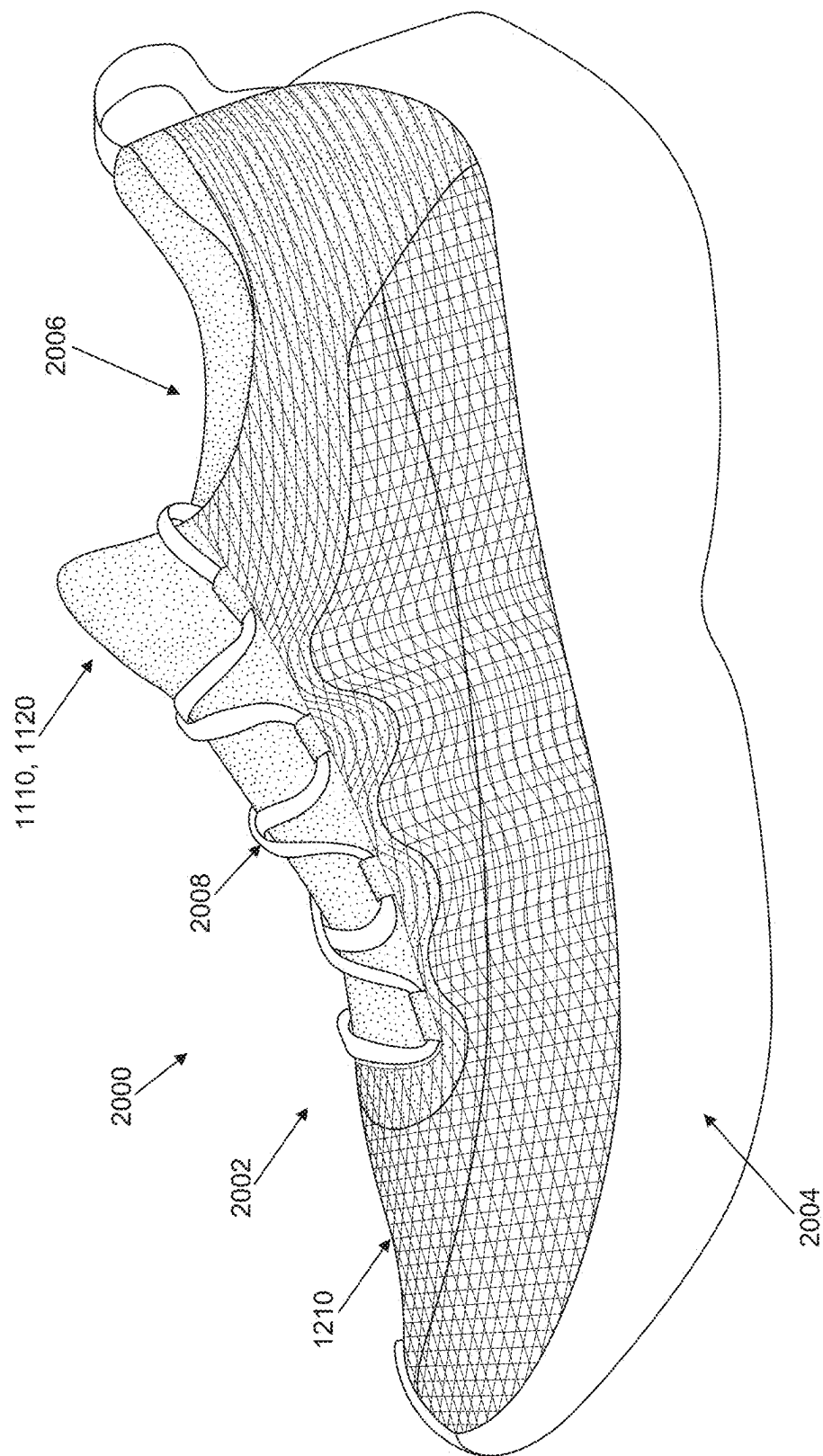


FIG. 7A

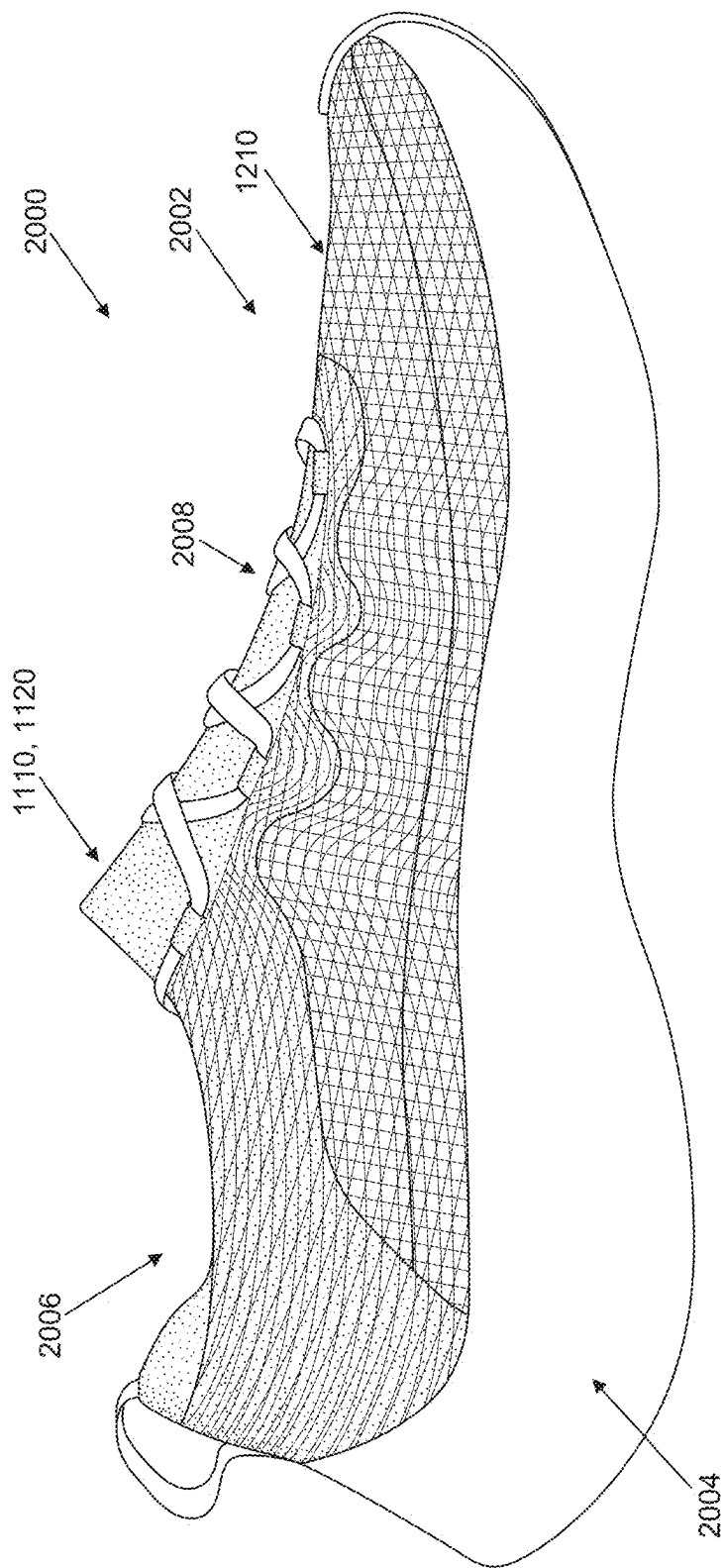


FIG. 7B

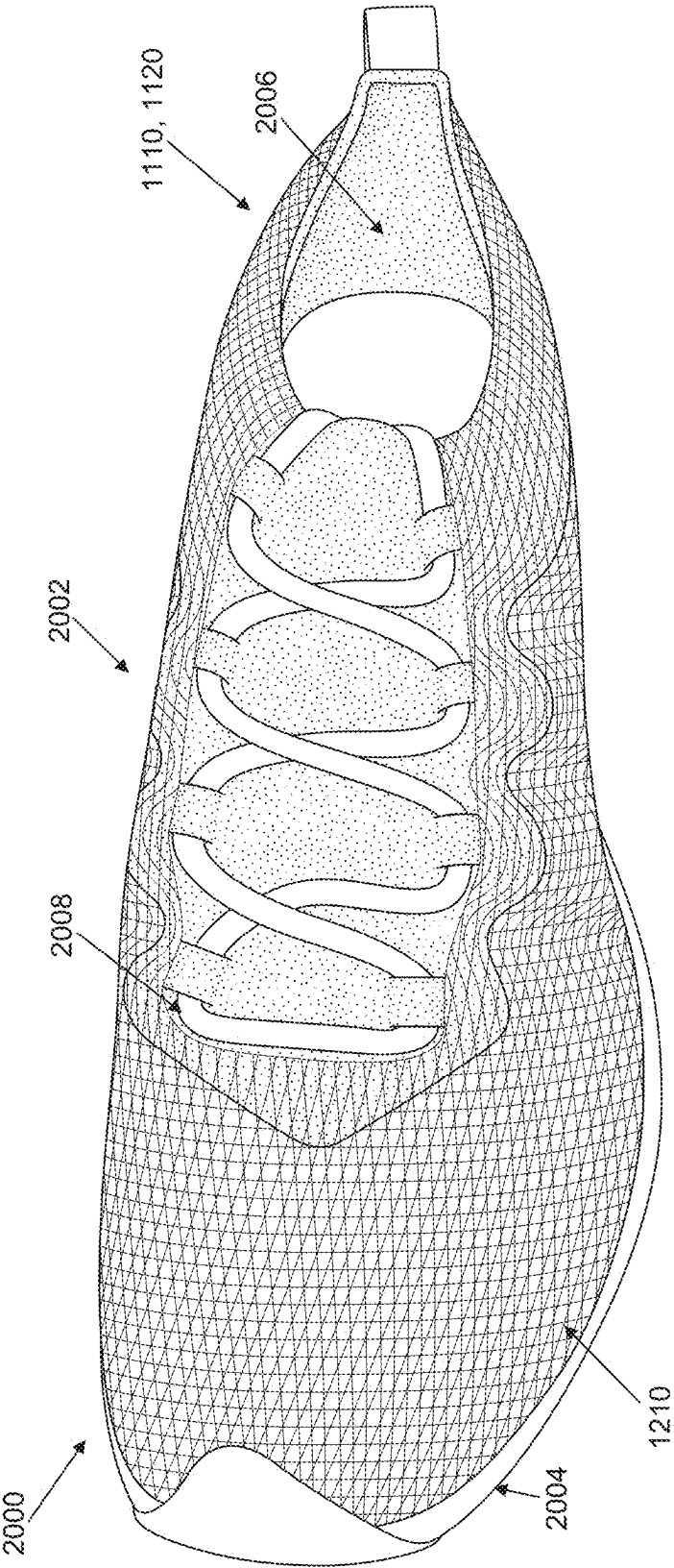


FIG. 7C

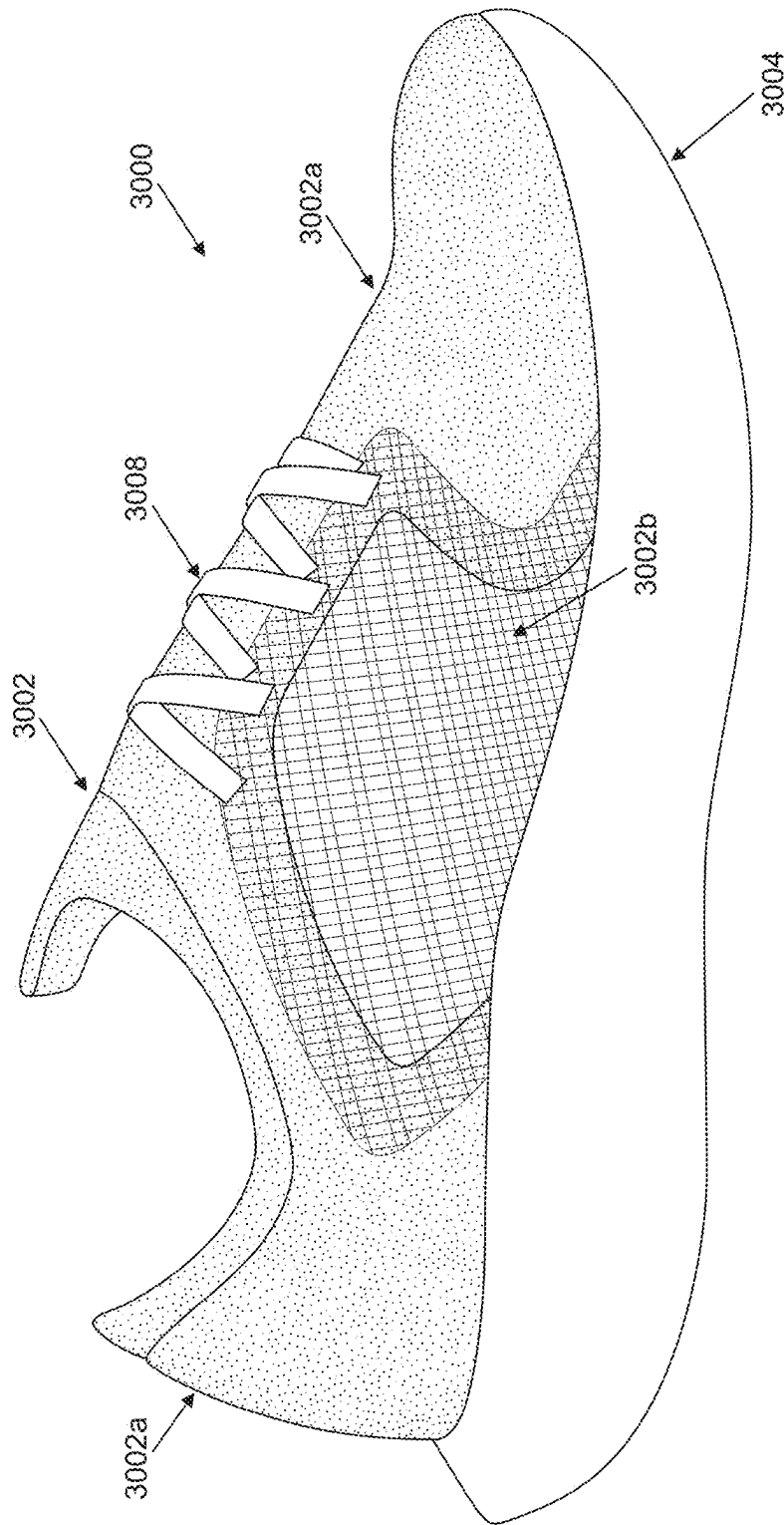


FIG. 8A

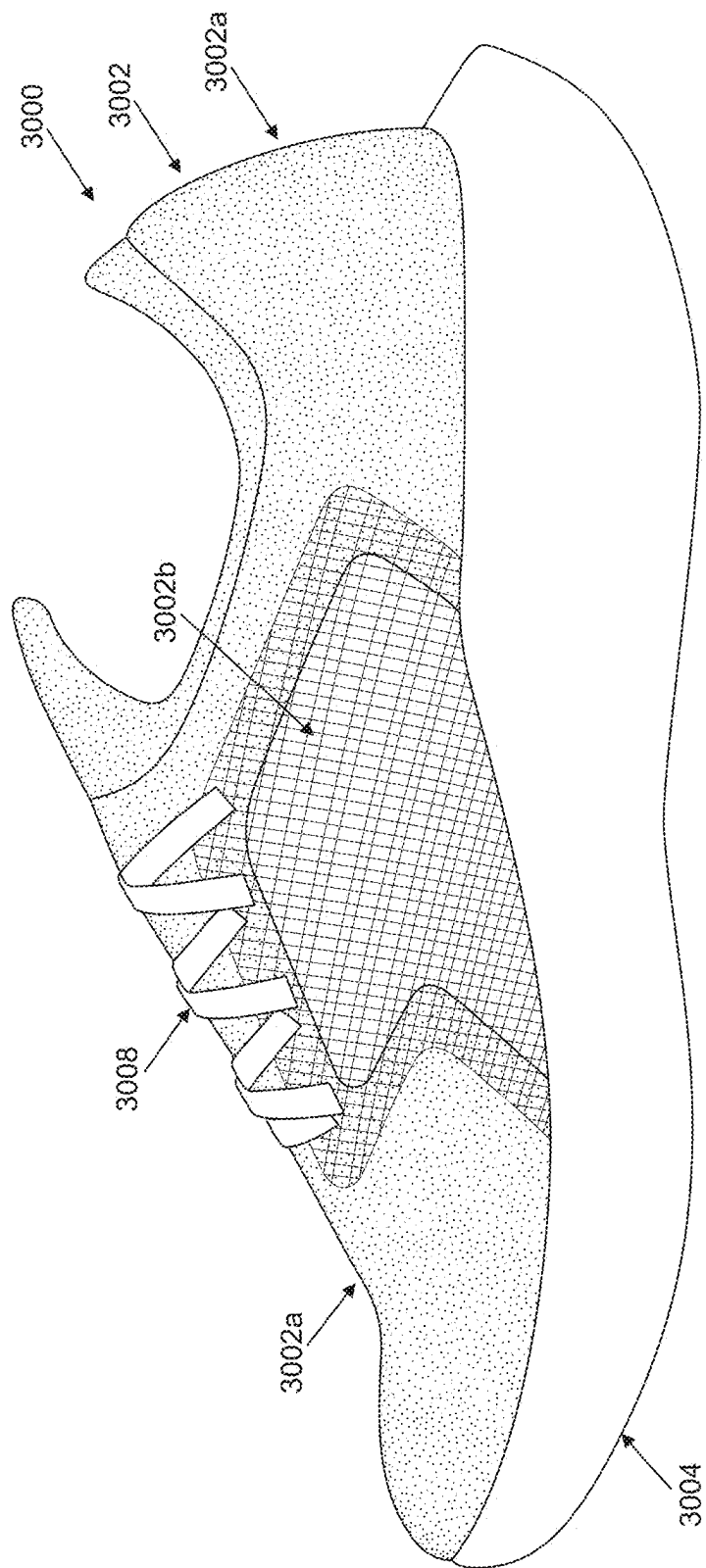


FIG. 8B

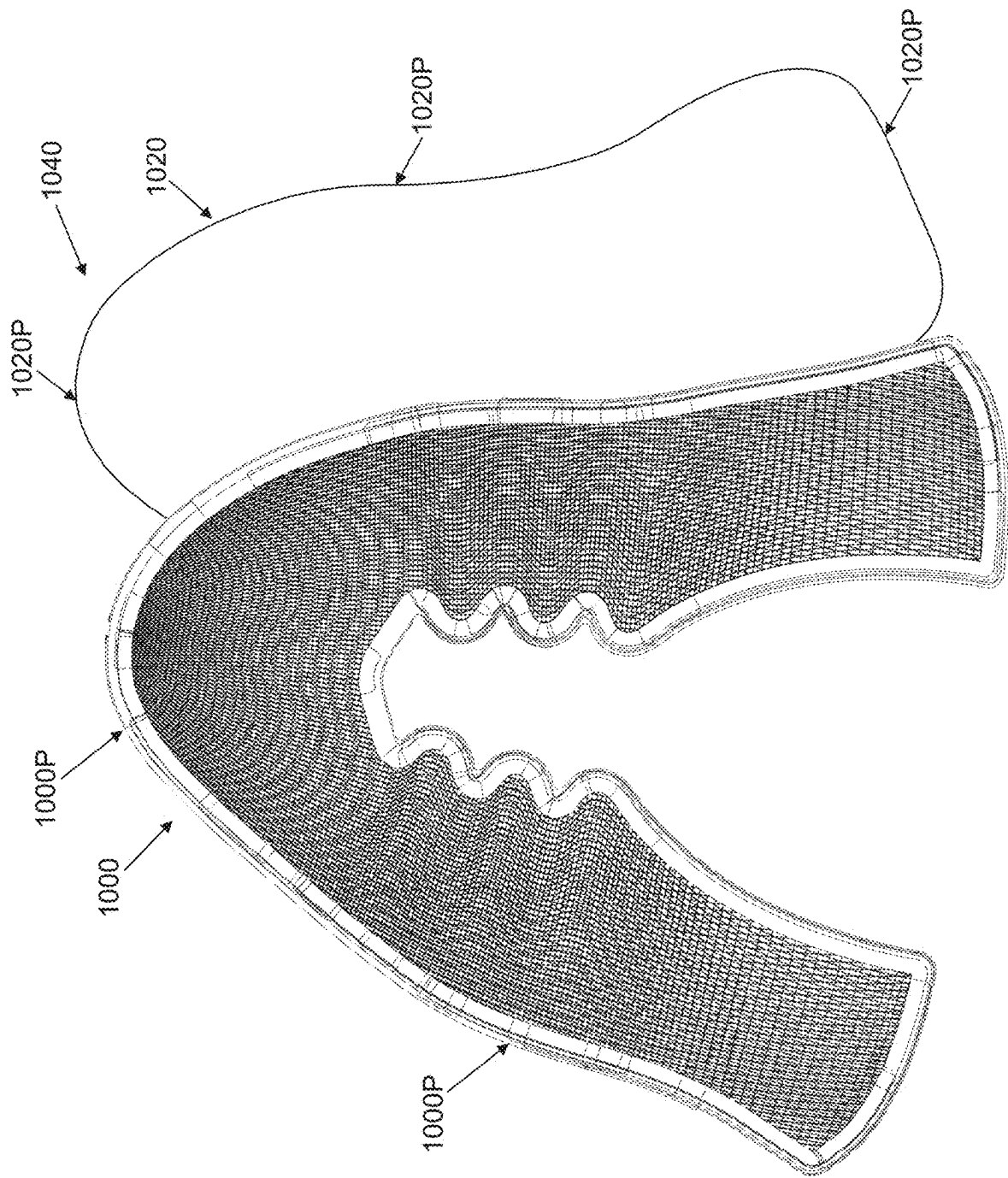


FIG. 9

MULTI-LAYER EXTRUDED UPPERS FOR ARTICLES OF FOOTWEAR AND OTHER FOOT-RECEIVING DEVICES

CROSS-REFERENCE TO RELATED APPLICATION

This application is a continuation of U.S. patent application Ser. No. 16/380,219 filed Apr. 10, 2019, which claims priority benefits based on (a) U.S. Provisional Patent Application No. 62/655,519 filed Apr. 10, 2018 and (b) U.S. Provisional Patent Application No. 62/655,539 filed Apr. 10, 2018, each of which is entirely incorporated herein by reference.

FIELD OF THE INVENTION

The present invention relates to the field of footwear and other foot-receiving devices. More specifically, aspects of the present invention pertain to uppers for articles of footwear and other foot-receiving devices and methods of making the uppers.

BACKGROUND

Conventional articles of athletic footwear include two primary elements, an upper and a sole structure. The upper provides a covering for the foot that securely receives and positions the foot with respect to the sole structure. In addition, the upper may have a configuration that protects the foot and provides ventilation, thereby cooling the foot and removing perspiration. The sole structure is secured to a lower surface of the upper and is generally positioned between the foot and any contact surface. In addition to attenuating ground reaction forces and absorbing energy, the sole structure may provide traction and control potentially harmful foot motion, such as over pronation. General features and configurations of uppers and sole structures are discussed in greater detail below.

The upper forms a void on the interior of the footwear for receiving the foot. The void has the general shape of the foot, and access to the void is provided at an ankle or foot-insertion opening. Accordingly, the upper extends over the instep and toe areas of the foot, along the medial and lateral sides of the foot, and around the heel area of the foot.

A lacing system often is incorporated into the upper to selectively change the size of the ankle opening and to permit the wearer to modify certain dimensions of the upper, particularly girth, to accommodate feet with varying proportions. In addition, the upper may include a tongue that extends under the lacing system to enhance the comfort of the footwear (e.g., to modulate pressure applied to the foot by the laces), and the upper also may include a heel counter to limit or control movement of the heel.

The sole structure generally incorporates multiple layers that are conventionally referred to as an “insole,” a “midsole,” and an “outsole.” The insole (which also may constitute a sock liner) is a thin member located within the upper and adjacent the plantar (lower) surface of the foot to enhance footwear comfort, e.g., to wick away moisture. The midsole, which is traditionally attached to the upper along the upper’s entire length, forms the middle layer of the sole structure and serves a variety of purposes that include controlling foot motions and attenuating impact forces. The outsole forms the ground-contacting element of footwear

and usually is fashioned from a durable, wear-resistant material that includes texturing or other features to improve traction.

Terminology/General Information

First, some general terminology and information is provided that will assist in understanding various portions of this specification and the invention(s) as described herein. As noted above, the present invention relates to the field of footwear and other foot-receiving devices. “Foot-receiving device” means any device into which a user places at least some portion of his or her foot. In addition to all types of footwear (described below), foot-receiving devices include, but are not limited to: bindings and other devices for securing feet in snow skis, cross country skis, water skis, snowboards, and the like; bindings, clips, or other devices for securing feet in pedals for use with bicycles, exercise equipment, and the like; bindings, clips, or other devices for receiving feet during play of video games or other games; and the like. “Foot-receiving devices” may include one or more “foot-covering members” (e.g., akin to footwear upper components), which help position the foot with respect to other components or structures, and one or more “foot-supporting members” (e.g., akin to footwear sole structure components), which support at least some portion(s) of a plantar surface of a user’s foot. “Securing systems” may help position and/or securely hold the user’s foot in place with respect to the foot-covering member(s) and/or the foot-supporting member(s). “Footwear” means any type of wearing apparel for the feet, and this term includes, but is not limited to: all types of shoes, boots, sneakers, sandals, thongs, flip-flops, mules, scuffs, slippers, sport-specific shoes (such as cross-country shoes, golf shoes, tennis shoes, baseball cleats, soccer or football cleats, ski boots, basketball shoes, cross training shoes, track shoes, track field event shoes (e.g., for high jump, triple jump, etc.), etc.), and the like. “Foot-supporting members” may include components for and/or functioning as midsoles and/or outsoles for articles of footwear (or components providing corresponding functions in non-footwear type foot-receiving devices).

The terms “forward” or “forward direction” as used herein, unless otherwise noted or clear from the context, mean toward or in a direction toward a forward-most toe area of the footwear or foot-receiving device structure or component. The terms “rearward” or “rearward direction” as used herein, unless otherwise noted or clear from the context, mean toward or in a direction toward a rear-most heel area of the footwear or foot-receiving device structure or component. The terms “lateral” or “lateral side” as used herein, unless otherwise noted or clear from the context, mean the outside or “little toe” side of the footwear or foot-receiving device structure or component. The terms “medial” or “medial side” as used herein, unless otherwise noted or clear from the context, mean the inside or “big toe” side of the footwear or foot-receiving device structure or component.

The term “moiré effect,” as used herein, means a visual perception that occurs when viewing a set of lines or dots that is superimposed on another set of lines or dots, where the sets differ in relative size, angle, or spacing. In some examples, the “moiré effect” can be seen when two sets of lines (e.g., path segments) of equal thickness and equal spacing are superimposed, but one set is angled (e.g., at a few degrees) with respect to the lines (e.g., path segments)

of the other set. The “moiré effect” can be seen in that case as a set of thick, ill-defined bars.

BRIEF DESCRIPTION OF THE DRAWINGS

The following Detailed Description, will be better understood when read in conjunction with the accompanying drawings in which like reference numerals refer to the same or similar elements in all of the various views in which that reference number appears.

FIG. 1 shows a multi-layer upper blank made from multiple layers of extruded filaments in accordance with one example of this invention;

FIGS. 2A-2F show various features of filament paths and filament path segments in upper layers in accordance with some examples of this invention;

FIGS. 3A-3W show various layers of extruded filament, steps in making a multi-layer upper component from extruded filaments, and various features/properties of multi-layer upper components in accordance with examples of this invention;

FIGS. 4A-4C shows various features of a filament based upper component engaged with another upper component by an adhesive;

FIGS. 5A-5F shows various features of a filament based upper component engaged with another upper component in an adhesive-free manner;

FIGS. 6A-6E illustrate example steps of engaging a filament based upper component with another upper component;

FIGS. 7A-7C illustrate an article of footwear including a multi-layer extruded filament component in accordance with one example of this invention;

FIGS. 8A-8B illustrate an article of footwear including a multi-layer extruded filament component in accordance with another example of this invention; and

FIG. 9 illustrates a multi-layer upper blank integrally formed with a strobil member in accordance with some examples of this invention.

The reader should understand that the attached drawings are not necessarily drawn to scale.

DETAILED DESCRIPTION

In the following description of various examples of footwear and foot-receiving device structures and components according to the present invention, reference is made to the accompanying drawings, which form a part hereof, and in which are shown by way of illustration various example structures and environments in which aspects of the invention may be practiced. It is to be understood that other structures and environments may be utilized and that structural and functional modifications may be made from the specifically described structures and functions without departing from the scope of the present invention.

I. DETAILED DESCRIPTION OF EXAMPLE UPPERS OR OTHER FOOT-COVERING COMPONENTS ACCORDING TO THIS INVENTION

Referring to the figures and following discussion, various articles of footwear/foot-receiving devices and features thereof in accordance with aspects of the present invention are disclosed. The footwear depicted and discussed are athletic shoes (e.g., cross country, running, or track shoes), but the concepts disclosed with respect to this footwear may

be applied to a wide range of athletic footwear styles, including, but not limited to: walking shoes, tennis shoes, soccer shoes, football shoes, basketball shoes, running shoes, track shoes, shoes for track field events (e.g., high jump, triple jump, etc.) and cross-training shoes. In addition, the concepts of the present invention may be applied to a wide range of non-athletic footwear, including work boots, sandals, loafers, and dress shoes, as well as to other foot-receiving devices.

Uppers for articles of footwear (or foot-covering components for other foot-receiving devices) in accordance with at least some examples and aspects of this invention may include: an upper component having: (a) a first layer comprising a first filament including first plural, non-intersecting, spaced apart path segments, wherein the first filament has a width dimension of less than 3 mm wide (and in some examples, less than 2 mm wide, less than 1.5 mm wide, less than 1 mm wide, or even less than 0.75 mm wide); and (b) a second layer comprising a second filament including second plural, non-intersecting, spaced apart path segments, wherein the second filament has a width dimension of less than 3 mm wide (and in some examples, less than 2 mm wide, less than 1.5 mm wide, less than 1 mm wide, or even less than 0.75 mm wide). The second layer may at least partially overlay the first layer and may be fused to the first layer at locations where the second layer contacts the first layer. Additional layers of material, including additional layers of filament, e.g., of the types described above, may be included in the upper. The filament material in the different layers may be the same or different from one another (e.g., a thermoplastic material, a thermoplastic polyurethane material, a hydrophobic material, a water-repelling material, a non-water absorbing material, etc.), and it may be extruded, e.g., formed in a solid deposition modeling process. The filament material may comprise any material as are conventionally known and used in solid deposition modeling arts as the fusible material (e.g., including thermoplastics such as acrylonitrile butadiene styrene (ABS), polylactic acid (PLA), high-impact polystyrene (HIPS), thermoplastic polyurethane (TPU), aliphatic polyamides (nylon), and/or other materials as are conventionally known and used in the solid deposition modeling arts. The term “solid deposition modeling” as used herein includes processes known in the art as “fused filament fabrication” and “fused deposition modeling.”

Upper blanks for articles of footwear (or foot-covering components for other foot-receiving devices) in accordance with at least some examples and aspects of this invention may include: (a) a first layer comprising a first filament formed as a first path (e.g., a first continuous path) of extruded filament, wherein the first path (e.g., the first continuous path) of the first filament forms a first lateral rear heel portion, a first lateral midfoot portion, a first forefoot portion, a first medial midfoot portion, and a first medial rear heel portion of the first layer, and wherein the first filament has a width dimension of less than 3 mm wide (and in some examples, less than 2 mm wide, less than 1.5 mm wide, less than 1 mm wide, or even less than 0.75 mm wide); and (b) a second layer comprising a second filament formed as a second path (e.g., a second continuous path) of extruded filament, wherein the second path (e.g., the second continuous path) of the second filament forms a second lateral rear heel portion, a second lateral midfoot portion, a second forefoot portion, a second medial midfoot portion, and a second medial rear heel portion, wherein the second filament has a width dimension of less than 3 mm wide (and in some examples, less than 2 mm wide, less than 1.5 mm wide, less

5

than 1 mm wide, or even less than 0.75 mm wide), and wherein the second layer is fused to the first layer at locations where the second layer contacts the first layer. Additional layers of material, including additional layers of filament, e.g., of the types described above, may be included in the upper blank. The filament material in the different layers may be the same or different from one another (e.g., as described above). The filament layers may be extruded, e.g., in a solid deposition modeling process.

Uppers for articles of footwear (or foot-covering components for other foot-receiving devices) in accordance with at least some examples and aspects of this invention may include: (a) a first upper component that includes a first layer including a first material as a first filament including first plural, non-intersecting, spaced apart path segments, wherein the first filament has a width dimension of less than 3 mm wide (and in some examples, less than 2 mm wide, less than 1.5 mm wide, less than 1 mm wide, or even less than 0.75 mm wide); and (b) a second upper component including a fabric element formed at least in part of a fusible material, wherein the fusible material of the second upper component is fused to the first material of the first upper component (e.g., in an adhesive-free manner). Additional layers of material, including additional layers of filament and/or additional fabric elements, e.g., of the types described above, may be included in the upper. The filament or fabric material in the different layers may be the same or different from one another (e.g., as described above).

Methods of forming uppers for articles of footwear (or foot-covering components for other foot-receiving devices) in accordance with at least some examples and aspects of this invention (and/or of the types described above) may include: (a) extruding a first material to form a first layer comprising a first extruded filament including first plural, non-intersecting, spaced apart path segments, wherein the first extruded filament has a width dimension of less than 3 mm wide (and in some examples, less than 2 mm wide, less than 1.5 mm wide, less than 1 mm wide, or even less than 0.75 mm wide); and (b) extruding a second material to form a second layer comprising a second extruded filament including second plural, non-intersecting, spaced apart path segments, wherein the second extruded filament has a width dimension of less than 3 mm wide (and in some examples, less than 2 mm wide, less than 1.5 mm wide, less than 1 mm wide, or even less than 0.75 mm wide), and wherein the step of extruding the second material includes fusing the second layer to the first layer at locations where the second layer contacts the first layer. The second layer at least partially overlaps the first layer. The filament may be deposited in a solid deposition modeling process.

Methods of forming uppers for articles of footwear (or foot-covering components for other foot-receiving devices) in accordance with at least some examples and aspects of this invention (and/or of the types described above) may include: (a) extruding a first material to form a first layer comprising a first extruded filament as a first path (e.g., a first continuous path), wherein the first path (e.g., the first continuous path) of the first extruded filament forms a first lateral rear heel portion, a first lateral midfoot portion, a first forefoot portion, a first medial midfoot portion, and a first medial rear heel portion of the first layer, and wherein the first extruded filament has a width dimension of less than 3 mm wide (and in some examples, less than 2 mm wide, less than 1.5 mm wide, less than 1 mm wide, or even less than 0.75 mm wide); and (b) extruding a second material to form a second layer comprising a second extruded filament as a second path (e.g., a second continuous path), wherein the

6

second path (e.g., the second continuous path) of the second extruded filament forms a second lateral rear heel portion, a second lateral midfoot portion, a second forefoot portion, a second medial midfoot portion, and a second medial rear heel portion of the second layer, wherein the second extruded filament has a width dimension of less than 3 mm wide (and in some examples, less than 2 mm wide, less than 1.5 mm wide, less than 1 mm wide, or even less than 0.75 mm wide), and wherein the step of extruding the second material includes fusing the second layer to the first layer at locations where the second layer contacts the first layer. The second layer at least partially overlaps the first layer, and these layers may be deposited in a solid deposition modeling process. More layers of extruded filament may be included in the upper, if desired.

Methods of forming uppers for articles of footwear (or foot-covering components for other foot-receiving devices) in accordance with at least some examples and aspects of this invention (and/or of the types described above) may include: (a) extruding a first material to form a first layer comprising a first extruded filament including first plural, non-intersecting, spaced apart path segments, wherein the first extruded filament has a width dimension of less than 3 mm wide (and in some examples, less than 2 mm wide, less than 1.5 mm wide, less than 1 mm wide, or even less than 0.75 mm wide), and wherein the first layer comprising the first extruded filament forms at least a portion of a first upper component; and (b) fusing a second upper component to the first upper component, wherein the second upper component includes a fabric element formed at least in part of a fusible material, wherein fusible material of the second upper component is fused to the first material of the first upper component, e.g., by application of heat and/or pressure, optionally in an adhesive free manner. The first upper component may include multiple layers of filament material. The extruded filament layer(s) may be deposited in a solid deposition modeling process.

Methods of forming uppers for articles of footwear (or foot-covering components for other foot-receiving devices) in accordance with at least some examples and aspects of this invention (and/or of the types described above) may include: (a) extruding a first material to form a first layer comprising a first extruded filament including first plural, non-intersecting, spaced apart path segments, wherein the first extruded filament has a width dimension of less than 3 mm wide (and in some examples, less than 2 mm wide, less than 1.5 mm wide, less than 1 mm wide, or even less than 0.75 mm wide), and wherein the first layer comprising the first extruded filament forms at least a portion of a first upper component; (b) covering a portion of the first layer with a release liner (e.g., a portion of the first layer extending inwardly from a peripheral edge of the first layer); (c) extruding a second material to form a second layer comprising a second extruded filament including second plural, non-intersecting, spaced apart path segments, wherein the second extruded filament has a width dimension of less than 3 mm wide (and in some examples, less than 2 mm wide, less than 1.5 mm wide, less than 1 mm wide, or even less than 0.75 mm wide), wherein the step of extruding the second material includes: (i) applying a first portion of the second layer to the release liner such that the release liner extends between a first portion of the first layer and the first portion of the second layer and (ii) fusing a second portion of the second layer to a second portion of the first layer at locations where the second layer contacts the first layer (e.g., at locations away from the release liner), and wherein the second layer forms a portion of the first upper component;

7

(d) removing the release liner from between the first portion of the first layer and the first portion of the second layer; (e) optionally, placing a portion of a second upper component between the first portion of the first layer and the first portion of the second layer, wherein the portion of the second upper component optionally includes a fabric element formed at least in part of a fusible material; and (f) optionally, engaging the second upper component with the first upper component. In examples where the second upper component includes a fabric element formed at least in part by a fusible material, the fusible material of the second upper component may be fused to the first material of the first upper component and/or to the second material of the first upper component, e.g., in an adhesive free manner. Multiple layers of filament material may be provided on either or both sides of the release liner (and the second upper component), if desired. The layers of filament material may be deposited in a solid deposition modeling process.

Two or more layers of fused filament materials in footwear uppers in accordance with examples of this invention may provide several options for designers to control properties and/or performance characteristics of a footwear upper and/or several options for designers to control properties and/or performance characteristics in different regions or zones of an individual upper. Many features or properties of an upper can be controlled or altered, including one or more of: (a) filament size (e.g., extruded diameter, extruded width, or extruded thickness) in one or more filament layers of an upper and/or in one or more zones or regions in a single layer of an upper; (b) filament material in one or more filament layers of an upper and/or in one or more zones or regions in a single layer of an upper (e.g., a filament material's elasticity, stretchability, strength, etc.); (c) filament spacing in one or more filament layers of an upper and/or in one or more zones or regions in a single layer of an upper; (d) extent of filament overlap between layers of an upper (e.g., overlap in the filament width direction and/or the filament axial direction); (e) filament ordering layer in layers of an upper; (f) the number of filament path segments in one or more filament layers of an upper and/or in one or more zones or regions in a single layer of an upper; (g) filament path direction in one or more filament layers of an upper and/or in one or more zones or regions in a single layer of an upper etc. Fusion at the intersections of the filament layers provides different connections and interactions between layers as compared to connections between strands or yarns of knitted or woven fabric materials. In general, filaments extending in a medio-lateral direction of the upper (e.g., from side-to-side and/or from a top edge (e.g., by the foot-receiving and/or instep opening(s)) to a bottom edge (e.g., where the upper will engage the sole) will provide enhanced "lock-down" effect on the foot (e.g., hold the foot down onto the footwear sole more securely). Filaments arranged in a more curved and/or serpentine pattern and/or in diamond or parallelogram shapes may provide directional stretch features (e.g., more stretch in one direction as compared to an opposite direction). Tighter filament spacings in or more layers and/or within an individual layer of an upper will tend to provide decreased flexibility, decreased stretch, decreased permeability (e.g., for air, water, or other materials), and/or decreased breathability for that layer and/or zone (and greater filament spacings will tend to increase these properties for that layer and/or zone).

Given the above background and general description of aspects and examples of this invention, a more detailed description of specific examples of uppers, upper compo-

8

nents, upper blanks, and/or articles of footwear in accordance with at least some examples of this invention follows.

II. DETAILED DESCRIPTION OF SPECIFIC EXAMPLE UPPERS, UPPER COMPONENTS, UPPER BLANKS, AND ARTICLES OF FOOTWEAR ACCORDING TO THIS INVENTION

FIG. 1 illustrates an upper blank **1000** for forming an upper for an article of footwear (or a foot-covering component for another type of foot-receiving device) in accordance with one example of this invention. The upper blank **1000** of this example is formed from multiple layers of extruded filament. One or more of the filament layers (and optionally each individual layer of the filament layers) of upper blank **1000** may be extruded as a continuous path of extruded filament, although one or more (or even all) of the individual layers need not be extruded as a continuous path in some examples of this invention. The extruded filament path(s) in any one or more of the layers of the upper blank **1000** may extend to form one or more of (and optionally all of): a lateral rear heel portion **1002** (e.g., extending along a lateral side **1002s** of the ankle/foot opening **1014** of the upper blank **1000**); a lateral midfoot portion **1004** (e.g., adjacent a lateral side **1004s** of an instep opening **1012** of the upper blank **1000**, which may include one or more structures to engage a shoe lace); a forefoot portion **1006** (e.g., which bridges from a lateral side to a medial side of the upper blank **1000**, forward of the midfoot portions); a medial midfoot portion **1008** (e.g., adjacent a medial side **1008s** of the instep opening **1012** of the upper blank **1000**, which may include one or more structures to engage a shoe lace); and a medial rear heel portion **1010** (e.g., extending along a medial side **1010s** of the ankle/foot opening **1014** of the upper blank **1000**). The vertical dashed lines shown in FIG. 1 generally define and break the upper blank **1000** into three portions or regions: (a) a posterior third (containing the lateral rear heel portion **1002** and the medial rear heel portion **1010**), (b) a central third (containing the lateral midfoot portion **1004** and the medial midfoot portion **1008**), and (c) an anterior third (containing the forefoot portion **1006**). In some examples of this invention, the upper blank **1000** will consist essentially of, or even consist of, the multi-layer filament structure. The white space visible in FIG. 1 for this example upper blank **1000** constitutes open space between filament path segments (e.g., where one can see completely through the upper blank **1000**).

Example features of individual layers of this example multi-layer upper blank **1000** now will be described in more detail in conjunction with FIGS. 2A-2F. FIGS. 2A and 2D generally show an extruded path segment **100** as may be laid down by an extruder **102** during an upper formation process in accordance with some examples of this invention (e.g., in a solid deposition modeling or a fused deposition modeling process). As shown in these figures, an individual path segment **100** of an extruded filament generally will have an axial length **L** that is much greater than the width **W** and/or thickness **T** of the individual filament path segment. As some more specific examples, an individual filament (and/or at least one or more path segments **100** thereof) may have an extruded width dimension **W** of less than 3 mm wide, and in some examples, less than 2 mm wide, less than 1.5 mm wide, less than 1 mm wide, or even less than 0.75 mm wide. Additionally or alternatively, an individual filament (and/or at least one or more path segments **100** thereof) may have an extruded thickness dimension **T** of less than 3 mm thick, and

in some examples, less than 2 mm thick, less than 1.5 mm thick, less than 1.25 mm thick, less than 1 mm thick, or less than 0.75 mm thick, or even less than 0.5 mm thick. For at least some path segments **100** (and optionally all path segments **100** in an upper layer and/or upper blank **1000**), the width dimension W may be greater than the thickness dimension T. The path segment length dimension L and/or overall continuous path length may be at least 10 times greater (and in some examples, at least 20 times greater, at least 50 times greater, at least 75 times greater, a least 100 times greater, or even at least 150 times greater) than the width dimension W and/or the thickness dimension T of the filament/filament path. Also, as described above, an individual layer of an upper component may include plural, non-intersecting, spaced apart path segments. As some more examples, as shown in the figures, an individual layer may include at least 5 non-intersecting path segments over path segment lengths of at least 25 mm, and in some examples, at least 5 non-intersecting path segments over path segment lengths of at least 50 mm, at least 75 mm, at least 100 mm, at least 150 mm, or even more. As additional examples, an individual layer may include at least 10 non-intersecting path segments over any of the above path segment length dimensions, at least 15 non-intersecting path segments over any of the above path segment length dimensions, or even at least 20 non-intersecting path segments over any of the above path segment length dimensions.

As shown in FIG. 2D, the material **108** for forming the filament path segment **100** may be forced through the nozzle **104** of the extruder **102** onto a substrate **106**, which may be formed of glass or other appropriate material. The nozzle **104** diameter may be somewhat narrower than the final extruded width W of the path segment **100**, e.g., because the heated filament material **108** may tend to flatten out after being deposited as a path segment **100** (or even may be pushed downward by the extruder **102** nozzle **104**). In general, increasing the temperature of the material **108** being extruded may cause the deposited path segment **100** to flatten out more (and generally increase in width W and decrease in thickness T). In one more specific example, the nozzle **104** diameter may be about 0.4 mm, although the nozzle **104** diameter may range, for example, from 0.25 mm to 2.5 mm (and in some examples, from 0.3 mm to 2 mm). The surface **106s** of the substrate **106** may be smooth or otherwise textured, and the characteristics of the bottom surface **100s** of the filament path segment **100** may form to and take the shape (e.g., smooth or textured characteristics) of the substrate **106** surface **106s** on which it contacts and is formed.

FIG. 2B illustrates a portion of an upper blank **1000** where two layers of filament are present. First, the first filament path segment **100** is extruded (e.g., having any one or more of the filament dimensions and/or structural features described above), and then a second filament path segment **200** (e.g., as a second layer of the upper/upper blank **1000**) is extruded in a direction to cross or intersect the first filament path segment **100**. The second filament path segment **200** may directly contact the first filament path segment **100** (at contact area **202**) as it is being extruded. Heat from the material of the second filament path segment **200** during the extrusion thereof (and/or another heat source) causes the second filament path segment **200** to fuse together with the first filament path segment **100** at location(s) **202** where they contact one another (e.g., the material of the second filament path segment **200** may polymerize with and seamlessly join the material of the first filament path segment **100**, and heat from the extruded second filament path

segment **200** as it is being deposited can support this fusion feature). In this manner, the first layer of the upper blank **1000** (including the first filament path segment(s) **100**) can be fixedly joined to the second layer of the upper blank **1000** (including the second filament path segment(s) **200**) in an adhesive free manner at contact location(s) **202**.

FIGS. 2C and 2E show another manner in which two (or more) layers of an upper blank **1000** may be engaged with one another. Rather than simply intersecting (as shown in FIG. 2B), the second extruded path segment(s) **200** may be extruded at locations that generally overlap (and optionally extend in parallel) with the first extruded path segment(s) **100** over at least a portion of their respective axial lengths L. This action produces an axially extending contact area **202** between path segments **200** and **100**. While other options are possible, after the first path segments **100** are extruded, the second path segments **200** (optionally formed in a second layer of the upper blank **1000**) can be extruded at locations offset slightly from the extrusion path(s) of the first path segments **100**. As shown in FIG. 2E, when the first path segment **100** was extruded, the nozzle **104** center was located at line **120a**. Then, when the second path segment **200** was extruded (e.g., with a second upper blank layer), the nozzle **104** was shifted by an offset distance D to center at line **120b**. This offset distance D may be any desired amount, and in some examples of this invention, may be between $0.5 D_N$ to $0.9 D_N$, and in some examples, between $0.625 D_N$ to $0.85 D_N$, or even about $0.75 D_N$, wherein D_N represents the nozzle **104** diameter.

The overlapping (and substantially parallel) contact area **202** of the type shown in FIGS. 2C and 2E may extend any desired axial length L without departing from the invention. In some examples, the second path segment(s) **200** of the second filament (or second layer) may extend parallel to and/or partially overlap with the first path segment(s) of the first filament (or first layer) over a path segment length **100** of at least 25 mm, and in some examples, at least 50 mm, at least 75 mm, at least 100 mm, at least 150 mm, or even more. The overlapping contact area **202** may follow along curved path segment(s) as well. Additionally or alternatively, the overall layer path of the second layer (including the second path segment(s) **200**): (a) may extend parallel to and/or partially overlap with the overall layer path of the first layer (including the first path segment(s) **100**) over at least 10%, at least 25%, at least 50%, at least 75%, at least 85%, at least 90%, or even at least 95% of an overall path length of the second layer and/or (b) may extend parallel to and/or partially overlap with the overall layer path of the first layer (including the first path segment(s) **100**) over at least 10%, at least 25%, at least 50%, at least 75%, at least 85%, at least 90%, or even at least 95% of an overall path length of the first layer.

In at least some examples of this aspect of the invention, the second path segment(s) **200** of the second filament will overlap with the first path segment(s) **100** of the first filament by an overlapped width WO that is from 5% to 50% of an overall combined width WC of the second filament and the first filament at the location(s) of overlap. See FIG. 2C. In some examples, this overlapped width WO may be from 10% to 45% or even 15% to 40% of the overall combined width WC at the location(s) of overlap. When the individual filament layers are formed as plural, non-intersecting, spaced part path segments, the plurality of the second plural, non-intersecting, spaced apart path segments (e.g., path segments **200**) of the second filament may overlap with the plurality of the first plural, non-intersecting, spaced apart path segments (e.g., path segments **100**) of the first filament

11

by an overlapped width WO that is from 5% to 50% (or from 10% to 45% or even from 15% to 40%) of an overall combined width WC of: (a) the second plural, non-intersecting, spaced apart path segment and (b) the first plural, non-intersecting, spaced apart path segments at the overlapping path segment location(s). Additionally or alternatively, if desired, the overlapped width WO of one or more filaments of the second path segment(s) **200** with corresponding first path segment(s) **100** may be within 10% to 75%, within 15% to 60%, or even within 25% to 50% of the width W of the second path segment **200** (or of the width W of the first path segment **100**) at the location of the overlap. Thus, $WO=0.1\ W$ to $0.75\ W$, or even $0.15\ W$ to $0.6\ W$ or $0.25\ W$ to $0.5\ W$, where W is the width of either path segment **100** or **200**.

While FIGS. 2C and 2E show two layers of overlapping path segments **100** and **200**, if desired, a third layer and/or additional layers may be deposited to partially overlap the first segment **100** and/or the second segment **200** at or adjacent locations where the first segment **100** and the second segment **200** overlap. This feature is shown as layer path segments **300** in dot-dash lines in FIGS. 2C and 2E. The third layer of path segments **300** may overlap the first layer of path segments **100** and/or the second layer of path segments **200** over any of the overlapping width and/or length ranges described above. The overlapping and substantially parallel path segments, e.g., each of path segments **100**, **200**, and/or **300** shown in FIGS. 2C and 2E, may have the same or different colors. In some examples, two or more of the overlapping and substantially parallel path segments may have the same general color but different shades of that color. These color features, if desired, can contribute to the interesting aesthetic characteristics of the upper component.

FIG. 2F shows additional path segment and/or path layer features that may be provided in at least some upper blanks **1000** and/or uppers in accordance with aspects of this invention. As mentioned above, one or more of the filament layers of an upper or upper blank **1000** may be formed by the extruder **102** (optionally as a continuous path). This filament path may form one or more of a lateral rear heel portion, a lateral midfoot portion, a forefoot portion, a medial midfoot portion, and/or a medial rear heel portion of the upper, the layer, and/or the upper blank **1000**. In making these portions of the upper/upper blank **1000** from the thin extruded filaments, in some areas of the upper/upper blank **1000**, the path segments **100** of an individual layer may be extruded to locations that are relatively close to one another, optionally extending in parallel. As shown in FIG. 2F, in one layer, the filament may be extruded into plural, non-intersecting, spaced apart path segments including at least 3 first non-intersecting path segments (**100a-100e** shown in FIG. 2F). In this illustrated example, each non-intersecting path segment (**100a-100e**) of the set of non-intersecting path segments of the individual layer is spaced apart from each directly adjacent non-intersecting path segments in the same layer by a spacing distance (S1 to S4 in FIG. 2F) of less than 10 mm over a length dimension L of at least 25 mm. In some examples, the spacing distance(s) S may be less than 8 mm, less than 6 mm, less than 5 mm, or even less than 3 mm and/or the length dimension L may be at least 15 mm, at least 50 mm, at least 75 mm, at least 100 mm, or even at least 150 mm. The path segments **100a** to **100e** widths W1 to W5, respectively, shown in FIG. 2F may have any of the width characteristics described above, e.g., in conjunction with FIGS. 2A and 2D. In some examples of this invention, a second layer (or even a third or more layers) of path segments will be deposited overlapping and/or in parallel

12

with the segments **100a-100e** shown in FIG. 2F, e.g., to overlap as shown in FIGS. 2C and 2E.

The spacing distances S, the width dimensions W, and/or the overlapping length dimensions L in a given layer may be constant or changing over the overall layer of the path segments. As some more specific examples, a filament in a layer (and optionally a continuous path of filament) may have a first thickness at a first region of the upper or upper blank **1000** and a second thickness at a second region of the upper or upper blank **1000**, wherein the first thickness differs from the second thickness (and optionally may be within the ranges described above). Additionally or alternatively, if desired, a filament in a layer (and optionally a continuous path of filament) may have a first diameter and/or a first width at a first region of the upper or upper blank **1000** and a second diameter and/or a second width at a second region of the upper or upper blank **1000**, wherein the first diameter and/or first width differs from the second diameter and/or second width (and optionally may be within the ranges described above). The different thicknesses, widths, and/or diameters of the filament within a layer may help control the properties of the upper and/or upper blank **1000** (e.g., strength, durability, flexibility, stretchability, breathability, support, etc.).

Various features and examples of an upper or upper blank **1000** made from multiple layers of filament material, e.g., like that of FIG. 1, and method of making them are described in more detail below in conjunction with FIGS. 3A-3W. FIG. 3A shows an example first layer **300** of a first filament (e.g., having any of the filament features and characteristics described above) formed by extruding a first material into multiple path segments (e.g., having any of the path and/or path segment features described above), e.g., via a solid deposition modeling process. Optionally, this first layer **300** may be extruded as a first continuous path. In this illustrated example, the first path of the first filament (optionally as a continuous path) forms the following portions of the first layer **300**: (a) a first lateral rear heel portion **302** (e.g., extending along and/or adjacent a lateral side **302s** of the ankle/foot opening **314** of the first layer **300**); (b) a first lateral midfoot portion **304** (e.g., extending along and/or adjacent a lateral side **304s** (or an inner edge) of an instep opening **312** of the first layer **300**); (c) a first forefoot portion **306** (e.g., which bridges from a lateral side to a medial side of the first layer **300**, forward of the midfoot portions); (d) a first medial midfoot portion **308** (e.g., extending along and/or adjacent a medial side **308s** (or inner edge) of the instep opening **312** of the first layer **300**); and (e) a first medial rear heel portion **310** (e.g., extending along and/or adjacent a medial side **310s** of the ankle/foot opening **314** of the first layer **300**). The vertical dashed lines shown in FIG. 3A generally define and break the first layer **300** into three portions: (a) a posterior third (containing the lateral rear heel portion **302** and the medial rear heel portion **310**), (b) a central third (containing the lateral midfoot portion **304** and the medial midfoot portion **308**), and (c) an anterior third (containing the forefoot portion **306**). In at least some examples of this invention, the first layer **300** will consist essentially of, or even consist of, this filament structure (optionally formed as a continuous path and/or as a one piece construction). The white space visible in FIG. 3A for this example first layer **300** constitutes open space between filament path segments (e.g., where one can see completely through the first layer **300**).

While the path segments of the first layer **300** can be extruded in any desired order without departing from this invention, in some examples of this invention, the outer

13

perimeter (e.g., 300P) may be extruded first, and then the remainder of the layer 300 can be extruded, e.g., in a “raster” like fashion, to fill in the area within the perimeter 300P. In this illustrated example, the extruded overall path of the first layer 300 lays down the first filament over much of the overall surface area of the first layer 300 as first plural, non-intersecting, spaced apart path segments that extend in a substantially medio-lateral direction of the first layer 300. Medio-lateral oriented and/or extending filaments of this type can help enhance the “lock down” features of the upper (e.g., help securely hold the foot down on the sole structure) and may help control/decrease stretchability. Along the lateral rear heel area 302 and the medial rear heel area 310, the first filament path segments extend generally from the ankle opening 302s/310s to a bottom perimeter portion 302t/310t of the first layer 300 (e.g., where the first layer 300 will meet a sole structure in a final article of footwear structure), where adjacent path segments of layer 300 extend substantially in parallel. Similarly, along the lateral midfoot area 304 and the medial midfoot area 308, the first filament path segments extend generally from the instep opening 312’s inner edges 304s/308s to a bottom perimeter portion 304t/308t (outer edges) of the first layer 300 (e.g., where the first layer 300 will meet a sole structure in a final article of footwear structure), where adjacent path segments of the layer 300 extend substantially in parallel. At the forefoot region 306, the first filament path segments extend generally from the lateral bottom edge 306s to the medial bottom edge 306t of the first layer 300 (e.g., where the first layer 300 will meet a sole structure in a final article of footwear structure), where adjacent path segments of layer 300 extend substantially in parallel. The path segments in these various regions 302, 304, 306, 308, 310 may have any of the features and/or options described above for the path segments shown in FIGS. 2A-2F.

In the first layer 300, the path segments in one area need not have constant spacing from directly adjacent path segments at other areas of the first layer 300. For example, as shown in FIG. 3A, the plural, non-intersecting, spaced apart path segments in the forefoot region 306 and/or the midfoot regions 304/308 of the first layer 300 are spaced closer together than are the plural, non-intersecting, spaced apart path segments in the heel region(s) 302/310 of the first layer 300. The path segment spacings (e.g., S1 to S4 from FIG. 2E) can be selected to provide desired characteristics for individual regions of the layer 300, the upper, and/or the upper blank 1000 (e.g., desired stretchability, breathability, etc.).

After the first layer 300 is extruded (e.g., onto a substrate 106), a second layer 350 of the overall upper or upper blank 1000 then may be applied to the first layer 300. FIG. 3B shows the individual path segments of this example second layer 350, and FIG. 3C schematically shows the production of the second layer 350 onto the previously prepared first layer 300 to create the combined first and second layers 380 of the upper or upper blank 1000. More specifically, FIG. 3B shows a second layer 350 formed of a second filament (e.g., having any of the filament features and characteristics described above) and formed by extruding a second material into multiple path segments (e.g., having any of the path and/or path segment features described above), e.g., via a solid deposition modeling process. Optionally, this second layer 350 may be extruded as a second continuous path. In this illustrated example, the second path of the second filament (optionally as a continuous path) forms the following portions of second layer 350: (a) a second lateral rear heel portion 352 (e.g., extending along and/or adjacent a

14

lateral side 352s of the ankle/foot opening 364 of the second layer 350); (b) a second lateral midfoot portion 354 (e.g., extending along and/or adjacent a lateral side 354s of an instep opening 362 (or an inner edge) of the second layer 350); (c) a second forefoot portion 356 (e.g., which bridges from a lateral side to a medial side of the second layer 350, forward of the midfoot portions); (d) a second medial midfoot portion 358 (e.g., extending along and/or adjacent a medial side 358s of the instep opening 362 (or an inner edge) of the second layer 350); and (e) a second medial rear heel portion 360 (e.g., extending along and/or adjacent a medial side 360s of the ankle/foot opening 364 of the second layer 350). The vertical dashed lines shown in FIG. 3B generally define and break the second layer 350 into three portions: (a) a posterior third (containing the lateral rear heel portion 352 and the medial rear heel portion 360), (b) a central third (containing the lateral midfoot portion 354 and the medial midfoot portion 358), and (c) an anterior third (containing the forefoot portion 356). In at least some examples of this invention, the second layer 350 will consist essentially of, or even consist of, this filament structure (optionally formed as a continuous path and/or a one piece construction). The white space visible in FIG. 3B for this example second layer 350 constitutes open space between filament path segments (e.g., where one can see completely through the second layer 350).

While the path segments of the second layer 350 can be extruded in any desired order without departing from this invention, in some examples of this invention, the outer perimeter (e.g., 350P) may be extruded first, and then the remainder of the layer 350 can be extruded, e.g., in a “raster” like fashion, to fill in the area within the perimeter 350P. In this illustrated example, the extruded overall path of the second layer 350 lays down the second filament over much of the overall surface area of the second layer 350 as second plural, non-intersecting, spaced apart path segments that extend in a substantially medio-lateral direction of the second layer 350 (e.g., to help provide the “lock down” or other features describe above for layer 300). Along the lateral rear heel area 352 and the medial rear heel area 360, the second filament path segments extend generally from the ankle opening 352s/360s to a bottom perimeter portion 352t/360t of the second layer 350 (e.g., where the second layer 350 will meet a sole structure in a final article of footwear structure), where adjacent path segments of layer 350 extend substantially in parallel. Similarly, along the lateral midfoot area 354 and the medial midfoot area 358, the second filament path segments extend generally from the instep opening 362’s inner edges 354s/358s to a bottom perimeter portion 354t/358t (outer edges) of the second layer 350 (e.g., where the second layer 350 will meet a sole structure in a final article of footwear structure), where adjacent path segments of layer 350 extend substantially in parallel. At the forefoot region 356, the second filament path segments extend generally from the lateral bottom edge 356s to the medial bottom edge 356t of the second layer 350 (e.g., where the second layer 350 will meet a sole structure in a final article of footwear structure), where adjacent path segments of layer 350 extend substantially in parallel. The path segments in these various regions 352, 354, 356, 358, 360 may have any of the features and/or options described above for the path segments shown in FIGS. 2A-2F.

In the second layer 350, the path segments in one area need not have constant spacing from directly adjacent path segments at other areas of the second layer 350. For example, as shown in FIG. 3B, the plural, non-intersecting, spaced apart path segments in the forefoot region 356 and/or

15

the midfoot regions **354/358** of the second layer **350** are spaced closer together than are the plural, non-intersecting, spaced apart path segments in the heel region(s) **352/360** of the second layer **350**. The path segment spacings (e.g., **S1** to **S4** from FIG. 2E) can be selected to provide desired characteristics for individual regions of the layer **350**, the upper, and/or the upper blank **1000** (e.g., desired stretchability, breathability, etc.).

As evident from a comparison of FIGS. 3A and 3B, the path segments of first layer **300** and second layer **350** extend over a substantial portion of their overall paths in a generally parallel manner. Thus, the path segments of the second layer **350** may be extruded generally in parallel and/or to overlap the path segments of the first layer **300** over much of their overall path lengths in a manner as shown in FIGS. 2C and 2E. If desired: (a) at least 25% (and in some examples, at least 40%, at least 50%, at least 60%, at least 75%, at least 85%, or even at least 90%) of the overall path length of the second layer **350** will overlap with path segments of the first layer **300** in the manner shown in FIGS. 2C and 2E and/or (b) at least 25% (and in some examples, at least 40%, at least 50%, at least 60%, at least 75%, at least 85%, or even at least 90%) of the overall path length of the first layer **300** will overlap with path segments of the second layer **350** in the manner shown in FIGS. 2C and 2E. Thus, filaments of the second layer **350** will directly contact filaments of the first layer **300** (at overlapping contact area **202**) as the second layer **350** is being extruded. Heat from the second layer **350** as it is being extruded (and/or another heat source) causes the second filament path segments to fuse together with the first filament path segments at location(s) **202** where they contact one another (e.g., the filament material of the second layer **350** may polymerize with and seamlessly join the filament material of the first layer **300**). In this manner, the first layer **300** of the upper or upper blank **1000** can be fixedly joined to the second layer **350** of the upper or upper blank **1000** in an adhesive free manner at contact location(s) **202** to form the combined first and second layers **380** of FIG. 3C. The upper component or intermediate including the combined first and second layers **380** constitutes a unitary construction in which the first layer **300** and the second layer **350** are fixed together only in a non-adhesive fused manner. The upper component or intermediate including the combined first and second layers **380** may consist essentially of, or even consist of, the first layer **300** and the second layer **350**.

After the second layer **350** is extruded (e.g., onto first layer **300** and/or substrate **106**), a third layer **400** of the overall upper or upper blank **1000** then may be applied to the combined first and second layers **380**. FIG. 3D shows the individual path segments of this example third layer **400**, and FIG. 3E schematically shows the production of the third layer **400** onto the previously prepared combined first and second layers **380** to create the combined first through third layers **440** of the upper or upper blank **1000**. More specifically, FIG. 3D shows a third layer **400** formed of a third filament (e.g., having any of the filament features and characteristics described above) and formed by extruding a third material into multiple path segments (e.g., having any of the path and/or path segment features described above), e.g., via a solid deposition modeling process. Optionally, this third layer **400** may be extruded as a third continuous path. In this illustrated example, the third path of the third filament (optionally as a continuous path) forms the following portions of the third layer **400**: (a) a third lateral rear heel portion **402** (e.g., extending along and/or adjacent a lateral side **402s** of the ankle/foot opening **414** of the third layer

16

400); (b) a third lateral midfoot portion **404** (e.g., extending along and/or adjacent a lateral side **404s** (or an inner edge) of an instep opening **412** of the third layer **400**); (c) a third forefoot portion **406** (e.g., which bridges from a lateral side to a medial side of the third layer **400**, forward of the midfoot portions); (d) a third medial midfoot portion **408** (e.g., extending along and/or adjacent a medial side **408s** (or an inner edge) of the instep opening **412** of the third layer **400**); and (e) a third medial rear heel portion **410** (e.g., extending along and/or adjacent a medial side **410s** of the ankle/foot opening **414** of the third layer **400**). The vertical dashed lines shown in FIG. 3D generally define and break the third layer **400** into three portions: (a) a posterior third (containing the lateral rear heel portion **402** and the medial rear heel portion **410**), (b) a central third (containing the lateral midfoot portion **404** and the medial midfoot portion **408**), and (c) an anterior third (containing the forefoot portion **406**). In at least some examples of this invention, the third layer **400** will consist essentially of, or even consist of, this filament structure (optionally formed as a continuous path and/or one piece construction). The white space visible in FIG. 3D for this example third layer **400** constitutes open space between filament path segments (e.g., where one can see completely through the third layer **400**).

While the path segments of the third layer **400** can be extruded in any desired order without departing from this invention, in some examples of this invention, the outer perimeter (e.g., **400P**) may be extruded first, and then the remainder of the third layer **400** can be extruded, e.g., in a “raster” like fashion, to fill in the area within the perimeter **400P**. In this illustrated example, the extruded overall path of the third layer **400** lays down the third filament over much of the overall surface area of the third layer **400** as third plural, non-intersecting, spaced apart path segments that extend in a substantially anterior-posterior direction of the third layer **400**. As shown in FIG. 3D, in the lateral rear heel portion **402**, the lateral midfoot portion **404**, the forefoot portion **406**, and the medial rear heel portion **410**, the third plural, non-intersecting path segments of the third layer **400** extend in a gently curved manner in a direction from the rear heel portion **402/410** forward. In the medial midfoot portion **408**, however, at least some of the third plural, non-intersecting, spaced apart path segments extend in a serpentine configuration including at least two peaks (**408P**) and at least two valleys (**408V**). The path segments in these various regions **402**, **404**, **406**, **408**, **410** may have any of the features and/or options described above for the path segments shown in FIGS. 2A-2F.

As further shown in FIG. 3D, the third path of filament defines a first inner edge at lateral instep opening edge **404s**, a first outer edge **404t** at lateral midfoot region **404**, a second inner edge at medial instep opening edge **408s**, and a second outer edge **408t** at medial midfoot region **408**. An instep opening **412** for the third layer **400** is defined between the first inner edge at **404s** and the second inner edge at **408s**. The third path of the filament in this example layer **400** includes: (a) first plural, non-intersecting, spaced apart path segments located between the first inner edge at **404s** and the first outer edge **404t**, wherein the first plural, non-intersecting, spaced apart path segments of the third path are linear and/or curved without defining plural peaks and valleys and/or (b) second plural, non-intersecting, spaced apart path segments located between the second inner edge at **408s** and the second outer edge **408t**, wherein a plurality of the second plural, non-intersecting, spaced apart path segments of the third continuous path located between the second inner edge at **408s** and the second outer edge **408t** extend in a serpentine

17

configuration and have at least two peaks **408P** and at least two valleys **408V**. The third layer **400** of FIG. 3D may include at least 4 path segments, at least 6 path segments, at least 8 path segments, at least 10 path segments, or even at least 12 path segments that extend substantially in parallel and/or have the noted serpentine configuration.

In the third layer **400**, the path segments in one area need not have constant spacing from directly adjacent path segments at other areas of the third layer **400**. For example, as shown in FIG. 3D, the plural, non-intersecting, spaced apart path segments in the forefoot region **406** of the third layer **400** are spaced closer together than are the plural, non-intersecting, spaced apart path segments in the midfoot region(s) **404/408** and/or the plural, non-intersecting, spaced apart path segments in the forefoot region **406** and/or the midfoot region(s) **404/408** are spaced closer together than are the plural, non-intersecting, spaced apart path segments in the heel region(s) **402/410** of the third layer **400**.

As evident from a comparison of FIG. 3D with FIGS. 3A and 3B, the path segments of third layer **400** will substantially intersect the path segments of the first layer **300** and the second layer **350** over a substantial portion of their overall paths. The intersecting path segments form a grid or generally matrix pattern, which can be seen in the combined first through third layers **440** shown in FIG. 3E. The path segments of the third layer **400** may intersect the path segments of the first layer **300** and/or the second layer **350** at any desired angles, e.g., from 5° to 175°, and in some examples, from 15° to 165°, from 25° to 155°, from 35° to 145°, from 45° to 135°, from 55° to 125°, from 60° to 120°, from 65° to 90°, etc. In at least some examples of this invention: (a) the third path of the third layer **400** will overlap the first path of the first layer **300** over less than 50% (and in some examples less than 40%, less than 30%, less than 20%, or even less than 10%) of an overall length of the third path, (b) the third path of the third layer **400** will overlap the second path of the second layer **350** over less than 50% (and in some examples less than 40%, less than 30%, less than 20%, or even less than 10%) of an overall length of the third path, (c) the third path of the third layer **400** will overlap the first path of the first layer **300** over less than 50% (and in some examples less than 40%, less than 30%, less than 20%, or even less than 10%) of an overall length of the first path, and/or (d) the third path of the third layer will overlap the second path of the second layer **350** over less than 50% (and in some examples less than 40%, less than 30%, less than 20%, or even less than 10%) of an overall length of the second path.

Thus, filaments of the third layer **400** will directly contact filaments of the first layer **300** and the filaments of the second layer (at intersecting contact area **202**) as the third layer **400** is being extruded. Heat from the third layer **400** as it is being extruded (and/or another heat source) causes the third filament path segments to fuse together with either or both of the first filament path segments and/or the second filament path segments at location(s) **202** where the third filament path segments contact either or both of the first filament path segments and/or the second filament path segments (e.g., the filament material of the third layer **400** may polymerize with and seamlessly join the filament materials of the first layer **300** and/or the second layer **350**). In this manner, the third layer **400** of the upper or upper blank **1000** can be fixedly joined to the first layer **300** and the second layer **350** of the upper or upper blank **1000** in an adhesive free manner at contact location(s) **202** to form the combined first through third layers **440**. The upper component or intermediate including the combined first through

18

third layers **440** constitutes a unitary construction in which the first layer **300**, the second layer **350**, and the third layer **400** are fixed together only in a non-adhesive fused manner. The upper component or intermediate including the combined first and third layers **440** may consist essentially of, or even consist of, the first layer **300**, the second layer **350**, and the third layer **400**.

After the third layer **400** is extruded (e.g., onto first layer **300**, second layer **350**, and/or substrate **106**), a fourth layer **500** of the overall upper or upper blank **1000** then may be applied to the combined first through third layers **440**. FIG. 3F shows the individual path segments of this example fourth layer **500**, and FIG. 3G schematically shows the production of the fourth layer **500** onto the previously prepared combined first through third layers **440** to create the combined first through fourth layers **540** of the upper or upper blank **1000**. More specifically, FIG. 3F shows a fourth layer **500** formed of a fourth filament (e.g., having any of the filament features and characteristics described above) and formed by extruding a fourth material into multiple path segments (e.g., having any of the path and/or path segment features described above), e.g., via a solid deposition modeling process. Optionally, this fourth layer **500** may be extruded as a fourth continuous path. In this illustrated example, the fourth path of the fourth filament (optionally as a continuous path) forms the following portions of the fourth layer **500**: (a) a fourth lateral rear heel portion **502** (e.g., extending along and/or adjacent a lateral side **502s** of the ankle/foot opening **514** of the fourth layer **500**); (b) a fourth lateral midfoot portion **504** (e.g., extending along and/or adjacent a lateral side **504s** (or an inner edge) of an instep opening **512** of the fourth layer **500**); (c) a fourth forefoot portion **506** (e.g., which bridges from a lateral side to a medial side of the fourth layer **500**, forward of the midfoot portions); (d) a fourth medial midfoot portion **508** (e.g., extending along and/or adjacent a medial side **508s** (or an inner edge) of the instep opening **512** of the fourth layer **500**); and (e) a fourth medial rear heel portion **510** (e.g., extending along and/or adjacent a medial side **510s** of the ankle/foot opening **514** of the fourth layer **500**). The vertical dashed lines shown in FIG. 3F generally define and break the fourth layer **500** into three portions: (a) a posterior third (containing the lateral rear heel portion **502** and the medial rear heel portion **510**), (b) a central third (containing the lateral midfoot portion **504** and the medial midfoot portion **508**), and (c) an anterior third (containing the forefoot portion **506**). In at least some examples of this invention, the fourth layer **500** will consist essentially of, or even consist of, this filament structure (optionally formed as a continuous path and/or a one piece construction). The white space visible in FIG. 3F for this example fourth layer **500** constitutes open space between filament path segments (e.g., where one can see completely through the fourth layer **500**).

While the path segments of the fourth layer **500** can be extruded in any desired order without departing from this invention, in some examples of this invention, the outer perimeter (e.g., **500P**) may be extruded first, and then the remainder of the fourth layer **500** can be extruded, e.g., in a "raster" like fashion, to fill in the area within the perimeter **500P**. In this illustrated example, the extruded overall path of the fourth layer **500** lays down the fourth filament over much of the overall surface area of the fourth layer **500** as fourth plural, non-intersecting, spaced apart path segments that extend in a substantially anterior-posterior direction of the fourth layer **500**. As shown in FIG. 3F, in the lateral rear heel portion **502**, the lateral midfoot portion **504**, the forefoot portion **506**, and the medial rear heel portion **510**, the fourth

plural, non-intersecting path segments extend in a gently curved manner in a direction from the rear heel portion **502/510** forward. In the medial midfoot portion **508**, however, at least some of the fourth plural, non-intersecting, spaced apart path segments of the fourth layer **500** extend in a serpentine configuration including at least two peaks (**508P**) and at least two valleys (**508V**). The path segments in these various regions **502**, **504**, **506**, **508**, **510** may have any of the features and/or options described above for the path segments shown in FIGS. 2A-2F.

As further shown in FIG. 3F, the fourth path of filament defines a first inner edge at lateral instep opening edge **504s**, a first outer edge **504t** at lateral midfoot region **504**, a second inner edge at medial instep opening edge **508s**, and a second outer edge **508t** at medial midfoot region **508**. An instep opening **512** for the fourth layer **500** is defined between the first inner edge at **504s** and the second inner edge at **508s**. The fourth path of the filament in this example layer **500** includes: (a) first plural, non-intersecting, spaced apart path segments located between the first inner edge at **504s** and the first outer edge **504t**, wherein the first plural, non-intersecting, spaced apart path segments of the fourth path are linear and/or curved without defining plural peaks and valleys and/or (b) second plural, non-intersecting, spaced apart path segments located between the second inner edge at **508s** and the second outer edge **508t**, wherein a plurality of the second plural, non-intersecting, spaced apart path segments of the fourth path located between the second inner edge at **508s** and the second outer edge **508t** extend in a serpentine configuration and have at least two peaks **508P** and at least two valleys **508V**. The fourth layer **500** of FIG. 3F may include at least 4 path segments, at least 6 path segments, at least 8 path segments, at least 10 path segments, or even at least 12 path segments that extend substantially in parallel and/or have the noted serpentine configuration.

As evident from a comparison of FIGS. 3D and 3F, the path segments of third layer **400** and fourth layer **500** extend over a substantial portion of their overall paths in a generally parallel manner. Thus, the path segments of the fourth layer **500** may be extruded generally in parallel and/or to overlap the path segments of the third layer **400** over much of their overall path lengths in a manner as shown in FIGS. 2C and 2E. If desired: (a) at least 25% (and in some examples, at least 40%, at least 50%, at least 60%, at least 75%, at least 85%, or even at least 90%) of the overall path length of the fourth layer **500** will overlap with path segments of the third layer **400** in the manner shown in FIGS. 2C and 2E and/or (b) at least 25% (and in some examples, at least 40%, at least 50%, at least 60%, at least 75%, at least 85%, or even at least 90%) of the overall path length of the third layer **400** will overlap with path segments of the fourth layer **500** in the manner shown in FIGS. 2C and 2E. In at least some examples of this invention: (a) the fourth path of the fourth layer **500** will overlap the first path of the first layer **300** over less than 50% (and in some examples less than 40%, less than 30%, less than 20%, or even less than 10%) of an overall length of the fourth path, (b) the fourth path of the fourth layer **500** will overlap the second path of the second layer **350** over less than 50% (and in some examples less than 40%, less than 30%, less than 20%, or even less than 10%) of an overall length of the fourth path, (c) the fourth path of the fourth layer **500** will overlap the first path of the first layer **300** over less than 50% (and in some examples less than 40%, less than 30%, less than 20%, or even less than 10%) of an overall length of the first path, and/or (d) the fourth path of the fourth layer **500** will overlap the second path of the second layer **350** over less than 50% (and in some

examples less than 40%, less than 30%, less than 20%, or even less than 10%) of an overall length of the second path.

Thus, filaments of the fourth layer **500** will directly contact filaments of the third layer **400** (at overlapping contact area **202**) as the fourth layer **500** is being extruded. Heat from the fourth layer **500** as it is being extruded (and/or another heat source) causes the fourth filament path segments to fuse together with the third filament path segments at location(s) **202** where they contact one another (e.g., the filament material of the fourth layer **500** may polymerize with and seamlessly join the filament material of the third layer **400**). In this manner, the third layer **400** of the upper or upper blank **1000** can be fixedly joined to the fourth layer **500** of the upper or upper blank **1000** in an adhesive free manner at contact location(s) **202**.

As also evident from a comparison of FIG. 3F with FIGS. 3A and 3B, the path segments of fourth layer **500** will substantially intersect the path segments of the first layer **300** and the second layer **350** over a substantial portion of their overall paths. The intersecting path segments form a grid or generally matrix pattern, which can be seen in the combined first through fourth layers **540** shown in FIG. 3G. The path segments of the fourth layer **500** may intersect the path segments of the first layer **300** and/or the second layer **350** at any desired angles, e.g., from 5° to 175°, and in some examples, from 15° to 165°, from 25° to 155°, from 35° to 145°, from 45° to 135°, from 55° to 125°, from 60° to 120°, from 65° to 90°, etc. Thus, filaments of the fourth layer **500** will directly contact filaments of the first layer **300** and filaments of the second layer **350** (at intersecting contact area **202**) as the fourth layer **500** is being extruded. Heat from the fourth layer **500** as it is being extruded (and/or another heat source) causes the fourth filament path segments to fuse together with either or both of the first filament path segments and/or the second filament path segments at location(s) **202** where the fourth filament path segments contact either or both of the first filament path segments and/or the second filament path segments (e.g., the filament material of the fourth layer **500** may polymerize with and seamlessly join the filament materials of the first layer **300** and/or the second layer **350**).

In these manners, the fourth layer **500** of the upper or upper blank **1000** can be fixedly joined to the first layer **300**, the second layer **350**, and the third layer **400** of the upper or upper blank **1000** in an adhesive free manner at contact location(s) **202** to form the combined first through fourth layers **540**. The upper component or intermediate including the combined first through fourth layers **540** constitutes a unitary construction in which the first layer **300**, the second layer **350**, the third layer **400**, and the fourth layer **500** are fixed together only in a non-adhesive fused manner. The upper component or intermediate including the combined first through fourth layers **540** may consist essentially of, or even consist of, the first layer **300**, the second layer **350**, the third layer **400**, and the fourth layer **500**.

In the fourth layer **500**, the path segments in one area need not have constant spacing from directly adjacent path segments at other areas of the fourth layer **500**. For example, as shown in FIG. 3F, the plural, non-intersecting, spaced apart path segments in the forefoot region **506** of the fourth layer **500** are spaced closer together than are the plural, non-intersecting, spaced apart path segments in the midfoot region(s) **504/508** and/or the plural, non-intersecting, spaced apart path segments in the forefoot region **506** and/or the midfoot region(s) **504/508** are spaced closer together than are the plural, non-intersecting, spaced apart path segments in the heel region(s) **502/510** of the fourth layer **500**.

21

After the fourth layer **500** is extruded (e.g., onto first layer **300**, second layer **350**, third layer **400**, and/or substrate **106**), a fifth layer **600** of the overall upper or upper blank **1000** then may be applied to the combined first through fourth layers **540**. FIG. 3H shows the individual path segments of this example fifth layer **600**, and FIG. 3I schematically shows the production of the fifth layer **600** onto the previously prepared combined first through fourth layers **540** to create the combined first through fifth layers **640** of the upper or upper blank **1000**. More specifically, FIG. 3H shows a fifth layer **600** formed of a fifth filament (e.g., having any of the filament features and characteristics described above) and formed by extruding a fifth material into multiple path segments (e.g., having any of the path and/or path segment features described above), e.g., via a solid deposition modeling process. Optionally, this fifth layer **600** may be extruded as a fifth continuous path. In this illustrated example, the fifth path of the fifth filament (optionally as a continuous path) forms the following portions of the fifth layer **600**: (a) a fifth lateral rear heel portion **602** (e.g., extending along and/or adjacent a lateral side **602s** of the ankle/foot opening **614** of the fifth layer **600**); (b) a fifth lateral midfoot portion **604** (e.g., extending along and/or adjacent a lateral side **604s** (or an inner edge) of an instep opening **612** of the fifth layer **600**); (c) a fifth forefoot portion **606** (e.g., which bridges from a lateral side to a medial side of the fifth layer **600**, forward of the midfoot portions); (d) a fifth medial midfoot portion **608** (e.g., extending along and/or adjacent a medial side **608s** of the instep opening **612** of the fifth layer **600**); and (e) a fifth medial rear heel portion **610** (e.g., extending along and/or adjacent a medial side **610s** (or an inner edge) of the ankle/foot opening **614** of the fifth layer **600**). The vertical dashed lines shown in FIG. 3H generally define and break the fifth layer **600** into three portions: (a) a posterior third (containing the lateral rear heel portion **602** and the medial rear heel portion **610**), (b) a central third (containing the lateral midfoot portion **604** and the medial midfoot portion **608**), and (c) an anterior third (containing the forefoot portion **606**). In at least some examples of this invention, the fifth layer **600** will consist essentially of, or even consist of, this filament structure (optionally formed as a continuous path and/or one piece construction). The white space visible in FIG. 3H for this example fifth layer **600** constitutes open space between filament path segments (e.g., where one can see completely through the fifth layer **600**).

While the path segments of the fifth layer **600** can be extruded in any desired order without departing from this invention, in some examples of this invention, the outer perimeter (e.g., **600P**) may be extruded first, and then the remainder of the fifth layer **600** can be extruded, e.g., in a "raster" like fashion, to fill in the area within the perimeter **600P**. In this illustrated example, the extruded overall path of the fifth layer **600** lays down the fifth filament over much of the overall surface area of the fifth layer **600** as fifth plural, non-intersecting, spaced apart path segments that extend in a substantially anterior-posterior direction of the fifth layer **600**. As shown in FIG. 3H, in the lateral rear heel portion **602**, the lateral midfoot portion **604**, the forefoot portion **606**, and the medial rear heel portion **610**, the fifth plural, non-intersecting path segments of the fifth layer **600** extend in a gently curved manner in a direction from the rear heel portion **602/610** forward. In the medial midfoot portion **608**, however, at least some of the fifth plural, non-intersecting, spaced apart path segments extend in a serpentine configuration including at least two peaks (**608P**) and at least two valleys (**608V**). The path segments in these various regions

22

602, 604, 606, 608, 610 may have any of the features and/or options described above for the path segments shown in FIGS. 2A-2F.

As further shown in FIG. 3H, the fifth path of filament defines a first inner edge at lateral instep opening edge **604s**, a first outer edge **604t** at lateral midfoot region **604**, a second inner edge at medial instep opening edge **608s**, and a second outer edge **608t** at medial midfoot region **608**. An instep opening **612** for the fifth layer **600** is defined between the first inner edge at **604s** and the second inner edge at **608s**. The fifth path of the filament in this example layer **600** includes: (a) first plural, non-intersecting, spaced apart path segments located between the first inner edge at **604s** and the first outer edge **604t**, wherein the first plural, non-intersecting, spaced apart path segments of the fifth path are linear and/or curved without defining plural peaks and valleys and/or (b) second plural, non-intersecting, spaced apart path segments located between the second inner edge at **608s** and the second outer edge **608t**, wherein a plurality of the second plural, non-intersecting, spaced apart path segments of the fifth path located between the second inner edge at **608s** and the second outer edge **608t** extend in a serpentine configuration and have at least two peaks **608P** and at least two valleys **608V**. The fifth layer **600** of FIG. 3H may include at least 4 path segments, at least 6 path segments, at least 8 path segments, at least 10 path segments, or even at least 12 path segments that extend substantially in parallel and/or have the noted serpentine configuration.

As evident from a comparison of FIGS. 3D, 3F, and 3H, the path segments of third layer **400**, fourth layer **500**, and fifth layer **600** extend over a substantial portion of their overall paths in a generally parallel manner. Thus, the path segments of the fifth layer **600** may be extruded generally in parallel and/or to overlap the path segments of the third layer **400** and/or the fourth layer **500** over much of their overall path lengths in a manner as shown in FIGS. 2C and 2E. If desired: (a) at least 25% (and in some examples, at least 40%, at least 50%, at least 60%, at least 75%, at least 85%, or even at least 90%) of the overall path length of the fifth layer **600** will overlap with path segments of at least one of the third layer **400** and/or the fourth layer **500** in the manner shown in FIGS. 2C and 2E and/or (b) at least 25% (and in some examples, at least 40%, at least 50%, at least 60%, at least 75%, at least 85%, or even at least 90%) of the overall path length of the third layer **400** and/or the fourth layer **500** will overlap with path segments of the fifth layer **600** in the manner shown in FIGS. 2C and 2E. In at least some examples of this invention: (a) the fifth path of the fifth layer **600** will overlap the first path of the first layer **300** over less than 50% (and in some examples less than 40%, less than 30%, less than 20%, or even less than 10%) of an overall length of the fifth path, (b) the fifth path of the fifth layer **600** will overlap the second path of the second layer **350** over less than 50% (and in some examples less than 40%, less than 30%, less than 20%, or even less than 10%) of an overall length of the fifth path, (c) the fifth path of the fifth layer **600** will overlap the first path of the first layer **300** over less than 50% (and in some examples less than 40%, less than 30%, less than 20%, or even less than 10%) of an overall length of the first path, and/or (d) the fifth path of the fifth layer **600** will overlap the second path of the second layer **350** over less than 50% (and in some examples less than 40%, less than 30%, less than 20%, or even less than 10%) of an overall length of the second path.

Thus, filaments of the fifth layer **600** will directly contact filaments of the third layer **400** and/or the fourth layer **500** (at overlapping contact area **202**) as the fifth layer **600** is

being extruded. Heat from the fifth layer **600** as it is being extruded (and/or another heat source) causes the fifth filament path segments to fuse together with either or both of the third filament path segments and/or the fourth filament path segments at location(s) **202** where the fifth filament path segments contact either or both of the third filament path segments and/or the fourth filament path segments (e.g., the filament material of the fifth layer **600** may polymerize with and seamlessly join the filament material of the third layer **400** and/or the fourth layer **500**). In this manner, the third layer **400** and/or fourth layer **500** of the upper or upper blank **1000** can be fixedly joined to the fifth layer **500** of the upper or upper blank **1000** in an adhesive free manner at contact location(s) **202**.

As also evident from a comparison of FIG. 3H with FIGS. 3A and 3B, the path segments of fifth layer **600** will substantially intersect the path segments of the first layer **300** and the second layer **350** over a substantial portion of their overall paths. The intersecting path segments form a grid or generally matrix pattern, which can be seen in the combined first through fifth layers **640** shown in FIG. 3I. The path segments of the fifth layer **600** may intersect the path segments of the first layer **300** and/or the second layer **350** at any desired angles, e.g., from 5° to 175°, and in some examples, from 15° to 165°, from 25° to 155°, from 35° to 145°, from 45° to 135°, from 55° to 125°, from 60° to 120°, from 65° to 90°, etc. Thus, filaments of the fifth layer **600** will directly contact filaments of the first layer **300** and filaments of the second layer **350** (at intersecting contact area **202**) as the fifth layer **600** is being extruded. Heat from the fifth layer **600** as it is being extruded (and/or another heat source) causes the fifth filament path segments to fuse together with either or both of the first filament path segments and/or the second filament path segments at location(s) **202** where the fifth filament path segments contact either or both of the first filament path segments and/or the second filament path segments (e.g., the filament material of the fifth layer **600** may polymerize with and seamlessly join the filament materials of the first layer **300** and/or the second layer **350**).

In these manners, the fifth layer **600** of the upper or upper blank **1000** can be fixedly joined to the first layer **300**, the second layer **350**, the third layer **400**, and the fourth layer **500** of the upper or upper blank **1000** in an adhesive free manner at contact location(s) **202** to form the combined first through fifth layers **640**. The upper component or intermediate including the combined first through fifth layers **640** constitutes a unitary construction in which the first layer **300**, the second layer **350**, the third layer **400**, the fourth layer **500**, and the fifth layer **600** are fixed together only in a non-adhesive fused manner. The upper component or intermediate including the combined first through fifth layers **640** may consist essentially of, or even consist of, the first layer **300**, the second layer **350**, the third layer **400**, the fourth layer **500**, and the fifth layer **600**.

In the fifth layer **600**, the path segments in one area need not have constant spacing from directly adjacent path segments at other areas of the fifth layer **600**. For example, as shown in FIG. 3H, the plural, non-intersecting, spaced apart path segments in the forefoot region **606** of the fifth layer **600** are spaced closer together than are the plural, non-intersecting, spaced apart path segments in the midfoot region(s) **604/608** and/or the plural, non-intersecting, spaced apart path segments in the forefoot region **606** and/or the midfoot region(s) **604/608** are spaced closer together than are the plural, non-intersecting, spaced apart path segments in the heel region(s) **602/610** of the fifth layer **600**.

After the fifth layer **600** is extruded (e.g., onto first layer **300**, second layer **350**, third layer **400**, fourth layer **500**, and/or substrate **106**), a sixth layer **700** of the overall upper or upper blank **1000** then may be applied to the combined first through fifth layers **640**. FIG. 3J shows the individual path segments of this example sixth layer **700**, and FIG. 3K schematically shows the production of the sixth layer **700** onto the previously prepared combined first through fifth layers **640** to create the combined first through sixth layers **740** of the upper or upper blank **1000**. More specifically, FIG. 3J shows a sixth layer **700** formed of a sixth filament (e.g., having any of the filament features and characteristics described above) and formed by extruding a sixth material into multiple path segments (e.g., having any of the path and/or path segment features described above), e.g., via a solid deposition modeling process. Optionally, this sixth layer **700** may be extruded as a sixth continuous path. In this illustrated example, the sixth path of the sixth filament (optionally as a continuous path) forms the following portions of the sixth layer **700**: (a) a sixth lateral rear heel portion **702** (e.g., extending along and/or adjacent a lateral side **702s** of the ankle/foot opening **714** of the sixth layer **700**); (b) a sixth lateral midfoot portion **704** (e.g., extending along and/or adjacent a lateral side **704s** (or an inner edge) of an instep opening **712** of the sixth layer **700**); (c) a sixth forefoot portion **706** (e.g., which bridges from a lateral side to a medial side of the sixth layer **700**, forward of the midfoot portions); (d) a sixth medial midfoot portion **708** (e.g., extending along and/or adjacent a medial side **708s** (or an inner edge) of the instep opening **712** of the sixth layer **700**); and (e) a sixth medial rear heel portion **710** (e.g., extending along and/or adjacent a medial side **710s** of the ankle/foot opening **714** of the sixth layer **700**). The vertical dashed lines shown in FIG. 3J generally define and break the sixth layer **700** into three portions: (a) a posterior third (containing the lateral rear heel portion **702** and the medial rear heel portion **710**), (b) a central third (containing the lateral midfoot portion **704** and the medial midfoot portion **708**), and (c) an anterior third (containing the forefoot portion **706**). In at least some examples of this invention, the sixth layer **700** will consist essentially of, or even consist of, this filament structure (optionally formed as a continuous path and/or a one piece construction). The white space visible in FIG. 3J for this example sixth layer **700** constitutes open space between filament path segments (e.g., where one can see completely through the sixth layer **700**).

While the path segments of the sixth layer **700** can be extruded in any desired order without departing from this invention, in some examples of this invention, the outer perimeter (e.g., **700P**) may be extruded first, and then the remainder of the sixth layer **700** can be extruded, e.g., in a "raster" like fashion, to fill in the area within the perimeter **700P**. In this illustrated example, the extruded overall path of the sixth layer **700** lays down the sixth filament over much of the overall surface area of the sixth layer **700** as sixth plural, non-intersecting, spaced apart path segments that extend in a substantially anterior-posterior direction of the sixth layer **700**. As shown in FIG. 3J, in the lateral rear heel portion **702**, the forefoot portion **706**, the medial midfoot portion **708**, and the medial rear heel portion **710**, the sixth plural, non-intersecting path segments of the sixth layer **700** extend in a gently curved manner in a direction from the rear heel portion **702/710** forward. In the lateral midfoot portion **704**, however, at least some of the sixth plural, non-intersecting, spaced apart path segments extend in a serpentine configuration including at least two peaks (**708P**) and at least two valleys (**708V**). The path segments in these various

25

regions **702**, **704**, **706**, **708**, **710** may have any of the features and/or options described above for the path segments shown in FIGS. 2A-2F.

As further shown in FIG. 3J, the sixth path of filament defines a first inner edge at lateral instep opening edge **704s**, a first outer edge **704t** at lateral midfoot region **704**, a second inner edge at medial instep opening edge **708s**, and a second outer edge **708t** at medial midfoot region **708**. An instep opening **712** for the sixth layer **700** is defined between the first inner edge at **704s** and the second inner edge at **708s**. The sixth path of the filament in this example layer **700** includes: (a) first plural, non-intersecting, spaced apart path segments located between the first inner edge at **704s** and the first outer edge **704t**, wherein a plurality of the first plural, non-intersecting, spaced apart path segments of the sixth path located between the first inner edge at **704s** and the first outer edge **704t** extend in a serpentine configuration and have at least two peaks **708P** and at least two valleys **708V** and (b) second plural, non-intersecting, spaced apart path segments located between the second inner edge at **708s** and the second outer edge **708t**, wherein the second, plural, non-intersecting, spaced apart path segments of the sixth path are linear and/or curved without defining plural peaks and valleys. The sixth layer **700** of FIG. 3J may include at least 4 path segments, at least 6 path segments, at least 8 path segments, at least 10 path segments, or even at least 12 path segments that extend substantially in parallel and/or have the noted serpentine configuration.

In the sixth layer **700**, the path segments in one area need not have constant spacing from directly adjacent path segments at other areas of the sixth layer **700**. For example, as shown in FIG. 3J, the plural, non-intersecting, spaced apart path segments in the forefoot region **706** of the sixth layer **700** are spaced closer together than are the plural, non-intersecting, spaced apart path segments in the midfoot region(s) **704/708** and/or the plural, non-intersecting, spaced apart path segments in the forefoot region **706** and/or the midfoot region(s) **704/708** are spaced closer together than are the plural, non-intersecting, spaced apart path segments in the heel region(s) **702/710** of the sixth layer **700**.

As evident from a comparison of FIG. 3J with FIGS. 3A, 3B, 3D, 3F, and 3H, the path segments of sixth layer **700** will substantially intersect the path segments of the first layer **300**, the second layer **350**, the third layer **400**, the fourth layer **500**, and the fifth layer **600** over a substantial portion of their overall paths. The intersecting path segments form generally diamond shapes, which can be seen in the combined first through sixth layers **740** shown in FIG. 3K. The path segments of the sixth layer **700** may intersect the path segments of the first layer **300**, the second layer **350**, the third layer **400**, the fourth layer **500**, and/or the fifth layer **600** at any desired angles, e.g., from 5° to 175°, and in some examples, from 15° to 165°, from 25° to 155°, from 35° to 145°, from 45° to 135°, from 55° to 125°, from 60° to 120°, from 65° to 90°, etc. In at least some examples of this invention: (a) the sixth path of the sixth layer **700** will overlap with one or more of the first path of the first layer **300**, the second path of the second layer **350**, the third path of the third layer **400**, the fourth path of the fourth layer **500**, and/or the fifth path of the fifth layer **600** over less than 50% (and in some examples less than 40%, less than 30%, less than 20%, or even less than 10%) of an overall length of the sixth path, and/or (b) the sixth path of the sixth layer **700** will overlap one or more of the first path of the first layer **300**, the second path of the second layer **350**, the third path of the third layer **400**, the fourth path of the fourth layer **500**, and/or the fifth path of the fifth layer **600** over less than 50%

26

(and in some examples less than 40%, less than 30%, less than 20%, or even less than 10%) of an overall length of the respective first path, second path, third path, fourth path, and/or fifth path.

Thus, filaments of the sixth layer **700** will directly contact filaments of the first layer **300**, filaments of the second layer **350**, filaments of the third layer **400**, filaments of the fourth layer **500**, and/or filaments of the fifth layer **600** (at intersecting contact area **202**) as the sixth layer **700** is being extruded. Heat from the sixth layer **700** as it is being extruded (and/or another heat source) causes the sixth filament path segments to fuse together with any one of or any combination of the first filament path segments, the second filament path segments, the third filament path segments, the fourth filament path segments, and/or the fifth filament path segments at location(s) **202** where the sixth filament path segments contact any one of or any combination of the first filament path segments, the second filament path segments, the third filament path segments, the fourth filament path segments, and/or the fifth filament path segments (e.g., the filament material of the sixth layer **700** may polymerize with and seamlessly join the filament materials of the first layer **300**, the second layer **350**, the third layer **400**, the fourth layer **500**, and/or the fifth layer **600**). In this manner, the sixth layer **700** of the upper or upper blank **1000** can be fixedly joined to the first layer **300**, the second layer **350**, the third layer **400**, the fourth layer **500**, and/or the fifth layer **600** of the upper or upper blank **1000** in an adhesive free manner at contact location(s) **202** to form the combined first through sixth layers **740**. The upper component or intermediate including the combined first through sixth layers **740** constitutes a unitary construction in which the first layer **300**, the second layer **350**, the third layer **400**, the fourth layer **500**, the fifth layer **600**, and the sixth layer **700** are fixed together only in a non-adhesive fused manner. The upper component or intermediate including the combined first through sixth layers **740** may consist essentially of, or even consist of, the first layer **300**, the second layer **350**, the third layer **400**, the fourth layer **500**, the fifth layer **600**, and the sixth layer **700**.

After the sixth layer **700** is extruded (e.g., onto first layer **300**, second layer **350**, third layer **400**, fourth layer **500**, fifth layer **600**, and/or substrate **106**), a seventh layer **800** of the overall upper or upper blank **1000** then may be applied to the combined first through sixth layers **740**. FIG. 3L shows the individual path segments of this example seventh layer **800**, and FIG. 3M schematically shows the production of the seventh layer **800** onto the previously prepared combined first through sixth layers **740** to create the combined first through seventh layers **840** of the upper or upper blank **1000**. More specifically, FIG. 3L shows a seventh layer **800** formed of a seventh filament (e.g., having any of the filament features and characteristics described above) and formed by extruding a seventh material into multiple path segments (e.g., having any of the path and/or path segment features described above), e.g., by a solid deposition modeling process. Optionally, this seventh layer **800** may be extruded as a seventh continuous path. In this illustrated example, the seventh path of the seventh filament (optionally as a continuous path) forms the following portions of the seventh layer **800**: (a) a seventh lateral rear heel portion **802** (e.g., extending along and/or adjacent a lateral side **802s** of the ankle/foot opening **814** of the seventh layer **800**); (b) a seventh lateral midfoot portion **804** (e.g., extending along and/or adjacent a lateral side **804s** (or an inner edge) of an instep opening **812** of the seventh layer **800**); (c) a seventh forefoot portion **806** (e.g., which bridges from a lateral side

to a medial side of the seventh layer **800**, forward of the midfoot portions); (d) a seventh medial midfoot portion **808** (e.g., extending along and/or adjacent a medial side **808s** (or an inner edge) of the instep opening **812** of the seventh layer **800**); and (e) a seventh medial rear heel portion **810** (e.g., extending along and/or adjacent a medial side **810s** of the ankle/foot opening **814** of the seventh layer **800**). The vertical dashed lines shown in FIG. 3L generally define and break the seventh layer **800** into three portions: (a) a posterior third (containing the lateral rear heel portion **802** and the medial rear heel portion **810**), (b) a central third (containing the lateral midfoot portion **804** and the medial midfoot portion **808**), and (c) an anterior third (containing the forefoot portion **806**). In at least some examples of this invention, the seventh layer **800** will consist essentially of, or even consist of, this filament structure (optionally formed as a continuous path and/or as a one piece construction). The white space visible in FIG. 3L for this example seventh layer **800** constitutes open space between filament path segments (e.g., where one can see completely through the seventh layer **800**).

While the path segments of the seventh layer **800** can be extruded in any desired order without departing from this invention, in some examples of this invention, the outer perimeter (e.g., **800P**) may be extruded first, and then the remainder of the seventh layer **800** can be extruded, e.g., in a "raster" like fashion, to fill in the area within the perimeter **800P**. In this illustrated example, the extruded overall path of the seventh layer **800** lays down the seventh filament over much of the overall surface area of the seventh layer **800** as seventh plural, non-intersecting, spaced apart path segments that extend in a substantially anterior-posterior direction of the seventh layer **800**. As shown in FIG. 3L, in the lateral rear heel portion **802**, the forefoot portion **806**, the medial midfoot portion **808**, and the medial rear heel portion **810**, the seventh plural, non-intersecting path segments extend in a gently curved manner in a direction from the rear heel portion **802/810** forward. In the lateral midfoot portion **804**, however, at least some of the seventh plural, non-intersecting, spaced apart path segments extend in a serpentine configuration including at least two peaks (**808P**) and at least two valleys (**808V**). The path segments in these various regions **802**, **804**, **806**, **808**, **810** may have any of the features and/or options described above for the path segments shown in FIGS. 2A-2F.

As further shown in FIG. 3L, the seventh path of filament defines a first inner edge at lateral instep opening edge **804s**, a first outer edge **804t** at lateral midfoot region **804**, a second inner edge at medial instep opening edge **808s**, and a second outer edge **808t** at medial midfoot region **808**. An instep opening **812** for the seventh layer **800** is defined between the first inner edge at **804s** and the second inner edge at **808s**. The seventh path of the filament in this example layer **800** includes: (a) first plural, non-intersecting, spaced apart path segments located between the first inner edge at **804s** and the first outer edge **804t**, wherein a plurality of the first plural, non-intersecting, spaced apart path segments of the seventh path located between the first inner edge at **804s** and the first outer edge **804t** extend in a serpentine configuration and have at least two peaks **808P** and at least two valleys **808V** and (b) second plural, non-intersecting, spaced apart path segments located between the second inner edge at **808s** and the second outer edge **808t**, wherein the second, plural, non-intersecting, spaced apart path segments of the seventh path are linear and/or curved without defining plural peaks and valleys. The seventh layer **800** of FIG. 3L may include at least 4 path segments, at least 6 path segments, at least 8

path segments, at least 10 path segments, or even at least 12 path segments that extend substantially in parallel and/or have the noted serpentine configuration.

As evident from a comparison of FIGS. 3J and 3L, the path segments of sixth layer **700** and seventh layer **800** extend over a substantial portion of their overall paths in a generally parallel manner. Thus, the path segments of the seventh layer **800** may be extruded generally in parallel and/or to overlap the path segments of the sixth layer **700** over much of the overall path lengths in a manner as shown in FIGS. 2C and 2E. If desired: (a) at least 25% (and in some examples, at least 40%, at least 50%, at least 60%, at least 75%, at least 85%, or even at least 90%) of the overall path length of the seventh layer **800** will overlap with path segments of the sixth layer **700** in the manner shown in FIGS. 2C and 2E and/or (b) at least 25% (and in some examples, at least 40%, at least 50%, at least 60%, at least 75%, at least 85%, or even at least 90%) of the overall path length of the sixth layer **700** will overlap with path segments of the seventh layer **800** in the manner shown in FIGS. 2C and 2E. Thus, filaments of the seventh layer **800** will directly contact filaments of the sixth layer **700** (at overlapping contact area **202**) as the seventh layer **800** is being extruded. Heat from the seventh layer **800** while it is being extruded (and/or another heat source) causes the seventh filament path segments to fuse together with the sixth filament path segments at location(s) **202** where they contact one another (e.g., the filament material of the seventh layer **800** may polymerize with and seamlessly join the filament material of the sixth layer **700**). In this manner, the sixth layer **700** of the upper or upper blank **1000** can be fixedly joined to the seventh layer **800** of the upper or upper blank **1000** in an adhesive free manner at contact location(s) **202**.

As also evident from a comparison of FIG. 3L with FIGS. 3A, 3B, 3D, 3F, and 3H, the path segments of seventh layer **800** will substantially intersect the path segments of the first layer **300**, the second layer **350**, the third layer **400**, the fourth layer **500**, and the fifth layer **600** over a substantial portion of their overall paths. The intersecting path segments form generally diamond shapes, which can be seen in the combined first through seventh layers **840** shown in FIG. 3M. The path segments of the seventh layer **800** may intersect the path segments of the first layer **300**, the second layer **350**, the third layer **400**, the fourth layer **500**, and/or the fifth layer **600** at any desired angles, e.g., from 5° to 175°, and in some examples, from 15° to 165°, from 25° to 155°, from 35° to 145°, from 45° to 135°, from 55° to 125°, from 60° to 120°, from 65° to 90°, etc. In at least some examples of this invention: (a) the seventh path of the seventh layer **800** will overlap with one or more of the first path of the first layer **300**, the second path of the second layer **350**, the third path of the third layer **400**, the fourth path of the fourth layer **500**, and/or the fifth path of the fifth layer **600** over less than 50% (and in some examples less than 40%, less than 30%, less than 20%, or even less than 10%) of an overall length of the seventh path, and/or (b) the seventh path of the seventh layer **800** will overlap one or more of the first path of the first layer **300**, the second path of the second layer **350**, the third path of the third layer **400**, the fourth path of the fourth layer **500**, and/or the fifth path of the fifth layer **600** over less than 50% (and in some examples less than 40%, less than 30%, less than 20%, or even less than 10%) of an overall length of the respective first path, second path, third path, fourth path, and/or fifth path.

Thus, filaments of the seventh layer **800** will directly contact filaments of one or more of the first layer **300**, the second layer **350**, the third layer **400**, the fourth layer **500**,

and/or the fifth layer **600** (at intersecting contact area **202**) as the seventh layer **800** is being extruded. Heat from the seventh layer **800** as it is being extruded (and/or another heat source) causes the seventh filament path segments to fuse together with any one of or any combination of the first filament path segments, the second filament path segments, the third filament path segments, the fourth filament path segments, and/or the fifth filament path segments at location(s) **202** where the seventh filament path segments contact any one of or any combination of the first filament path segments, the second filament path segments, the third filament path segments, the fourth filament path segments, and/or the fifth filament path (e.g., the filament material of the seventh layer **800** may polymerize with and seamlessly join the filament materials of the first layer **300**, the second layer **350**, the third layer **400**, the fourth layer **500**, and/or the fifth layer **600**).

In these manners, the seventh layer **800** of the upper or upper blank **1000** can be fixedly joined to the first layer **300**, the second layer **350**, the third layer **400**, the fourth layer **500**, the fifth layer **600**, and/or the sixth layer **700** of the upper or upper blank **1000** in an adhesive free manner at contact location(s) **202** to form the combined first through seventh layers **840**. The upper component or intermediate including the combined first through seventh layers **840** constitutes a unitary construction in which the first layer **300**, the second layer **350**, the third layer **400**, the fourth layer **500**, the fifth layer **600**, the sixth layer **700**, and the seventh layer **800** are fixed together only in a non-adhesive fused manner. The upper component or intermediate including the combined first through seventh layers **840** may consist essentially of, or even consist of, the first layer **300**, the second layer **350**, the third layer **400**, the fourth layer **500**, the fifth layer **600**, the sixth layer **700**, and the seventh layer **800**.

In the seventh layer **800**, the path segments in one area need not have constant spacing from directly adjacent path segments at other areas of the seventh layer **800**. For example, as shown in FIG. 3L, the plural, non-intersecting, spaced apart path segments in the forefoot region **806** of the seventh layer **800** are spaced closer together than are the plural, non-intersecting, spaced apart path segments in the midfoot region(s) **804/808** and/or the plural, non-intersecting, spaced apart path segments in the forefoot region **806** and/or the midfoot region(s) **804/808** are spaced closer together than are the plural, non-intersecting, spaced apart path segments in the heel region(s) **802/810** of the seventh layer **800**.

After the seventh layer **800** is extruded (e.g., onto first layer **300**, second layer **350**, third layer **400**, fourth layer **500**, fifth layer **600**, sixth layer **700**, and/or substrate **106**), an eighth layer **900** of the overall upper or upper blank **1000** then may be applied to the combined first through seventh layers **840**. FIG. 3N shows the individual path segments of this example eighth layer **900**, and FIG. 3O schematically shows the production of the eighth layer **900** onto the previously prepared combined first through seventh layers **840** to create the combined first through eighth layers **1000** of the upper or upper blank **1000**. More specifically, FIG. 3N shows an eighth layer **900** formed of an eighth filament (e.g., having any of the filament features and characteristics described above) and formed by extruding an eighth material into multiple path segments (e.g., having any of the path and/or path segment features described above), e.g., by a solid deposition modeling process. Optionally, this eighth layer **900** may be extruded as an eighth continuous path. In this illustrated example, the eighth path of the eighth fila-

ment (optionally as a continuous path) forms the following portions of the eighth layer **900**: (a) an eighth lateral rear heel portion **902** (e.g., extending along and/or adjacent a lateral side **902s** of the ankle/foot opening **914** of the eighth layer **900**); (b) an eighth lateral midfoot portion **904** (e.g., extending along and/or adjacent a lateral side **904s** (or an inner edge) of an instep opening **912** of the eighth layer **900**); (c) an eighth forefoot portion **906** (e.g., which bridges from a lateral side to a medial side of the eighth layer **900**, forward of the midfoot portions); (d) an eighth medial midfoot portion **908** (e.g., extending along and/or adjacent a medial side **908s** (or an inner edge) of the instep opening **912** of the eighth layer **900**); and (e) an eighth medial rear heel portion **910** (e.g., extending along and/or adjacent a medial side **910s** of the ankle/foot opening **914** of the eighth layer **900**). The vertical dashed lines shown in FIG. 3N generally define and break the eighth layer **900** into three portions: (a) a posterior third (containing the lateral rear heel portion **902** and the medial rear heel portion **910**), (b) a central third (containing the lateral midfoot portion **904** and the medial midfoot portion **908**), and (c) an anterior third (containing the forefoot portion **906**). In at least some examples of this invention, the eighth layer **900** will consist essentially of, or even consist of, this filament structure (optionally formed as a continuous path and/or a one piece construction). The white space visible in FIG. 3N for this example eighth layer **900** constitutes open space between filament path segments (e.g., where one can see completely through the eighth layer **900**).

While the path segments of the eighth layer **900** can be extruded in any desired order without departing from this invention, in some examples of this invention, the outer perimeter (e.g., **900P**) may be extruded first, and then the remainder of the eighth layer **900** can be extruded, e.g., in a "raster" like fashion, to fill in the area within the perimeter **900P**. In this illustrated example, the extruded overall path of the eighth layer **900** lays down the eighth filament over much of the overall surface area of the eighth layer **900** as eighth plural, non-intersecting, spaced apart path segments that extend in a substantially anterior-posterior direction of the eighth layer **900**. As shown in FIG. 3N, in the lateral rear heel portion **902**, the forefoot portion **906**, the medial midfoot portion **908**, and the medial rear heel portion **910**, the eighth plural, non-intersecting path segments extend in a gently curved manner in a direction from the rear heel portion **902/910** forward. In the lateral midfoot portion **904**, however, at least some of the eighth plural, non-intersecting, spaced apart path segments extend in a serpentine configuration including at least two peaks (**908P**) and at least two valleys (**908V**). The path segments in these various regions **902**, **904**, **906**, **908**, **910** may have any of the features and/or options described above for the path segments shown in FIGS. 2A-2F.

As further shown in FIG. 3N, the eighth path of filament defines a first inner edge at lateral instep opening edge **904s**, a first outer edge **904t** at lateral midfoot region **904**, a second inner edge at medial instep opening edge **908s**, and a second outer edge **908t** at medial midfoot region **908**. An instep opening **912** for the eighth layer **900** is defined between the first inner edge at **904s** and the second inner edge at **908s**. The eighth path of the filament in this example layer **900** includes: (a) first plural, non-intersecting, spaced apart path segments located between the first inner edge at **904s** and the first outer edge **904t**, wherein a plurality of the first plural, non-intersecting, spaced apart path segments of the eighth path located between the first inner edge at **904s** and the first outer edge **904t** extend in a serpentine configuration and

31

have at least two peaks **908P** and at least two valleys **908V** and (b) second plural, non-intersecting, spaced apart path segments located between the second inner edge at **908s** and the second outer edge **908t**, wherein the second, plural, non-intersecting, spaced apart path segments of the eighth continuous path are linear and/or curved without defining plural peaks and valleys. The eighth layer **900** of FIG. 3N may include at least 4 path segments, at least 6 path segments, at least 8 path segments, at least 10 path segments, or even at least 12 path segments that extend substantially in parallel and/or have the noted serpentine configuration.

As evident from a comparison of FIGS. 3J, 3L, and 3N, the path segments of sixth layer **700**, seventh layer **800**, and eighth layer **900** extend over a substantial portion of their overall paths in a generally parallel manner. Thus, the path segments of the eighth layer **900** may be extruded generally in parallel and/or to overlap the path segments of the sixth layer **700** and/or the seventh layer **800** over much of their overall path lengths in a manner as shown in FIGS. 2C and 2E. If desired: (a) at least 25% (and in some examples, at least 40%, at least 50%, at least 60%, at least 75%, at least 85%, or even at least 90%) of the overall path length of the eighth layer **900** will overlap with path segments of at least one of the sixth layer **700** and/or the seventh layer **800** in the manner shown in FIGS. 2C and 2E and/or (b) at least 25% (and in some examples, at least 40%, at least 50%, at least 60%, at least 75%, at least 85%, or even at least 90%) of the overall path length of the sixth layer **700** and/or the seventh layer **800** will overlap with path segments of the eighth layer **900** in the manner shown in FIGS. 2C and 2E. Thus, filaments of the eighth layer **900** will directly contact filaments of the sixth layer **700** and/or the seventh layer **800** (at overlapping contact area **202**) as the eighth layer **900** is being extruded. Heat from the eighth layer **900** while it is being extruded (and/or another heat source) causes the eighth filament path segments to fuse together with either or both of the sixth filament path segments and/or the seventh filament path segments at location(s) **202** where the eighth filament path segments contact either or both of the sixth filament path segments and/or the seventh filament path segments (e.g., the filament material of the eighth layer **900** may polymerize with and seamlessly join the filament material of the sixth layer **700** and/or the seventh layer **800**). In this manner, the sixth layer **700** and/or seventh layer **800** of the upper or upper blank **1000** can be fixedly joined to the eighth layer **900** of the upper or upper blank **1000** in an adhesive free manner at contact location(s) **202**.

As also evident from a comparison of FIG. 3N with FIGS. 3A, 3B, 3D, 3F, and 3H, the path segments of eighth layer **900** will substantially intersect the path segments of the first layer **300**, the second layer **350**, the third layer **400**, the fourth layer **500**, and the fifth layer **600** over a substantial portion of their overall paths. The intersecting path segments form generally diamond shapes, which can be seen in the combined first through eighth layers **1000** shown in FIGS. 3O and 3P. The path segments of the eighth layer **900** may intersect the path segments of the first layer **300**, the second layer **350**, the third layer **400**, the fourth layer **500**, and/or the fifth layer **600** at any desired angles, e.g., from 5° to 175°, and in some examples, from 15° to 165°, from 25° to 155°, from 35° to 145°, from 45° to 135°, from 55° to 125°, from 60° to 120°, from 65° to 90°, etc. In at least some examples of this invention: (a) the eighth path of the eighth layer **900** will overlap with one or more of the first path of the first layer **300**, the second path of the second layer **350**, the third path of the third layer **400**, the fourth path of the fourth layer **500**, and/or the fifth path of the fifth layer **600** over less than

32

50% (and in some examples less than 40%, less than 30%, less than 20%, or even less than 10%) of an overall length of the eighth path, and/or (b) the eighth path of the eighth layer **900** will overlap one or more of the first path of the first layer **300**, the second path of the second layer **350**, the third path of the third layer **400**, the fourth path of the fourth layer **500**, and/or the fifth path of the fifth layer **600** over less than 50% (and in some examples less than 40%, less than 30%, less than 20%, or even less than 10%) of an overall length of the respective first path, second path, third path, fourth path, and/or fifth path.

Thus, filaments of the eighth layer **900** will directly contact filaments of one or more of the first layer **300**, the second layer **350**, the third layer **400**, the fourth layer **500**, and/or the fifth layer **600** (at intersecting contact area **202**) as the eighth layer **900** is being extruded. Heat from the eighth layer **900** while it is being extruded (and/or another heat source) causes the eighth filament path segments to fuse together with any one of or any combination of the first filament path segments, the second filament path segments, the third filament path segments, the fourth filament path segments, and/or the fifth filament path segments at location(s) **202** where the eighth filament path segments contact any one of or any combination of the first filament path segments, the second filament path segments, the third filament path segments, the fourth filament path segments, and/or the fifth filament path segments (e.g., the filament material of the eighth layer **900** may polymerize with and seamlessly join the filament materials of the first layer **300**, the second layer **350**, the third layer **400**, the fourth layer **500**, and/or the fifth layer **600**).

In these manners, the eighth layer **900** of the upper or upper blank **1000** can be fixedly joined to the first layer **300**, the second layer **350**, the third layer **400**, the fourth layer **500**, the fifth layer **600**, the sixth layer **700**, and/or the seventh layer **800** of the upper or upper blank **1000** in an adhesive free manner at contact location(s) **202** to form the combined first through eighth layers **1000**. The upper component or intermediate including the combined first through eighth layers **1000** constitutes a unitary construction in which the first layer **300**, the second layer **350**, the third layer **400**, the fourth layer **500**, the fifth layer **600**, the sixth layer **700**, the seventh layer **800**, and the eighth layer **900** are fixed together only in a non-adhesive fused manner. The upper component or upper blank **1000** may consist essentially of, or even consist of, the first layer **300**, the second layer **350**, the third layer **400**, the fourth layer **500**, the fifth layer **600**, the sixth layer **700**, the seventh layer **800**, and the eighth layer **900**.

In the eighth layer **900**, the path segments in one area need not have constant spacing from directly adjacent path segments at other areas of the eighth layer **900**. For example, as shown in FIG. 3N, the plural, non-intersecting, spaced apart path segments in the forefoot region **906** of the eighth layer **900** are spaced closer together than are the plural, non-intersecting, spaced apart path segments in the midfoot region(s) **904/908** and/or the plural, non-intersecting, spaced apart path segments in the forefoot region **906** and/or the midfoot region(s) **904/908** are spaced closer together than are the plural, non-intersecting, spaced apart path segments in the heel region(s) **902/910** of the eighth layer **900**.

After all desired layers of the upper or upper blank **1000** are formed, the extruded and fused layers form a fused upper component (e.g., an upper blank **1000**). Then, the fused upper component **1000** may be removed from the base substrate **106** onto which it was extruded, optionally trimmed (e.g., around its perimeter edges, if needed) or

33

otherwise processed (e.g., coated, painted, etc.), optionally engaged with another upper component, and/or incorporated into a footwear structure, as will be explained in more detail below. As described above, if desired, the surface 106s of the substrate 106 onto which the layer(s) of filament are extruded may be smooth or otherwise textured. The surface (e.g., individual filaments) of the upper blank 1000 that contacted the substrate 106 surface 106s then may take on the smoothness (or textured) characteristics of the substrate 106 surface 106s on which it contacts and is formed. Thus, in some examples of this invention, one surface (e.g., the inner surface or bottom surface) of the upper blank 1000 may be smooth or textured to correspond to texture on the surface 106s of the substrate 106 while the opposite surface (e.g., the outer surface or top surface) of the upper blank 1000 may have texture corresponding to the multiple overlapping layers of filament.

FIGS. 3Q to 3W provide enlarged views of portions of the upper blank 1000 at areas A-G shown in FIG. 3P to show more details of these example structures. FIGS. 3Q and 3R provide enlarged views of the upper blank 1000 at the lateral rear heel portion 1002 and the medial rear heel portion 1010 (areas A and B, respectively); FIGS. 3S and 3T provide enlarged views of the upper blank 1000 at the lateral midfoot portion 1004 and the medial midfoot portion 1008 (areas C and D, respectively); and FIGS. 3U, 3V, and 3W provide enlarged views of the upper blank 1000 at the lateral forefoot portion, the medial forefoot portion, and the extreme forefoot portion (areas E, F, and G, respectively).

As shown in FIG. 3Q, the filaments of the first layer 300 and the second layer 350 at this location of the upper blank 1000 (which may be substantially parallel and/or overlapping over much of their lengths) generally extend in a direction between the ankle opening 1014 and the bottom perimeter edge of the upper blank 1000 (e.g., in a medio-lateral direction). The filaments of the third layer 400, the fourth layer 500, and the fifth layer 600 generally curve (and are substantially in parallel and/or overlapping) and extend in an anterior-posterior direction. Similarly, the filaments of the sixth layer 700, the seventh layer 800, and the eighth layer 900 generally curve (and are substantially in parallel and/or overlapping) and extend in an anterior-posterior direction. The filaments of the first layer 300 and second layer 350 intersect the filaments of the third layer 400, fourth layer 500, and fifth layer 600 and form generally parallelogram or diamond shapes (e.g., with angles between 60° and 120°). Similarly, the filaments of the first layer 300 and second layer 350 intersect the filaments of the sixth layer 700, seventh layer 800, and eighth layer 900 and form generally parallelogram or diamond shapes (e.g., with angles between 60° and 120°). The filaments of the third layer 400, fourth layer 500, and fifth layer 600 intersect the filaments of the sixth layer 700, seventh layer 800, and eighth layer 900 and form generally parallelogram or diamond shapes (e.g., with two angles between 5° and 60° and two angles between 120° and 175°). When formed as parallelogram and/or diamond shapes, a long axis of the parallelogram/diamond shapes formed by the filaments of layers 400/500/600 and the filaments of layers 700/800/900 may extend in generally an anterior-posterior direction of the upper blank 1000 (e.g., as shown by diamond DIA highlighted in FIG. 3Q).

As shown in FIG. 3R, the filaments of the first layer 300 and the second layer 350 at this location of the upper blank 1000 (which may be substantially parallel and/or overlapping over much of their lengths) generally extend in a direction between the ankle opening 1014 and the bottom

34

perimeter edge of the upper blank 1000 (e.g., in a medio-lateral direction). The filaments of the third layer 400, the fourth layer 500, and the fifth layer 600 generally curve (and are substantially in parallel and/or overlapping) and extend in an anterior-posterior direction. Similarly, the filaments of the sixth layer 700, the seventh layer 800, and the eighth layer 900 generally curve (and are substantially in parallel and/or overlapping) and extend in an anterior-posterior direction. The filaments of the first layer 300 and second layer 350 intersect the filaments of the third layer 400, fourth layer 500, and fifth layer 600 and form a generally parallelogram or diamond shaped pattern (e.g., with angles between 60° and 120°). Similarly, the filaments of the first layer 300 and second layer 350 intersect the filaments of the sixth layer 700, seventh layer 800, and eighth layer 900 and form generally parallelogram or diamond shapes (e.g., with angles between 60° and 120°). The filaments of the third layer 400, fourth layer 500, and fifth layer 600 intersect the filaments of the sixth layer 700, seventh layer 800, and eighth layer 900 and form generally parallelogram or diamond shapes (e.g., with two angles between 5° and 60° and two angles between 120° and 175°). When formed as parallelogram and/or diamond shapes, a long axis of the parallelogram/diamond shapes formed by the filaments of layers 400/500/600 and the filaments of layers 700/800/900 may extend in generally an anterior-posterior direction of the upper blank 1000 (e.g., as shown by diamond DIA highlighted in FIG. 3R).

As shown in FIG. 3S, the filaments of the first layer 300 and the second layer 350 at this location of the upper blank 1000 (which may be substantially parallel and/or overlapping over much of their lengths) generally extend in a direction between the instep opening 1012 and the bottom perimeter edge of the upper blank 1000 (e.g., in a medio-lateral direction). The filaments of the third layer 400, the fourth layer 500, and the fifth layer 600 generally curve (and are substantially in parallel and/or overlapping) and extend in an anterior-posterior direction. The filaments of the sixth layer 700, the seventh layer 800, and the eighth layer 900 in this area of the upper blank 1000 form the serpentine pattern described above (and are substantially in parallel and/or overlapping), and generally extend in an anterior-posterior direction. The intersecting filaments of the various layers may have any of the angular properties described above in conjunction with FIGS. 3Q and 3R. At this particular area of the upper blank 1000, the intersecting filaments define some more “square” or rectangular openings through the upper blank 1000.

As shown in FIG. 3T, the filaments of the first layer 300 and the second layer 350 at this location of the upper blank 1000 (which may be substantially parallel and/or overlapping over much of their lengths) generally extend in a direction between the instep opening 1012 and the bottom perimeter edge of the upper blank 1000 (e.g., in a medio-lateral direction). The filaments of the third layer 400, the fourth layer 500, and the fifth layer 600 in this area of the upper blank 1000 form the serpentine pattern described above (and are substantially in parallel and/or overlapping), and generally extend in an anterior-posterior direction. The filaments of the sixth layer 700, the seventh layer 800, and the eighth layer 900 generally curve (and are substantially in parallel and/or overlapping) and extend in an anterior-posterior direction. The intersecting filaments of the various layers may have any of the angular properties described above in conjunction with FIGS. 3Q and 3R. At this particular area of the upper blank 1000, the intersecting fila-

35

ments define some more “square” or rectangular openings through the upper blank 1000.

As shown in FIG. 3U, the filaments of the first layer 300 and the second layer 350 at this location of the upper blank 1000 (which may be substantially parallel and/or overlapping over much of their lengths) generally extend in a medio-lateral direction across the upper blank 1000 (from the medial side toward the lateral side). The filaments of the third layer 400, the fourth layer 500, and the fifth layer 600 generally curve (and are substantially in parallel and/or overlapping) and extend in an anterior-posterior direction. Similarly, the filaments of the sixth layer 700, the seventh layer 800, and the eighth layer 900 generally curve (and are substantially in parallel and/or overlapping) and extend in an anterior-posterior direction. The filaments of the first layer 300 and second layer 350 intersect the filaments of the third layer 400, fourth layer 500, and fifth layer 600 and form generally parallelogram or diamond shapes (e.g., with angles between 60° and 120°). Similarly, the filaments of the first layer 300 and second layer 350 intersect the filaments of the sixth layer 700, seventh layer 800, and eighth layer 900 and form generally parallelogram or diamond shapes (e.g., with angles between 60° and 120°). The filaments of the third layer 400, fourth layer 500, and fifth layer 600 intersect the filaments of the sixth layer 700, seventh layer 800, and eighth layer 900 and form generally parallelogram or diamond shapes (e.g., with two angles between 5° and 60° and two angles between 120° and 175°). When formed as parallelogram and/or diamond shapes, a long axis of the parallelogram/diamond shapes formed by the filaments of layers 400/500/600 and the filaments of layers 700/800/900 may extend in generally an anterior-posterior direction of the upper blank 1000 (e.g., as shown by diamond DIA highlighted in FIG. 3U).

As shown in FIG. 3V, the filaments of the first layer 300 and the second layer 350 at this location of the upper blank 1000 (which may be substantially parallel and/or overlapping over much of their lengths) generally extend in a medio-lateral direction across the upper blank 1000 (from the medial side toward the lateral side). The filaments of the third layer 400, the fourth layer 500, and the fifth layer 600 generally curve (and are substantially in parallel and/or overlapping) and extend in an anterior-posterior direction. Similarly, the filaments of the sixth layer 700, the seventh layer 800, and the eighth layer 900 generally curve (and are substantially in parallel and/or overlapping) and extend in an anterior-posterior direction. The filaments of the first layer 300 and second layer 350 intersect the filaments of the third layer 400, fourth layer 500, and fifth layer 600 and form generally parallelogram or diamond shapes (e.g., with angles between 60° and 120°). Similarly, the filaments of the first layer 300 and second layer 350 intersect the filaments of the sixth layer 700, seventh layer 800, and eighth layer 900 and form generally parallelogram or diamond shapes (e.g., with angles between 60° and 120°). The filaments of the third layer 400, fourth layer 500, and fifth layer 600 intersect the filaments of the sixth layer 700, seventh layer 800, and eighth layer 900 and form generally parallelogram or diamond shapes pattern (e.g., with two angles between 5° and 60° and two angles between 120° and 175°). When formed as parallelogram and/or diamond shapes, a long axis of the parallelogram/diamond shapes formed by the filaments of layers 400/500/600 and the filaments of layers 700/800/900 may extend in generally an anterior-posterior direction of the upper blank 1000 (e.g., as shown by diamond DIA highlighted in FIG. 3V).

36

As shown in FIG. 3W, the filaments of the first layer 300 and the second layer 350 at this location of the upper blank 1000 (which may be substantially parallel and/or overlapping over much of their lengths) generally extend in a medio-lateral direction across the upper blank 1000 (from the medial side toward the lateral side). The filaments of the third layer 400, the fourth layer 500, and the fifth layer 600 generally curve (and are substantially in parallel and/or overlapping) and extend in an anterior-posterior direction. Similarly, the filaments of the sixth layer 700, the seventh layer 800, and the eighth layer 900 generally curve (and are substantially in parallel and/or overlapping) and extend in an anterior-posterior direction. The filaments of the first layer 300 and second layer 350 intersect the filaments of the third layer 400, fourth layer 500, and fifth layer 600 and form generally parallelogram or diamond shapes (e.g., with angles between 60° and 120°). Similarly, the filaments of the first layer 300 and second layer 350 intersect the filaments of the sixth layer 700, seventh layer 800, and eighth layer 900 and form generally parallelogram or diamond shapes (e.g., with angles between 60° and 120°). The filaments of the third layer 400, fourth layer 500, and fifth layer 600 intersect the filaments of the sixth layer 700, seventh layer 800, and eighth layer 900 and form generally parallelogram or diamond shapes (e.g., with two angles between 5° and 60° and two angles between 120° and 175°). When formed as parallelogram and/or diamond shapes, a long axis of the parallelogram/diamond shapes formed by the filaments of layers 400/500/600 and the filaments of layers 700/800/900 may extend in generally an anterior-posterior direction of the upper blank 1000 (e.g., as shown by diamond DIA highlighted in FIG. 3W).

As evident from a comparison of FIGS. 3Q-3V with 3W, the spacings between directly adjacent filaments in a given layer generally become smaller at the forefoot area of this example upper blank 1000 as compared to the spacings between directly adjacent filaments in the same layer at the rear heel and/or midfoot portions of the upper blank 1000. Also, as shown in these figures, the diamond shaped openings formed by intersections between (a) the filaments of the third layer 400, fourth layer 500, and fifth layer 600 and (b) the filaments of the sixth layer 700, seventh layer 800, and eighth layer 900 become smaller toward the forefoot area of the upper blank 1000 as compared to the corresponding diamond shaped openings provided at the rear heel and/or midfoot portions of the upper blank 1000. These features provide improved support, durability, lock-down, and less stretchability at the forefoot area of the upper blank 1000 as compared to at least some other areas. As further evident from the enlarged views of FIGS. 3Q-3W, much of the space within the perimeter of the upper or upper blank constitutes open space between intersections of the filaments in the layers of the upper or upper blank 1000. In at least some examples of this invention, the upper or upper blank 1000 may constitute at least 15% open space between filaments of the various layers, and in some examples, at least 25%, at least 30%, at least 40%, or even at least 50% open space may be present between filaments in the upper or upper blank 1000.

Many variations in the upper or upper blank 1000 structure and/or individual upper layers of filament are possible without departing from this invention. As some examples, an upper or upper blank 1000 could include more or fewer upper layers than the eight layers described above (e.g., from 2 to 24 layers, and in some examples, from 3 to 20 layers, from 4 to 16 layers, from 5 to 12 layers, etc.). As additional or alternative examples, the layering order of the filament

layers described above could be changed in some examples of this invention. Additionally or alternatively, while FIGS. 3D, 3F, and 3H show the serpentine configurations formed at the medial midfoot portions of the third layer 400, the fourth layer 500, and the fifth layer 600, respectively, and FIGS. 3J, 3L, and 3N show the serpentine configurations formed at the lateral midfoot portions of the sixth layer 700, the seventh layer 800, and the eighth layer 900, respectively, an upper or upper blank in accordance with some examples of this invention may include more of fewer of this same type of "serpentine" configurations, e.g., in one or more of the forefoot area, the rear heel areas, etc. More or fewer layers of an upper blank 1000 also may include this type of "serpentine" configuration. The curved and serpentine configurations of the filament layers provide enhanced flexibility.

The filament materials provided in the various layers of a single upper or upper blank 1000 may be the same or different without departing from this invention. As some more specific examples, the upper or upper blank and/or the individual layers thereof may have one or more of the following properties or characteristics: (a) the filament materials on all layers of an upper or upper blank may be the same material; (b) the filament material in one layer of an upper or upper blank may differ from the filament material in one or more other layers of the upper or upper blank; (c) the filament material may be different in each layer of an upper or upper blank; (d) the filaments on all layers of an upper or upper blank may be the same color; (e) the filament color in one layer of an upper or upper blank may differ from the filament color in one or more other layers of the upper or upper blank; (f) the filament colors may be different in each layer of an upper or upper blank; (g) the filament in one or more layers of an upper or upper blank may be made from a transparent or translucent material; (h) the filament in one or more layers of the upper or upper blank (and optionally in each layer of the upper or upper blank) may be made from a thermoplastic material (e.g., a thermoplastic polyurethane material); (i) the filament in one or more layers of the upper or upper blank (and optionally in each layer of the upper or upper blank) may be made from a material that does not substantially absorb water; (j) the filament in one or more layers of the upper or upper blank (and optionally in each layer of the upper or upper blank) may be made from a hydrophobic material; (k) the filament in one or more layers of the upper or upper blank (and optionally in each layer of the upper or upper blank) may be made from a material capable of fusing to material(s) of the other layers, e.g., in an adhesive free manner, such as via a solid deposition modeling technique; (l) the filament materials on all layers of an upper or upper blank may have the same diameter, width, and/or thickness (or other dimensions); and/or (m) the filament material in one layer of an upper or upper blank may differ in diameter, width, and/or thickness (or other dimensions) from the filament material in one or more other layers of the upper or upper blank. In some examples of this invention, the upper or upper blank (e.g., two or more layers thereof cooperatively) may produce a moiré effect, e.g., as shown in FIGS. 1 and 3A-3P.

As described above, in at least some examples of this invention, one or more of the various upper layers (e.g., layers 300, 350, 400, 500, 600, 700, 800, and/or 900) may be formed as a continuous path of extruded filament. For example, in some examples of this invention, the entire upper layer (e.g., as shown in each of one or more of FIG. 3A (first layer 300), FIG. 3B (second layer 350), FIG. 3D (third layer 400), FIG. 3F (fourth layer 500), FIG. 3H (fifth

layer 600), FIG. 3J (sixth layer 700), FIG. 3L (seventh layer 800), and FIG. 3N (eighth layer 900)) may be formed by beginning extrusion at one location of the individual upper layer and extruding material continuously (including through all necessary turns of the extruder nozzle 104/head) until the entire layer is completely extruded (e.g., into the form shown in one or more of FIGS. 3A, 3B, 3D, 3F, 3H, 3J, 3L, and 3N). This type of continuous extrusion path (with one start and one stop in extruding an entire layer), however, is not a requirement. Rather, in at least some examples of this invention, an individual upper layer (e.g., like those shown in FIGS. 3A, 3B, 3D, 3F, 3H, 3J, 3L, and 3N) may be formed in a discontinuous manner (e.g., with two or more pairs of extrusion start/stop actions per individual layer). The extrusion nozzle 104/head may be moved between the extrusion starts and stops (while extrusion is stopped) to another desired location of the upper layer so that the extrusion path segments of the layer can begin/end at any desired locations.

Further, FIGS. 3A-3P show that each individual layer 300, 350, 400, 500, 600, 700, 800, 900 in this illustrated upper or upper blank 1000 includes individual path segments throughout each of the lateral rear heel portion, lateral midfoot portion, forefoot portion, medial midfoot portion, and medial rear heel portion. This is not a requirement. Rather, if desired, in accordance with some examples of this invention, an individual upper layer (e.g., one or more of 300, 350, 400, 500, 600, 700, 800, and/or 900) need not extend into and/or throughout all of the noted regions. As some more specific examples, if desired: (a) one or more layers of the upper may be present in only the forefoot region of the overall upper; (b) one or more layers may be present in only the lateral midfoot region and/or medial midfoot region of the overall upper; (c) one or more layers may be present in the forefoot region and one or both of the lateral midfoot region and/or medial midfoot region of the overall upper; (d) one or more layers may be omitted at one or both of the lateral rear heel region and/or the medial rear heel region; (e) only a small amount of filament may be present in some portions or regions of an individual layer; etc. Selective inclusion of filament at less than all regions of an upper layer can allow one to selectively control properties in that region of the overall upper or upper blank (e.g., control breathability, stretchability, permeability, lock down, etc.). When one or more layers are omitted in an individual area or region, the layers of filament that are included in that individual area or region need not be extruded consecutively in the overall upper production process. For example, if desired, an individual area or region of an upper or upper blank may include the first layer 300, the second layer 350, the fourth layer 500, the sixth layer 700, and the seventh layer 800 (or any desired number and/or combination of layers, including layers having different paths from the specific examples shown in the figures).

FIGS. 3A-3P further show an example with upper layers laid down as filaments in a specific order, i.e., layer 300, then layer 350, then layer 400, then layer 500, then layer 600, then layer 700, then layer 800, then layer 900. The order in which layers are put down can influence the characteristics and/or properties of the overall upper or upper blank 1000. For example, if layers with larger filament spacings are laid down closer to the upper interior and layers with smaller filament spacings are laid down closer to the upper exterior, this may enhance the overall upper's ability to shed water (or other materials) and/or resist introduction of water (or other materials) inside and/or within the upper as compared to an upper with the same layers formed in a different order with

larger filament spacings located closer to the upper exterior. Changing layer orders also may affect zonal or overall performance of the upper (e.g., affect lockdown, directional stretch, breathability, permeability, etc.) By changing the layer orders, the intersections of the different layers may be altered and/or changed in locations, and thus the fusion of one layer to the other layers may be altered and/or changed to different locations and/or positions. These changes may affect overall properties, performance, and/or feel of the upper (e.g., lockdown, directional stretch, breathability, permeability, etc.).

As mentioned above, if desired, the upper blank **1000** or an upper component built from one or more layers of extruded filaments of the types described above may be engaged with one or more other upper components. FIGS. 4A through 4C illustrate examples in which the filament(s) **1102**, **1104**, **1106**, **1108** of one or more layers of an extruded filament based upper component **1100** are engaged with another upper component **1110**, e.g., by an adhesive or cement **1112**. The extruded filament based upper component **1100** may be made of one or more layers of extruded filament, e.g., including one of more layers, including any one or more of the various layers **300**, **350**, **400**, **500**, **600**, **700**, **800**, and/or **900** described above. The upper component **1110** may be any desired type of upper component, including upper components as conventionally known and used in the footwear arts, such as fabrics, textiles, elastic materials, knitted components, woven components, leathers (natural or synthetic), thermoplastic materials, thermoplastic polyurethanes, other polymer materials, etc. The extruded filament based upper component **1100**: (a) may be located outside the other upper component **1110** toward the footwear exterior and away from the wearer's foot (FIG. 4A); (b) may be located inside the other upper component **1110** toward the footwear interior and closer to (and optionally adjacent or in contact with) the wearer's foot (FIG. 4B); and/or (c) may be both inside and outside the other upper component **1110** (FIG. 4C). The additional upper component(s) **1110** may provide additional support at desired areas (e.g., a heel counter, a toe box, structural support, shape support, lace engaging support along the instep opening **1012**, structure for engaging a sole component, etc.); a soft foot contact surface (e.g., around the ankle opening **1014**, as a "tongue" member across the instep opening **1012**, etc.); support for engaging another footwear component; etc. The combined upper components **1100/1110** may be incorporated into an overall footwear structure, as will be described in more detail below.

FIGS. 5A through 5C, however, illustrate engagement of an extruded filament based upper component **1100** with another upper component **1120** in an adhesive free manner, by fusing (or fuse bonding) the filament(s) in the upper component **1100** directly with the material of the other upper component **1120**. Such combined upper components **1100/1120** may include: (a) a first upper component **1100** that includes at least a first layer formed to include a first material as a first filament **1102** (four filaments **1102**, **1104**, **1106**, and **1108** are shown in FIGS. 5A-5C), optionally including first plural, non-intersecting, spaced apart path segments of the filament, wherein the first extruded filament has a width dimension of less than 3 mm wide (and in some examples, less than 2 mm wide, less than 1.5 mm wide, less than 1 mm wide, or even less than 0.75 mm wide); and (b) a second upper component **1120** including a fabric element (e.g., of any of the types described above for upper component **1110**, such as fabrics, textiles, elastic materials, thermoplastic materials, etc.) formed at least in part of a fusible material.

The extruded filament based upper component **1100** may be made of one or more layers of extruded filament, e.g., including one of more layers like layers **300**, **350**, **400**, **500**, **600**, **700**, **800**, and/or **900** described above. In such structures, the fusible material of the second upper component **1120** is fused to the filaments **1102**, **1104**, **1106**, and/or **1108** of the first upper component **1100**. If necessary or desired, heat and/or pressure may be applied to the combined upper components **1100/1120** to fuse the filament containing upper component **1100** to the fabric element upper component **1120**. The fusing between components **1100** and **1120** may be a result of the material of one or more filaments **1102-1108** polymerizing with and seamlessly joining the fusible material of the fabric element **1120**.

FIGS. 5A through 5C illustrate examples in which the filament(s) **1102**, **1104**, **1106**, **1108** of one or more layers of an extruded filament upper component **1100** are engaged with an upper component **1120** that includes a fusible material as part of a fabric element. The extruded filament upper component **1100**: (a) may be located outside the other upper component **1120** toward the footwear exterior and away from the wearer's foot (FIG. 5A); (b) may be located inside the other upper component **1120** toward the footwear interior and closer to (and optionally adjacent or in contact with) the wearer's foot (FIG. 5B); and/or (c) may be both inside and outside the other upper component **1120** (FIG. 5C). The additional upper component(s) **1120** may provide additional support at desired areas (e.g., a heel counter, a toe box, structural support, shape support, lace engaging support along the instep opening **1012**, structure for engaging a sole component, etc.); a soft foot contact surface (e.g., around the ankle opening **1014**, as a "tongue" member across the instep opening **1012**, etc.); structure for engaging another footwear component; etc. The combined upper components **1100/1120** may be incorporated into an overall footwear structure, as will be described in more detail below.

FIGS. 5D through 5F provide examples of manners in which a fusible material may be incorporated into a fabric element **1120**. As one example, as shown in FIG. 5D, a fabric element **1120** may be formed as a woven, knitted, or unwoven structure in which at least one strand (e.g., yarn) of the fabric element **1120** is formed of a fusible material compatible to fuse with the material of one or more of the filaments of upper component **1100**. In FIG. 5D, the yarn **1122** of one color is a traditional fabric material (e.g., made from a polyester, cotton, elastomeric material, etc.) and the yarn **1124** of the other color is made from the fusible material of the types described above for the fusible material in the filaments, such as a thermoplastic polyurethane material or other thermoplastic material. In this arrangement, the yarn **1124** can directly fuse to the filament(s) **1102**, **1104**, **1106**, **1108** of the filament based upper component **1100**.

As another option, as shown in FIG. 5E, a fabric element **1120** may be formed as a woven, knitted, or unwoven structure that includes at least one yarn of the fabric element structure **1120** formed of: (a) a strand **1132** made of a traditional fabric material (e.g., made from a polyester, cotton, elastomeric material, etc.) intertwined with (b) a strand **1134** made from the fusible material of the types described above for the fusible material in the filaments, such as a thermoplastic polyurethane material or other thermoplastic material. In this arrangement, the fusible yarn strand **1134** can directly fuse to the filament(s) **1102**, **1104**, **1106**, **1108** of the filament based upper component **1100**.

As yet another option, FIG. 5F provides a cross sectional view of a yarn or strand that may be used to make a fabric element **1120**, e.g., as a woven, knitted, or unwoven struc-

41

ture. This strand or yarn is formed as a coaxial element over at least a portion of its axial length, including: (a) an inner core **1142** made of a traditional fabric material (e.g., made from a polyester, cotton, elastomeric material, etc.) coated (e.g., co-extruded or otherwise formed) with (b) an outer cover **1144** made from the fusible material of the types described above for the fusible material in the filaments, such as a thermoplastic polyurethane material or other thermoplastic material. In this arrangement, the fusible outer cover **1144** can directly fuse to the filament(s) **1102**, **1104**, **1106**, **1108** of the filament based upper component **1100**.

FIGS. 6A through 6E illustrate an example of a manner in which a conventional fabric element (or other footwear component) **1110** and/or a fabric element **1120** including a fusible material (e.g., of the types described above in conjunction with FIGS. 5D-5F) can be engaged with a multi-layered fusible upper component **1100** (e.g., of the types described above in conjunction with FIGS. 1A-3W). First, an upper blank intermediate **1200** can be made including one or more layers of extruded filament formed as described above (e.g., by a solid deposition modeling technique, as a continuous path, with plural, non-intersecting path segments, etc.). FIG. 6A shows an upper blank intermediate **1200** formed to include the first four layers **300**, **350**, **400**, **500** of filament as described above, e.g., which can be formed in the same manners as described above. Any desired number of filament layers, having any desired filament paths and arrangements, may be provided as upper blank intermediate **1200** without departing from this invention (including one or more filament based layers and/or any individual layer and/or combination of layers described above).

Once the desired upper blank intermediate **1200** has been prepared including one or more filament layers (e.g., by extrusion, solid deposition modeling, etc.) on the extruder substrate **106**, a release liner **1202** may be applied, e.g., to cover a portion of the top surface of the upper blank intermediate **1200** (e.g., a portion of the first layer extending inwardly from a peripheral edge of the first layer may be covered with the release liner **1202**, such as a portion around one or both sides of the instep opening, a portion around one or both sides of the ankle opening, a portion around the bottom edge where the upper will meet a sole structure, etc.). The release liner **1202** may be made from paper, plastic, or any type of material, provided that the release liner **1202** does not become permanently fixed to the filament material(s) to be included in the upper component in the steps to follow.

With the release liner **1202** in place, one or more additional layers of filament may be extruded to form one or more additional layers of the upper component. For example, as shown in FIG. 6B, the fifth layer **600**, sixth layer **700**, seventh layer **800**, and/or the eighth layer **900** described above may be extruded onto the release liner **1202** and the upper intermediate **1200** (which may include the first layer **300**, second layer **350**, third layer **400**, and/or fourth layer **500** described above). In this illustrated example, one or more additional materials are extruded to form one or more additional layers of extruded filament, optionally including additional plural, non-intersecting, spaced apart path segments in the individual layer(s). The additional extruded filament(s) may have any of the dimensional and/or other filament characteristics described above. The additional layer extruding step(s) of this example may include: (a) applying a first portion of the additional filament layer(s) to the release liner **1202** such that the release liner **1202** extends between a first portion of the previously extruded filament layer(s) (in upper intermediate **1200**) and the first

42

portion of the additional layer(s) of filament and (b) fusing a second portion of the additional layer(s) of filament to a second portion of the previously extruded filament layer(s) (in upper intermediate **1200**) at locations where the more recently extruded filament layer(s) contact the previously extruded filament layers. In this manner, the additional filament layer(s) applied after the release liner **1202** is positioned will join to (in an adhesive free manner) to the previously extruded filament layer(s) at areas away from the release liner **1202** to provide a unitary upper component (e.g., an overall upper component, like upper blank **1000** described above).

Once all of the desired additional filament layers are extruded and joined to the previously extruded filament layers of upper intermediate **1200**, a combined upper component **1210** and release liner **1202** combination is present, e.g., as shown in FIG. 6C. At this time, one or more of the upper component **1210** filament layers are located below a bottom surface of the release liner **1202** and one or more of the upper component **1210** filament layers are located above a top surface of the release liner **1202**. At this time, the upper component **1210** and release liner **1202** can be removed from the extruder substrate **106**, and the release liner **1202** can be removed from its location between the first portion of the lower filament layer(s) of the upper component **1210** (e.g., layers **300**, **350**, **400**, **500**) and the first portion of the upper filament layer(s) of the upper component **1210** (e.g., layers **600**, **700**, **800**, **900**). Because of the previous presence of the release liner **1202**, the first portion of the lower filament layer(s) of the upper component **1210** (e.g., layers **300**, **350**, **400**, **500**) will remain unfixed to the first portion of the upper filament layer(s) of the upper component **1210** (e.g., layers **600**, **700**, **800**, **900**) at the location where the release liner **1202** was present (although all layers **300-900** may be fixed together at the "second portions" thereof, where the release liner **1202** was not present). In other words, the presence and removal of the release liner **1202** forms a "pocket" between layers of the upper component **1210**.

At this time, as shown in FIG. 6D, another upper component (e.g., fabric component **1110**, **1120** like those described above) may be placed into the "pocket" formed between layers of the upper component **1210** where the release liner **1202** was removed. The upper component **1110** could be engaged with the upper component **1210** via adhesives, as described above in conjunction with FIG. 4C. Alternatively, when upper component **1120** includes a fusible material (e.g., a fabric element of the types described above in conjunction with FIGS. 5D-5F), then upper component **1120** can be engaged with the upper component **1210** in an adhesive free manner, e.g., by fusing the fusible material of the upper component **1120** with the extruded filament material(s) of the upper component **1210**. This action can place and fix the upper component **1120** between individual layers of the upper component **1210**, e.g., as shown in the cross section of FIG. 5C. If needed, this fusing step can include, for example, applying heat and/or pressure to the combined upper component **1120**/upper component **1210**, e.g., in the form shown in FIG. 6E. If necessary, the upper component **1120** (and/or **1210**) can be trimmed, combined with other upper components, and/or otherwise processed, if necessary, e.g., in preparation for incorporation into a footwear structure. While FIGS. 6A to 6E show the upper component **1110/1120** between central layers of the filament based upper component **1210**, this is not a requirement. Rather, the upper component **1110/1120** may be placed between any desired filament layers of upper com-

ponent **1210** without departing from this invention (e.g., between layers closer to the upper interior and/or between layers closer to the upper exterior).

The upper component **1110**, **1120** may form any desired portion or proportion of the footwear upper structure without departing from this invention. As some more specific examples, the upper component **1110**, **1120** may provide or overlap with less than 50%, and in some examples, less than 40%, less than 30%, less than 20%, or even less than 10% of a total surface area of the upper component **1210** (the term “total surface area” as used herein in this context means the complete area defined within the outermost perimeter of the upper component **1210**, including open space defined between individual filaments of the upper component **1210**). The second (e.g., fabric) upper component **1110**, **1120** may form any desired portion of the overall upper construction, including one or more of: an instep/tongue portion of the upper, a vamp portion of the upper, a lace-engaging portion of the upper, a foot-receiving opening of the upper, a collar of the upper, a rear heel engaging portion of the upper, etc.

FIGS. 7A-7C illustrate an example article of footwear **2000** including an upper **2002** having at least a portion formed as a multi-layered filament type upper blank **1000** of the types described above. As shown in these figures, the upper **2002** includes the combined upper component **1110**, **1120**/upper component **1210**, e.g., of the types described above in conjunction with FIGS. 6A-6E. The upper component **1110**, **1120** may be combined with a filament based upper component (e.g., like upper blank **1000**) in any desired manner, including the various manners described above in conjunction with FIGS. 4A-6E. As shown in these figures, upper component **1110**, **1120** of this example is one or more fabric elements that extend along a lace-engaging region and an instep region of the upper **2002** (e.g., forming a “tongue” like element and/or vamp portion of the upper **2002**), around the foot-insertion opening **2006** (e.g., to provide a comfort enhancing collar around the ankle), and down around a rear heel portion of the upper **2002** (e.g., to provide comfort at the heel).

Both the fabric upper component **1110**, **1120** and the filament based upper component **1210** of this illustrated example include openings and/or other structures for engaging a shoe lace **2008**. Alternatively, if desired, only one of fabric upper component **1110**, **1120** and the filament based upper component **1210** may include lace engaging openings or structures. As another option or alternative, if desired, the fabric upper component **1110**, **1120** may include lace engaging openings or structures in one or more areas of the upper **2002** and the filament based upper component **1210** may include lace engaging openings or structures in one or more other areas of the upper **2002**. If desired, when the filament based upper component **1210** engages the lace **2008**, the lace **2008** may extend through openings provided in the filament based upper component **1210** between individual filaments of a multi-layered upper structure.

The upper component **2002** may be engaged with a sole structure **2004** in any desired manner, including in conventional manners as are known and used in the footwear arts (e.g., adhesives, mechanical connectors, sewn seams, etc.). Any desired type of sole structure **2004** may be provided, including one or multi-part sole structures as are known and used in the footwear arts. In some examples of this invention, the sole structure **2004** will include a lightweight foam or fluid-filled bladder structure (optionally made from a hydrophobic, water-repelling, and/or non-water absorbing material(s)). The sole structure **2004** also may include track spikes, cleats, and/or other traction enhancing elements.

Footwear uppers in accordance with examples of this invention, including extruded filament based upper components of the types described above (including upper blank **1000**, upper component **1210**, and/or upper **2002**), can provide many desirable properties, particularly for athletic footwear constructions. The upper blank **1000**, upper component **1210**, and/or upper **2002** can be made very lightweight and/or breathable in view of the large number of openings between the various filaments in the layers, the thin structures of the filaments, and the relatively reduced volume of heavier materials present. As another potential advantage, as noted above, the filament(s) may be made from hydrophobic material(s) and/or material(s) that do not substantially absorb water. Thus, if the upper **2002** is exposed to wet conditions during use, the filament(s) may shed/repel the water, maintaining a lightweight condition. The ability to maintain a lightweight condition even when used in wet conditions can be further enhanced in upper structures in which any fabric based upper component(s) (e.g., component **1110**, **1120**) included in the upper **2002** also is/are formed (at least in part) from hydrophobic material(s) and/or material(s) that do not substantially absorb water (such as thermoplastic polyurethane materials and/or other thermoplastic materials and/or other hydrophobic and/or water repelling materials that may be provided as the fusible component, e.g., in the manners described in one or more of FIGS. 5D-5F) and/or if the sole structure **2004** is formed (at least in part) from hydrophobic material(s) and/or material(s) that do not substantially absorb water (such as thermoplastic polyurethane materials and/or other thermoplastic materials and/or other hydrophobic and/or water repelling materials). When the upper component **1210** is formed at least in part as a multi-layered filament construction, many different colors and color combinations can be used to provide a wide range of aesthetic and design opportunities, including production of a moiré effect, e.g., as shown in FIGS. 1 and 3A-3P.

Also, the extruded filament type structures and production methods allow seamless transitions between upper structural features that provide different functions and/or characteristics (e.g., seamless transitions between areas providing increased support and lock down with areas providing improved flexibility; seamless changes in breathability in different areas; etc., by changing the filament sizes, spacings, materials, etc.). Moreover, a relatively uniform overall upper thickness with the multi-layered filament construction can provide varying stretch and/or breathability characteristics in different areas of the upper with a seamless, one piece, unitary construction. As another potential advantage/feature, the upper blank **1000**/upper component **1210** can be formed by the multi-layer extrusion process without producing tensile stresses to the individual filaments of the upper construction.

In the specific examples described above, the filament based upper component forms an upper component (such as an upper blank) that substantially covers or surrounds the wearer's foot (from the rear heel area, through the midfoot area, and to a forward toe area). Other options are possible in which the filament based upper component (e.g., having any desired number of extruded layers, including from 1 to 20 layers, and in some examples, from 2 to 16 layers, 3 to 12 layers, 4-10 layers, or even 8 layers) form one or more discrete portions of an overall upper. FIGS. 8A and 8B illustrate an article of footwear **3000** having such an upper **3002** engaged with a sole structure **3004**. As shown in these figures, the upper **3002** of this example includes one or more fabric based upper components **3002a** and one or more

45

extruded filament based upper components **3002b** (which may include one or more layers of extruded filaments of the types described above). The extruded filament based upper components **3002b** of this example form a lateral side upper panel (FIG. 8A) and a separate medial side upper panel (FIG. 8B). These extruded filament based upper components **3002b** may be engaged with the other upper component(s) **3002a** in any desired manner, including any of the manners described above in conjunction with FIGS. 4A to 6E (including via adhesives and/or in an adhesive free manner).

As further shown in FIGS. 8A and 8B, in this illustrated example, both the fabric upper component **3002a** and the filament based upper component **3002b** include openings and/or other structures for engaging a shoe lace **3008**. Alternatively, if desired, only one of fabric upper component **3002a** and the filament based upper component **3002b** may include lace engaging openings or structures. As another option or alternative, if desired, the fabric upper component **3002a** may include lace engaging openings or structures in one or more areas of the upper **3002** and the filament based upper component **3002b** may include lace engaging openings or structures in one or more other areas of the upper **3002**. If desired, when the filament based upper component **3002b** engages the lace **3008**, the lace **3008** may extend through openings provided in the filament based upper component **3002b** between individual filaments of a multi-layered upper structure.

The upper component **3002a** may have any desired structure and/or may be made of any desired materials without departing from this invention, including conventional structures and/or materials as are known and used in the footwear arts and/or structures as described above in conjunction with FIGS. 5A-7C (including hydrophobic, water-repelling, and/or non-water absorbing material(s)). The sole structure **3004** may be any desired type of sole structure, including one or multi-part sole structures as are known and used in the footwear arts. In some examples of this invention, the sole structure **3004** will include a lightweight foam or fluid-filled bladder structure (optionally made from a hydrophobic, water-repelling, and/or non-water absorbing material(s)). The sole structure **3004** also may include track spikes, cleats, and/or other traction enhancing elements.

Even with only a portion of the upper **3002** formed by extruded filament based upper components **3002b**, enhanced breathability and/or lightweight characteristics can be realized. The extruded filament based upper components **3002b** can be placed at any locations where enhanced breathability may be required or desired. As to weight reduction, in the example structure of FIGS. 8A-8B, the use of the two extruded filament based upper components **3002b** reduces the overall weight of upper **3002** by about 24 grams as compared to a similar upper made fully with component **3002a** (i.e., with component **3002a** extending through the medial and lateral midfoot regions). In some examples of this invention, the filament based upper component(s) **3002b** may form from 15% to 100% of the overall upper surface area, and in some examples, from 25% to 100%, 35% to 100%, 50% to 100%, or 60% to 100%, or even 75% to 100% of the overall upper surface area.

If desired, as generally shown in FIG. 9, when one or more of the upper layers (e.g., one or more of layers **300**, **350**, **400**, **500**, **600**, **700**, **800**, and/or **900**) are formed by filament extrusion/solid deposition modeling (e.g., forming upper blank **1000**), a footwear strobil component (or sock liner) **1020** may be formed along with one or more of the filament layer(s) by the extrusion/solid deposition modeling technique (optionally, as part of a continuous path with one

46

or more of the filament layers). As shown in FIG. 9, the strobil component **1020** may be integrally formed with the upper component **1000**, e.g., along one or more edges of the upper component **1000** (e.g., at a lateral lower perimeter edge, a medial lower perimeter edge, a forefoot edge, one or more rear heel edges, etc.). The strobil component **1020** may be formed as filaments, e.g., having any of the characteristics, features, and/or options for the filaments described above (e.g., with a grid or matrix structure, in multiple layers, with intersecting/overlapping filament path segments, etc.). Once the overall upper/strobil combination **1040** is formed, the strobil **1020** can be folded along its integral edge with the upper component **1000** and attached at its free perimeter edge **1020P** with the bottom perimeter edge **1000P** of the upper component **1000** (e.g., by adhesives, by fusing, etc.). The strobil component **1020** could be formed as two or more parts, e.g., at two or more separated locations along the upper perimeter **1000P**.

III. CONCLUSION

The present invention is disclosed above and in the accompanying drawings with reference to a variety of embodiments and/or options. The purpose served by the disclosure, however, is to provide examples of various features and concepts related to the invention, not to limit the scope of the invention. One skilled in the relevant art will recognize that numerous variations and modifications may be made to the features of the invention described above without departing from the scope of the present invention, as defined by the appended claims.

For the avoidance of doubt, the present application includes at least the subject matter described in the following numbered Clauses:

Clause 1. An upper for an article of footwear, comprising: an upper component that includes:

- (a) a first layer comprising a first filament including first plural, non-intersecting, spaced apart path segments, wherein the first filament has a width dimension of less than 1 mm wide; and
- (b) a second layer comprising a second filament including second plural, non-intersecting, spaced apart path segments, wherein the second filament has a width dimension of less than 1 mm wide, and wherein the second layer is fused to the first layer at locations where the second layer contacts the first layer.

Clause 2. The upper according to Clause 1, wherein the first plural, non-intersecting, spaced apart path segments extend in a substantially medio-lateral direction of the upper and/or wherein the second plural, non-intersecting, spaced apart path segments extend in a substantially medio-lateral direction of the upper.

Clause 3. The upper according to Clause 2, wherein the first plural, non-intersecting, spaced apart path segments are located closer together in a forefoot region of the upper as compared to in a midfoot region and/or a heel region of the upper.

Clause 4. The upper according to Clause 2 or 3, wherein the second plural, non-intersecting, spaced apart path segments are located closer together in a forefoot region of the upper as compared to in a midfoot region and/or a heel region of the upper.

Clause 5. The upper according to Clause 1, wherein the first plural, non-intersecting, spaced apart path segments extend in a substantially medio-lateral direction of the upper,

and wherein the second plural, non-intersecting, spaced apart path segments extend in a substantially anterior-posterior direction of the upper.

Clause 6. The upper according to Clause 5, wherein the first plural, non-intersecting, spaced apart path segments are located closer together in a forefoot region of the upper as compared to in a midfoot region and/or a heel region of the upper.

Clause 7. The upper according to Clause 5 or 6, wherein the second plural, non-intersecting, spaced apart path segments are located closer together in a forefoot region of the upper as compared to in a midfoot region and/or a heel region of the upper.

Clause 8. The upper according to Clause 1, wherein the first plural, non-intersecting, spaced apart path segments extend in a serpentine configuration including at least two peaks and at least two valleys, and/or wherein the second plural, non-intersecting, spaced apart path segments extend in a serpentine configuration including at least two peaks and at least two valleys.

Clause 9. The upper according to Clause 1, wherein the first plural, non-intersecting, spaced apart path segments extend in a substantially anterior-posterior direction of the upper and include a serpentine configuration having at least two peaks and at least two valleys.

Clause 10. The upper according to Clause 8 or 9, wherein the serpentine configuration is formed at least at a medial midfoot portion of the upper component.

Clause 11. The upper according to Clause 8 or 9, wherein the serpentine configuration is formed at least at a lateral midfoot portion of the upper component.

Clause 12. The upper according to any one of Clauses 1 to 11, wherein the upper component includes a unitary construction in which the first layer and the second layer are fixed together only in a non-adhesive fused manner, and wherein the unitary construction of the upper component includes a medial heel portion, a medial midfoot portion, a lateral heel portion, and lateral midfoot portion, and a forefoot portion extending between the medial midfoot portion and the lateral midfoot portion.

Clause 13. The upper according to any one of Clauses 1 to 11, wherein the upper component includes a medial side upper panel.

Clause 14. The upper according to any one of Clauses 1 to 11, wherein the upper component includes a lateral side upper panel.

Clause 15. The upper according to any one of Clauses 1 to 14, wherein the upper component is engaged with another upper component.

Clause 16. The upper according to any one of Clauses 1 to 15, wherein the first plural, non-intersecting, spaced apart path segments include at least 5 first non-intersecting path segments, wherein each non-intersecting path segment of the at least 5 first non-intersecting path segments is spaced apart from each directly adjacent first non-intersecting path segments by less than 5 mm over a length dimension of at least 25 mm.

Clause 17. The upper according to any one of Clauses 1 to 16, wherein the second plural, non-intersecting, spaced apart path segments include at least 5 second non-intersecting path segments, wherein each non-intersecting path segment of the at least 5 second non-intersecting path segments is spaced apart from each directly adjacent second non-intersecting path segments by less than 5 mm over a length dimension of at least 25 mm.

Clause 18. The upper according to any one of Clauses 1 to 15, wherein the first plural, non-intersecting, spaced apart

path segments include at least 5 first non-intersecting path segments, wherein each non-intersecting path segment of the at least 5 first non-intersecting path segments is spaced apart from each directly adjacent first non-intersecting path segments by less than 5 mm over a length dimension of at least 50 mm.

Clause 19. The upper according to any one of Clauses 1 to 15 or Clause 18, wherein the second plural, non-intersecting, spaced apart path segments include at least 5 second non-intersecting path segments, wherein each non-intersecting path segment of the at least 5 second non-intersecting path segments is spaced apart from each directly adjacent second non-intersecting path segments by less than 5 mm over a length dimension of at least 50 mm.

Clause 20. The upper according to any one of Clauses 1 to 19, wherein the upper component further includes a third layer comprising a third filament including third plural, non-intersecting, spaced apart path segments, wherein the third filament has a width dimension of less than 1 mm wide, and wherein the third layer is fused to either or both of the first layer and to the second layer at locations where the third layer contacts either or both of the first layer and the second layer, respectively.

Clause 21. The upper according to Clause 20, wherein the upper component further includes a fourth layer comprising a fourth filament including fourth plural, non-intersecting, spaced apart path segments, wherein the extruded filament has a width dimension of less than 1 mm wide, and wherein the fourth layer is fused to any one of or any combination of the first layer, the second layer, and the third layer at locations where the fourth layer contacts any one of or any combination of the first layer, the second layer, and the third layer, respectively.

Clause 22. The upper according to Clause 21, wherein the upper component further includes a fifth layer comprising a fifth filament including fifth plural, non-intersecting, spaced apart path segments, wherein the fifth filament has a width dimension of less than 1 mm wide, and wherein the fifth layer is fused to any one of or any combination of the first layer, the second layer, the third layer, and the fourth layer at locations where the fifth layer contacts any one of or any combination of the first layer, the second layer, the third layer, and the fourth layer, respectively.

Clause 23. The upper according to Clause 22, wherein the upper component further includes a sixth layer comprising a sixth filament including sixth plural, non-intersecting, spaced apart path segments, wherein the sixth filament has a width dimension of less than 1 mm wide, and wherein the sixth layer is fused to any one of or any combination of the first layer, the second layer, the third layer, the fourth layer, and the fifth layer at locations where the sixth layer contacts any one of or any combination of the first layer, the second layer, the third layer, the fourth layer, and the fifth layer, respectively.

Clause 24. The upper according to Clause 23, wherein the upper component further includes a seventh layer comprising a seventh filament including seventh plural, non-intersecting, spaced apart path segments, wherein the seventh filament has a width dimension of less than 1 mm wide, and wherein the seventh layer is fused to any one of or any combination of the first layer, the second layer, the third layer, the fourth layer, the fifth layer, and the sixth layer at locations where the seventh layer contacts any one of or any combination of the first layer, the second layer, the third layer, the fourth layer, the fifth layer, and the sixth layer, respectively.

Clause 25. The upper according to Clause 24, wherein the upper component further includes an eighth layer comprising an eighth filament including eighth plural, non-intersecting, spaced apart path segments, wherein the eighth filament has a width dimension of less than 1 mm wide, and wherein the eighth layer is fused to any one of or any combination of the first layer, the second layer, the third layer, the fourth layer, the fifth layer, the sixth layer, and the seventh layer at locations where the eighth layer contacts any one of or any combination of the first layer, the second layer, the third layer, the fourth layer, the fifth layer, the sixth layer, and the seventh layer, respectively.

Clause 26. The upper according to any one of Clauses 1 to 25, wherein the first filament comprises a first material, wherein the second filament comprises a second material, and wherein the first material is the same as the second material.

Clause 27. The upper according to any one of Clauses 1 to 25, wherein the first filament comprises a first material, wherein the second filament comprises a second material, and wherein the first material is different from the second material.

Clause 28. The upper according to Clause 26 or 27, wherein the first filament is a first color and the second filament is a second color that differs from the first color.

Clause 29. The upper according to Clause 26 or 27, wherein at least one of the first filament or the second filament comprises a translucent or transparent material.

Clause 30. The upper according to any one of Clauses 1 to 25, wherein the filament in each layer of the upper component is made from the same material.

Clause 31. The upper according to any one of Clauses 1 to 25, wherein the filament in each layer of the upper component comprises a thermoplastic polyurethane material or other thermoplastic material.

Clause 32. The upper according to any one of Clauses 1 to 25, wherein the filament in each layer of the upper component comprises a material that does not substantially absorb water.

Clause 33. The upper according to any one of Clauses 1 to 25, wherein the filament in each layer of the upper component comprises a hydrophobic material.

Clause 34. The upper according to any one of Clauses 1 to 33, wherein two or more layers of the upper component collectively produce a moiré effect in at least a portion of the upper.

Clause 35. The upper according to any one of Clauses 1 to 4 or 8 to 34, wherein a plurality of the second plural, non-intersecting, spaced apart path segments of the second filament extend parallel to and partially overlap with a plurality of the first plural, non-intersecting, spaced apart path segments of the first filament over a path segment length of at least 25 mm.

Clause 36. The upper according to any one of Clauses 1 to 34, wherein a plurality of the second plural, non-intersecting, spaced apart path segments of the second filament intersect and form an angle with a plurality of the first plural, non-intersecting, spaced apart path segments of the first filament.

Clause 37. The upper according to Clause 36, wherein the angle is within a range of 65° to 90°.

Clause 38. The upper according to Clause 36 or 37, wherein the plurality of the second plural, non-intersecting, spaced apart path segments of the second filament and the plurality of the first plural, non-intersecting, spaced apart path segments of the first filament cooperate to form one or more diamond shapes in a forefoot region of the upper.

Clause 39. The upper according to Clause 38, wherein a long axis of each of the diamond shapes extends substantially in an anterior-posterior direction of the upper.

Clause 40. The upper according to any one of Clauses 1 to 39, wherein the first filament has a first thickness at a first region of the upper, wherein the first filament has a second thickness at a second region of the upper, and wherein the first thickness differs from the second thickness.

Clause 41. The upper according to any one of Clauses 1 to 39, wherein the first filament has a first diameter at a first region of the upper, wherein the first filament has a second diameter at a second region of the upper, and wherein the first diameter differs from the second diameter.

Clause 42. The upper according to any one of Clauses 1 to 41, wherein the first filament has a first thickness at a first region of the upper, wherein the second filament has a second thickness at a second region of the upper, and wherein the first thickness differs from the second thickness.

Clause 43. The upper according to any one of Clauses 1 to 41, wherein the first filament has a first diameter at a first region of the upper, wherein the second filament has a second diameter at a second region of the upper, and wherein the first diameter differs from the second diameter.

Clause 44. The upper according to any one of Clauses 1 to 43, wherein the first layer includes one or more path segments that extend along and form a perimeter of the upper.

Clause 45. An article of footwear, comprising:
an upper according to any one of Clauses 1 to 44; and
a sole structure engaged with the upper.

Clause 46. An upper blank for an article of footwear, comprising:

a first layer comprising a first filament formed as a first path, wherein the first path of the first filament forms a first lateral rear heel portion, a first lateral midfoot portion, a first forefoot portion, a first medial midfoot portion, and a first medial rear heel portion, and wherein the first filament has a width dimension of less than 1 mm wide; and

a second layer comprising a second filament formed as a second path, wherein the second path of the second filament forms a second lateral rear heel portion, a second lateral midfoot portion, a second forefoot portion, a second medial midfoot portion, and a second medial rear heel portion, wherein the second filament has a width dimension of less than 1 mm wide, and wherein the second layer is fused to the first layer at locations where the second layer contacts the first layer.

Clause 47. The upper blank according to Clause 46, wherein the second path overlaps the first path over at least 90% of an overall length of the second path and/or wherein the second path overlaps the first path over at least 90% of an overall length of the first path.

Clause 48. The upper blank according to Clause 46 or 47, wherein the first path extends in a substantially medio-lateral direction of the upper blank and/or wherein the second path extends in a substantially medio-lateral direction of the upper blank.

Clause 49. The upper blank according to any one of Clauses 46 to 48, further comprising:

a third layer comprising a third filament formed as a third path, wherein the third path of the third filament forms a third lateral rear heel portion, a third lateral midfoot portion, a third forefoot portion, a third medial midfoot portion, and a third medial rear heel portion, wherein the third filament has a width dimension of less than 1 mm wide, and wherein the third layer is fused to either

51

or both of the first layer and the second layer at locations where the third layer contacts either or both of the first layer and the second layer, respectively.

Clause 50. The upper blank according to Clause 49, wherein the third path extends in a substantially anterior-posterior direction of the upper blank.

Clause 51. The upper blank according to Clause 49 or 50, wherein the third path defines a first inner edge, a first outer edge, a second inner edge, and a second outer edge, wherein an instep opening is defined between the first inner edge and the second inner edge, wherein the third path includes: (a) first plural, non-intersecting, spaced apart path segments located between the first inner edge and the first outer edge, wherein the first plural, non-intersecting, spaced apart path segments of the third path are linear and/or curved without defining plural peaks and valleys and/or (b) second plural, non-intersecting, spaced apart path segments located between the second inner edge and the second outer edge, wherein a plurality of the second plural, non-intersecting, spaced apart path segments of the third path located between the second inner edge and the second outer edge each extends in a serpentine configuration and has at least two peaks and at least two valleys.

Clause 52. The upper blank according to any one of Clauses 49 to 51, wherein: (a) the third path overlaps the first path over less than 50% of an overall length of the third path, (b) the third path overlaps the second path over less than 50% of an overall length of the third path, (c) the third path overlaps the first path over less than 50% of an overall length of the first path, and/or (d) the third path overlaps the second path over less than 50% of an overall length of the second path.

Clause 53. The upper blank according to any one of Clauses 49 to 52, further comprising:

a fourth layer comprising a fourth filament formed as a fourth path, wherein the fourth path of the fourth filament forms a fourth lateral rear heel portion, a fourth lateral midfoot portion, a fourth forefoot portion, a fourth medial midfoot portion, and a fourth medial rear heel portion, wherein the fourth filament has a width dimension of less than 1 mm wide, and wherein the fourth layer is fused to any one of or any combination of the first layer, the second layer, and the third layer at locations where the fourth layer contacts any one of or any combination of the first layer, the second layer, and the third layer, respectively.

Clause 54. The upper blank according to Clause 53, wherein the fourth path overlaps the third path over at least 90% of an overall length of the fourth path and/or wherein the fourth path overlaps the third path over at least 90% of an overall length of the third path.

Clause 55. The upper blank according to Clause 53 or 54, wherein the fourth path defines a first inner edge, a first outer edge, a second inner edge, and a second outer edge, wherein an instep opening is defined between the first inner edge and the second inner edge, wherein the fourth path includes: (a) first plural, non-intersecting, spaced apart path segments located between the first inner edge and the first outer edge, wherein the first plural, non-intersecting, spaced apart path segments of the fourth path are linear and/or curved without defining plural peaks and valleys and/or (b) second plural, non-intersecting, spaced apart path segments located between the second inner edge and the second outer edge, wherein a plurality of the second plural, non-intersecting, spaced apart path segments of the fourth path located between the second inner edge and the second outer edge

52

each extends in a serpentine configuration and has at least two peaks and at least two valleys.

Clause 56. The upper blank according to any one of Clauses 53 to 55, wherein: (a) the fourth path overlaps the first path over less than 50% of an overall length of the fourth path, (b) the fourth path overlaps the second path over less than 50% of an overall length of the fourth path, (c) the fourth path overlaps the first path over less than 50% of an overall length of the first path, and/or (d) the fourth path overlaps the second path over less than 50% of an overall length of the second path.

Clause 57. The upper blank according to any one of Clauses 53 to 56, further comprising:

a fifth layer comprising a fifth filament formed as a fifth path, wherein the fifth path of the fifth filament forms a fifth lateral rear heel portion, a fifth lateral midfoot portion, a fifth forefoot portion, a fifth medial midfoot portion, and a fifth medial rear heel portion, wherein the fifth filament has a width dimension of less than 1 mm wide, and wherein the fifth layer is fused to any one of or any combination of the first layer, the second layer, the third layer, and the fourth layer at locations where the fifth layer contacts any one of or any combination of the first layer, the second layer, the third layer, and the fourth layer, respectively.

Clause 58. The upper blank according to Clause 57, wherein the fifth path overlaps the fourth path over at least 90% of an overall length of the fifth path and/or wherein the fifth path overlaps the fourth path over at least 90% of an overall length of the fourth path.

Clause 59. The upper blank according to Clause 57 or 58, wherein the fifth path overlaps the third path over at least 90% of an overall length of the fifth path and/or wherein the fifth path overlaps the third path over at least 90% of an overall length of the third path.

Clause 60. The upper blank according to any one of Clauses 57 to 59, wherein the fifth path defines a first inner edge, a first outer edge, a second inner edge, and a second outer edge, wherein an instep opening is defined between the first inner edge and the second inner edge, wherein the fifth path includes: (a) first plural, non-intersecting, spaced apart path segments located between the first inner edge and the first outer edge, wherein the first plural, non-intersecting, spaced apart path segments of the fifth path are linear and/or curved without defining plural peaks and valleys and/or (b) second plural, non-intersecting, spaced apart path segments located between the second inner edge and the second outer edge, wherein a plurality of the second plural, non-intersecting, spaced apart path segments of the fifth path located between the second inner edge and the second outer edge each extends in a serpentine configuration and has at least two peaks and at least two valleys.

Clause 61. The upper blank according to any one of Clauses 57 to 60, wherein: (a) the fifth path overlaps the first path over less than 50% of an overall length of the fifth path, (b) the fifth path overlaps the second path over less than 50% of an overall length of the fifth path, (c) the fifth path overlaps the first path over less than 50% of an overall length of the first path, and/or (d) the fifth path overlaps the second path over less than 50% of an overall length of the second path.

Clause 62. The upper blank according to any one of Clauses 57 to 61, further comprising:

a sixth layer comprising a sixth filament formed as a sixth path, wherein the sixth path of the sixth filament forms a sixth lateral rear heel portion, a sixth lateral midfoot portion, a sixth forefoot portion, a sixth medial midfoot

53

portion, and a sixth medial rear heel portion, wherein the sixth filament has a width dimension of less than 1 mm wide, and wherein the sixth layer is fused to any one of or any combination of the first layer, the second layer, the third layer, the fourth layer, and the fifth layer at locations where the sixth layer contacts any one of or any combination of the first layer, the second layer, the third layer, the fourth layer, and the fifth layer, respectively.

Clause 63. The upper blank according to Clause 62, wherein the sixth path defines a first inner edge, a first outer edge, a second inner edge, and a second outer edge, wherein an instep opening is defined between the first inner edge and the second inner edge, wherein the sixth path includes: (a) first plural, non-intersecting, spaced apart path segments located between the first inner edge and the first outer edge, wherein a plurality of the first plural, non-intersecting, spaced apart path segments of the sixth path located between the first inner edge and the first outer edge each extends in a serpentine configuration and has at least two peaks and at least two valleys and/or (b) second plural, non-intersecting, spaced apart path segments located between the second inner edge and the second outer edge, wherein the second plural, non-intersecting, spaced apart path segments of the sixth path are linear and/or curved without defining plural peaks and valleys.

Clause 64. The upper blank according to Clause 62 or 63, wherein: (a) the sixth path overlaps the first path over less than 50% of an overall length of the sixth path, (b) the sixth path overlaps the second path over less than 50% of an overall length of the sixth path, (c) the sixth path overlaps the first path over less than 50% of an overall length of the first path, and/or (d) the fifth path overlaps the second path over less than 50% of an overall length of the second path.

Clause 65. The upper blank according to any one of Clauses 62 to 64, further comprising:

a seventh layer comprising a seventh filament formed as a seventh path, wherein the seventh path of the seventh filament forms a seventh lateral rear heel portion, a seventh lateral midfoot portion, a seventh forefoot portion, a seventh medial midfoot portion, and a seventh medial rear heel portion, wherein the seventh filament has a width dimension of less than 1 mm wide, and wherein the seventh layer is fused to any one of or any combination of the first layer, the second layer, the third layer, the fourth layer, the fifth layer, and the sixth layer at locations where the seventh layer contacts any one of or any combination of the first layer, the second layer, the third layer, the fourth layer, the fifth layer, and the sixth layer, respectively.

Clause 66. The upper blank according to Clause 65, wherein the seventh path overlaps the sixth path over at least 90% of an overall length of the seventh path and/or wherein the seventh path overlaps the sixth path over at least 90% of an overall length of the sixth path.

Clause 67. The upper blank according to Clause 65 or 66, wherein the seventh path defines a first inner edge, a first outer edge, a second inner edge, and a second outer edge, wherein an instep opening is defined between the first inner edge and the second inner edge, wherein the seventh path includes: (a) first plural, non-intersecting, spaced apart path segments located between the first inner edge and the first outer edge, wherein a plurality of the first plural, non-intersecting, spaced apart path segments of the seventh path located between the first inner edge and the first outer edge each extends in a serpentine configuration and has at least two peaks and at least two valleys and/or (b) second plural,

54

non-intersecting, spaced apart path segments located between the second inner edge and the second outer edge, wherein the second plural, non-intersecting, spaced apart path segments of the seventh path are linear and/or curved without defining plural peaks and valleys.

Clause 68. The upper blank according to any one of Clauses 65 to 67, wherein: (a) the seventh path overlaps the first path over less than 50% of an overall length of the seventh path, (b) the seventh path overlaps the second path over less than 50% of an overall length of the seventh path, (c) the seventh path overlaps the first path over less than 50% of an overall length of the first path, and/or (d) the seventh path overlaps the second path over less than 50% of an overall length of the second path.

Clause 69. The upper blank according to any one of Clauses 65 to 68, further comprising:

an eighth layer comprising an eighth filament formed as an eighth path, wherein the eighth path of the eighth filament forms an eighth lateral rear heel portion, an eighth lateral midfoot portion, an eighth forefoot portion, an eighth medial midfoot portion, and an eighth medial rear heel portion, wherein the eighth filament has a width dimension of less than 1 mm wide, and wherein the eighth layer is fused to any one of or any combination of the first layer, the second layer, the third layer, the fourth layer, the fifth layer, the sixth layer, and the seventh layer at locations where the eighth layer contacts any one of or any combination of the first layer, the second layer, the third layer, the fourth layer, the fifth layer, the sixth layer, and the seventh layer, respectively.

Clause 70. The upper blank according to Clause 69, wherein the eighth path overlaps the seventh path over at least 90% of an overall length of the eighth path and/or wherein the eighth path overlaps the seventh path over at least 90% of an overall length of the seventh path.

Clause 71. The upper blank according to Clause 69 or Clause 70, wherein the eighth path overlaps the sixth path over at least 90% of an overall length of the eighth path and/or wherein the eighth path overlaps the sixth path over at least 90% of an overall length of the sixth path.

Clause 72. The upper blank according to any one of Clauses 69 to 71, wherein the eighth path defines a first inner edge, a first outer edge, a second inner edge, and a second outer edge, wherein an instep opening is defined between the first inner edge and the second inner edge, wherein the eighth path includes: (a) first plural, non-intersecting, spaced apart path segments located between the first inner edge and the first outer edge, wherein a plurality of the first plural, non-intersecting, spaced apart path segments of the eighth path located between the first inner edge and the first outer edge each extends in a serpentine configuration and has at least two peaks and at least two valleys and/or (b) second plural, non-intersecting, spaced apart path segments located between the second inner edge and the second outer edge, wherein the second plural, non-intersecting, spaced apart path segments of the eighth path are linear and/or curved without defining plural peaks and valleys.

Clause 73. The upper blank according to any one of Clauses 69 to 72, wherein: (a) the eighth path overlaps the first path over less than 50% of an overall length of the eighth path, (b) the eighth path overlaps the second path over less than 50% of an overall length of the eighth path, (c) the eighth path overlaps the first path over less than 50% of an overall length of the first path, and/or (d) the eighth path overlaps the second path over less than 50% of an overall length of the second path.

55

Clause 74. The upper blank according to Clause 46, wherein: (a) the first path extends in a substantially medio-lateral direction of the upper blank and the second path extends in a substantially anterior-posterior direction of the upper blank or (b) the first path extends in a substantially anterior-posterior direction of the upper blank and the second path extends in a substantially medio-lateral direction of the upper blank.

Clause 75. The upper blank according to Clause 46, wherein the upper blank includes a first inner edge, a first outer edge, a second inner edge, and a second outer edge, wherein an instep opening is defined between the first inner edge and the second inner edge, wherein the first path includes: (a) first plural, non-intersecting, spaced apart path segments located between the first inner edge and the first outer edge, wherein the first plural, non-intersecting, spaced apart path segments of the first path are linear and/or curved without defining plural peaks and valleys and/or (b) second plural, non-intersecting, spaced apart path segments located between the second inner edge and the second outer edge, wherein a plurality of the second plural, non-intersecting, spaced apart path segments of the first path located between the second inner edge and the second outer edge each extends in a serpentine configuration and has at least two peaks and at least two valleys.

Clause 76. The upper blank according to Clause 75, wherein the second path includes: (a) first plural, non-intersecting, spaced apart path segments located between the first inner edge and the first outer edge, wherein a plurality of the first plural, non-intersecting, spaced apart path segments of the second path located between the first inner edge and the first outer edge each extends in a serpentine configuration and has at least two peaks and at least two valleys and/or (b) second plural, non-intersecting, spaced apart path segments located between the second inner edge and the second outer edge, wherein the second plural, non-intersecting, spaced apart path segments of the second path are linear and/or curved without defining plural peaks and valleys.

Clause 77. The upper blank according to Clause 46, wherein the upper blank includes a first inner edge, a first outer edge, a second inner edge, and a second outer edge, wherein an instep opening is defined between the first inner edge and the second inner edge, wherein the first path extends in a substantially medio-lateral direction of the upper blank, and wherein the second path includes: (a) first plural, non-intersecting, spaced apart path segments located between the first inner edge and the first outer edge, wherein a plurality of the first plural, non-intersecting, spaced apart path segments of the second path located between the first inner edge and the first outer edge each extends in a serpentine configuration and has at least two peaks and at least two valleys and/or (b) second plural, non-intersecting, spaced apart path segments located between the second inner edge and the second outer edge, wherein the second plural, non-intersecting, spaced apart path segments of the second path are linear and/or curved without defining plural peaks and valleys.

Clause 78. The upper blank according to any one of Clauses 46 to 77, wherein the first path of the first filament includes at least 5 first non-intersecting path segments, wherein each non-intersecting path segment of the at least 5 first non-intersecting path segments is spaced apart from each directly adjacent first non-intersecting path segments by less than 5 mm over a length dimension of at least 25 mm.

Clause 79. The upper blank according to any one of Clauses 46 to 78, wherein the second path of the second filament includes at least 5 second non-intersecting path

56

segments, wherein each non-intersecting path segment of the at least 5 second non-intersecting path segments is spaced apart from each directly adjacent second non-intersecting path segments by less than 5 mm over a length dimension of at least 25 mm.

Clause 80. The upper blank according to any one of Clauses 46 to 77, wherein the first path of the first filament includes at least 5 first non-intersecting path segments, wherein each non-intersecting path segment of the at least 5 first non-intersecting path segments is spaced apart from each directly adjacent first non-intersecting path segments by less than 5 mm over a length dimension of at least 50 mm.

Clause 81. The upper according to any one of Clauses 46 to 77 or Clause 80, wherein the second path of the second filament includes at least 5 second non-intersecting path segments, wherein each non-intersecting path segment of the at least 5 second non-intersecting path segments is spaced apart from each directly adjacent second non-intersecting path segments by less than 5 mm over a length dimension of at least 50 mm.

Clause 82. An upper blank for an article of footwear, comprising:

- a first layer comprising a first filament formed as a first continuous path, wherein the first continuous path of the first filament forms a first lateral rear heel portion, a first lateral midfoot portion, a first forefoot portion, a first medial midfoot portion, and a first medial rear heel portion, and wherein the first filament has a width dimension of less than 1 mm wide; and

- a second layer comprising a second filament formed as a second continuous path, wherein the second continuous path of the second filament forms a second lateral rear heel portion, a second lateral midfoot portion, a second forefoot portion, a second medial midfoot portion, and a second medial rear heel portion, wherein the second filament has a width dimension of less than 1 mm wide, and wherein the second layer is fused to the first layer at locations where the second layer contacts the first layer.

Clause 83. The upper blank according to Clause 82, wherein the second continuous path overlaps the first continuous path over at least 90% of an overall length of the second continuous path and/or wherein the second continuous path overlaps the first continuous path over at least 90% of an overall length of the first continuous path.

Clause 84. The upper blank according to Clause 82 or 83, wherein the first continuous path extends in a substantially medio-lateral direction of the upper blank and/or wherein the second continuous path extends in a substantially medio-lateral direction of the upper blank.

Clause 85. The upper blank according to any one of Clauses 82 to 84, further comprising:

- a third layer comprising a third filament formed as a third continuous path, wherein the third continuous path of the third filament forms a third lateral rear heel portion, a third lateral midfoot portion, a third forefoot portion, a third medial midfoot portion, and a third medial rear heel portion, wherein the third filament has a width dimension of less than 1 mm wide, and wherein the third layer is fused to either or both of the first layer and the second layer at locations where the third layer contacts either or both of the first layer and the second layer, respectively.

Clause 86. The upper blank according to Clause 85, wherein the third continuous path extends in a substantially anterior-posterior direction of the upper blank.

57

Clause 87. The upper blank according to Clause 85 or 86, wherein the third continuous path defines a first inner edge, a first outer edge, a second inner edge, and a second outer edge, wherein an instep opening is defined between the first inner edge and the second inner edge, wherein the third continuous path includes: (a) first plural, non-intersecting, spaced apart path segments located between the first inner edge and the first outer edge, wherein the first plural, non-intersecting, spaced apart path segments of the third continuous path are linear and/or curved without defining plural peaks and valleys and/or (b) second plural, non-intersecting, spaced apart path segments located between the second inner edge and the second outer edge, wherein a plurality of the second plural, non-intersecting, spaced apart path segments of the third continuous path located between the second inner edge and the second outer edge each extends in a serpentine configuration and has at least two peaks and at least two valleys.

Clause 88. The upper blank according to any one of Clauses 85 to 87, wherein: (a) the third continuous path overlaps the first continuous path over less than 50% of an overall length of the third continuous path, (b) the third continuous path overlaps the second continuous path over less than 50% of an overall length of the third continuous path, (c) the third continuous path overlaps the first continuous path over less than 50% of an overall length of the first continuous path, and/or (d) the third continuous path overlaps the second continuous path over less than 50% of an overall length of the second continuous path.

Clause 89. The upper blank according to any one of Clauses 85 to 88, further comprising:

a fourth layer comprising a fourth filament formed as a fourth continuous path, wherein the fourth continuous path of the fourth filament forms a fourth lateral rear heel portion, a fourth lateral midfoot portion, a fourth forefoot portion, a fourth medial midfoot portion, and a fourth medial rear heel portion, wherein the fourth filament has a width dimension of less than 1 mm wide, and wherein the fourth layer is fused to any one of or any combination of the first layer, the second layer, and the third layer at locations where the fourth layer contacts any one of or any combination of the first layer, the second layer, and the third layer, respectively.

Clause 90. The upper blank according to Clause 89, wherein the fourth continuous path overlaps the third continuous path over at least 90% of an overall length of the fourth continuous path and/or wherein the fourth continuous path overlaps the third continuous path over at least 90% of an overall length of the third continuous path.

Clause 91. The upper blank according to Clause 89 or 90, wherein the fourth continuous path defines a first inner edge, a first outer edge, a second inner edge, and a second outer edge, wherein an instep opening is defined between the first inner edge and the second inner edge, wherein the fourth continuous path includes: (a) first plural, non-intersecting, spaced apart path segments located between the first inner edge and the first outer edge, wherein the first plural, non-intersecting, spaced apart path segments of the fourth continuous path are linear and/or curved without defining plural peaks and valleys and/or (b) second plural, non-intersecting, spaced apart path segments located between the second inner edge and the second outer edge, wherein a plurality of the second plural, non-intersecting, spaced apart path segments of the fourth continuous path located between the second inner edge and the second outer edge each extends in a serpentine configuration and has at least two peaks and at least two valleys.

58

Clause 92. The upper blank according to any one of Clauses 89 to 91, wherein: (a) the fourth continuous path overlaps the first continuous path over less than 50% of an overall length of the fourth continuous path, (b) the fourth continuous path overlaps the second continuous path over less than 50% of an overall length of the fourth continuous path, (c) the fourth continuous path overlaps the first continuous path over less than 50% of an overall length of the first continuous path, and/or (d) the fourth continuous path overlaps the second continuous path over less than 50% of an overall length of the second continuous path.

Clause 93. The upper blank according to any one of Clauses 89 to 92, further comprising:

a fifth layer comprising a fifth filament formed as a fifth continuous path, wherein the fifth continuous path of the fifth filament forms a fifth lateral rear heel portion, a fifth lateral midfoot portion, a fifth forefoot portion, a fifth medial midfoot portion, and a fifth medial rear heel portion, wherein the fifth filament has a width dimension of less than 1 mm wide, and wherein the fifth layer is fused to any one of or any combination of the first layer, the second layer, the third layer, and the fourth layer at locations where the fifth layer contacts any one of or any combination of the first layer, the second layer, the third layer, and the fourth layer, respectively.

Clause 94. The upper blank according to Clause 93, wherein the fifth continuous path overlaps the fourth continuous path over at least 90% of an overall length of the fifth continuous path and/or wherein the fifth continuous path overlaps the fourth continuous path over at least 90% of an overall length of the fourth continuous path.

Clause 95. The upper blank according to Clause 93 or 94, wherein the fifth continuous path overlaps the third continuous path over at least 90% of an overall length of the fifth continuous path and/or wherein the fifth continuous path overlaps the third continuous path over at least 90% of an overall length of the third continuous path.

Clause 96. The upper blank according to any one of Clauses 93 to 95, wherein the fifth continuous path defines a first inner edge, a first outer edge, a second inner edge, and a second outer edge, wherein an instep opening is defined between the first inner edge and the second inner edge, wherein the fifth continuous path includes: (a) first plural, non-intersecting, spaced apart path segments located between the first inner edge and the first outer edge, wherein the first plural, non-intersecting, spaced apart path segments of the fifth continuous path are linear and/or curved without defining plural peaks and valleys and/or (b) second plural, non-intersecting, spaced apart path segments located between the second inner edge and the second outer edge, wherein a plurality of the second plural, non-intersecting, spaced apart path segments of the fifth continuous path located between the second inner edge and the second outer edge each extends in a serpentine configuration and has at least two peaks and at least two valleys.

Clause 97. The upper blank according to any one of Clauses 93 to 96, wherein: (a) the fifth continuous path overlaps the first continuous path over less than 50% of an overall length of the fifth continuous path, (b) the fifth continuous path overlaps the second continuous path over less than 50% of an overall length of the fifth continuous path, (c) the fifth continuous path overlaps the first continuous path over less than 50% of an overall length of the first continuous path, and/or (d) the fifth continuous path overlaps the second continuous path over less than 50% of an overall length of the second continuous path.

Clause 98. The upper blank according to any one of Clauses 93 to 97, further comprising:

a sixth layer comprising a sixth filament formed as a sixth continuous path, wherein the sixth continuous path of the sixth filament forms a sixth lateral rear heel portion, a sixth lateral midfoot portion, a sixth forefoot portion, a sixth medial midfoot portion, and a sixth medial rear heel portion, wherein the sixth filament has a width dimension of less than 1 mm wide, and wherein the sixth layer is fused to any one of or any combination of the first layer, the second layer, the third layer, the fourth layer, and the fifth layer at locations where the sixth layer contacts any one of or any combination of the first layer, the second layer, the third layer, the fourth layer, and the fifth layer, respectively.

Clause 99. The upper blank according to Clause 98, wherein the sixth continuous path defines a first inner edge, a first outer edge, a second inner edge, and a second outer edge, wherein an instep opening is defined between the first inner edge and the second inner edge, wherein the sixth continuous path includes: (a) first plural, non-intersecting, spaced apart path segments located between the first inner edge and the first outer edge, wherein a plurality of the first plural, non-intersecting, spaced apart path segments of the sixth continuous path located between the first inner edge and the first outer edge each extends in a serpentine configuration and has at least two peaks and at least two valleys and/or (b) second plural, non-intersecting, spaced apart path segments located between the second inner edge and the second outer edge, wherein the second plural, non-intersecting, spaced apart path segments of the sixth continuous path are linear and/or curved without defining plural peaks and valleys.

Clause 100. The upper blank according to Clause 98 or 99, wherein: (a) the sixth continuous path overlaps the first continuous path over less than 50% of an overall length of the sixth continuous path, (b) the sixth continuous path overlaps the second continuous path over less than 50% of an overall length of the sixth continuous path, (c) the sixth continuous path overlaps the first continuous path over less than 50% of an overall length of the first continuous path, and/or (d) the fifth continuous path overlaps the second continuous path over less than 50% of an overall length of the second continuous path.

Clause 101. The upper blank according to any one of Clauses 98 to 100, further comprising:

a seventh layer comprising a seventh filament formed as a seventh continuous path, wherein the seventh continuous path of the seventh filament forms a seventh lateral rear heel portion, a seventh lateral midfoot portion, a seventh forefoot portion, a seventh medial midfoot portion, and a seventh medial rear heel portion, wherein the seventh filament has a width dimension of less than 1 mm wide, and wherein the seventh layer is fused to any one of or any combination of the first layer, the second layer, the third layer, the fourth layer, the fifth layer, and the sixth layer at locations where the seventh layer contacts any one of or any combination of the first layer, the second layer, the third layer, the fourth layer, the fifth layer, and the sixth layer, respectively.

Clause 102. The upper blank according to Clause 101, wherein the seventh continuous path overlaps the sixth continuous path over at least 90% of an overall length of the seventh continuous path and/or wherein the seventh continuous path overlaps the sixth continuous path over at least 90% of an overall length of the sixth continuous path.

Clause 103. The upper blank according to Clause 101 or 102, wherein the seventh continuous path defines a first inner edge, a first outer edge, a second inner edge, and a second outer edge, wherein an instep opening is defined between the first inner edge and the second inner edge, wherein the seventh continuous path includes: (a) first plural, non-intersecting, spaced apart path segments located between the first inner edge and the first outer edge, wherein a plurality of the first plural, non-intersecting, spaced apart path segments of the seventh continuous path located between the first inner edge and the first outer edge each extends in a serpentine configuration and has at least two peaks and at least two valleys and/or (b) second plural, non-intersecting, spaced apart path segments located between the second inner edge and the second outer edge, wherein the second plural, non-intersecting, spaced apart path segments of the seventh continuous path are linear and/or curved without defining plural peaks and valleys.

Clause 104. The upper blank according to any one of Clauses 101 to 103, wherein: (a) the seventh continuous path overlaps the first continuous path over less than 50% of an overall length of the seventh continuous path, (b) the seventh continuous path overlaps the second continuous path over less than 50% of an overall length of the seventh continuous path, (c) the seventh continuous path overlaps the first continuous path over less than 50% of an overall length of the first continuous path, and/or (d) the seventh continuous path overlaps the second continuous path over less than 50% of an overall length of the second continuous path.

Clause 105. The upper blank according to any one of Clauses 101 to 104, further comprising:

an eighth layer comprising an eighth filament formed as an eighth continuous path, wherein the eighth continuous path of the eighth filament forms an eighth lateral rear heel portion, an eighth lateral midfoot portion, an eighth forefoot portion, an eighth medial midfoot portion, and an eighth medial rear heel portion, wherein the eighth filament has a width dimension of less than 1 mm wide, and wherein the eighth layer is fused to any one of or any combination of the first layer, the second layer, the third layer, the fourth layer, the fifth layer, the sixth layer, and the seventh layer at locations where the eighth layer contacts any one of or any combination of the first layer, the second layer, the third layer, the fourth layer, the fifth layer, the sixth layer, and the seventh layer, respectively.

Clause 106. The upper blank according to Clause 105, wherein the eighth continuous path overlaps the seventh continuous path over at least 90% of an overall length of the eighth continuous path and/or wherein the eighth continuous path overlaps the seventh continuous path over at least 90% of an overall length of the seventh continuous path.

Clause 107. The upper blank according to Clause 105 or Clause 106, wherein the eighth continuous path overlaps the sixth continuous path over at least 90% of an overall length of the eighth continuous path and/or wherein the eighth continuous path overlaps the sixth continuous path over at least 90% of an overall length of the sixth continuous path.

Clause 108. The upper blank according to any one of Clauses 105 to 107, wherein the eighth continuous path defines a first inner edge, a first outer edge, a second inner edge, and a second outer edge, wherein an instep opening is defined between the first inner edge and the second inner edge, wherein the eighth continuous path includes: (a) first plural, non-intersecting, spaced apart path segments located between the first inner edge and the first outer edge, wherein a plurality of the first plural, non-intersecting, spaced apart

61

path segments of the eighth continuous path located between the first inner edge and the first outer edge each extends in a serpentine configuration and has at least two peaks and at least two valleys and/or (b) second plural, non-intersecting, spaced apart path segments located between the second inner edge and the second outer edge, wherein the second plural, non-intersecting, spaced apart path segments of the eighth continuous path are linear and/or curved without defining plural peaks and valleys.

Clause 109. The upper blank according to any one of Clauses 105 to 108, wherein: (a) the eighth continuous path overlaps the first continuous path over less than 50% of an overall length of the eighth continuous path, (b) the eighth continuous path overlaps the second continuous path over less than 50% of an overall length of the eighth continuous path, (c) the eighth continuous path overlaps the first continuous path over less than 50% of an overall length of the first continuous path, and/or (d) the eighth continuous path overlaps the second continuous path over less than 50% of an overall length of the second continuous path.

Clause 110. The upper blank according to Clause 82, wherein: (a) the first continuous path extends in a substantially medio-lateral direction of the upper blank and the second continuous path extends in a substantially anterior-posterior direction of the upper blank or (b) the first continuous path extends in a substantially anterior-posterior direction of the upper blank and the second continuous path extends in a substantially medio-lateral direction of the upper blank.

Clause 111. The upper blank according to Clause 82, wherein the upper blank includes a first inner edge, a first outer edge, a second inner edge, and a second outer edge, wherein an instep opening is defined between the first inner edge and the second inner edge, wherein the first continuous path includes: (a) first plural, non-intersecting, spaced apart path segments located between the first inner edge and the first outer edge, wherein the first plural, non-intersecting, spaced apart path segments of the first continuous path are linear and/or curved without defining plural peaks and valleys and/or (b) second plural, non-intersecting, spaced apart path segments located between the second inner edge and the second outer edge, wherein a plurality of the second plural, non-intersecting, spaced apart path segments of the first continuous path located between the second inner edge and the second outer edge each extends in a serpentine configuration and has at least two peaks and at least two valleys.

Clause 112. The upper blank according to Clause 111, wherein the second continuous path includes: (a) first plural, non-intersecting, spaced apart path segments located between the first inner edge and the first outer edge, wherein a plurality of the first plural, non-intersecting, spaced apart path segments of the second continuous path located between the first inner edge and the first outer edge each extends in a serpentine configuration and has at least two peaks and at least two valleys and/or (b) second plural, non-intersecting, spaced apart path segments located between the second inner edge and the second outer edge, wherein the second plural, non-intersecting, spaced apart path segments of the second continuous path are linear and/or curved without defining plural peaks and valleys.

Clause 113. The upper blank according to Clause 82, wherein the upper blank includes a first inner edge, a first outer edge, a second inner edge, and a second outer edge, wherein an instep opening is defined between the first inner edge and the second inner edge, wherein the first continuous path extends in a substantially medio-lateral direction of the

62

upper blank, and wherein the second continuous path includes: (a) first plural, non-intersecting, spaced apart path segments located between the first inner edge and the first outer edge, wherein a plurality of the first plural, non-intersecting, spaced apart path segments of the second continuous path located between the first inner edge and the first outer edge each extends in a serpentine configuration and has at least two peaks and at least two valleys and/or (b) second plural, non-intersecting, spaced apart path segments located between the second inner edge and the second outer edge, wherein the second plural, non-intersecting, spaced apart path segments of the second continuous path are linear and/or curved without defining plural peaks and valleys.

Clause 114. The upper blank according to any one of Clauses 82 to 113, wherein the first continuous path of the first filament includes at least 5 first non-intersecting path segments, wherein each non-intersecting path segment of the at least 5 first non-intersecting path segments is spaced apart from each directly adjacent first non-intersecting path segments by less than 5 mm over a length dimension of at least 25 mm.

Clause 115. The upper blank according to any one of Clauses 82 to 114, wherein the second continuous path of the second filament includes at least 5 second non-intersecting path segments, wherein each non-intersecting path segment of the at least 5 second non-intersecting path segments is spaced apart from each directly adjacent second non-intersecting path segments by less than 5 mm over a length dimension of at least 25 mm.

Clause 116. The upper blank according to any one of Clauses 82 to 113, wherein the first continuous path of the first filament includes at least 5 first non-intersecting path segments, wherein each non-intersecting path segment of the at least 5 first non-intersecting path segments is spaced apart from each directly adjacent first non-intersecting path segments by less than 5 mm over a length dimension of at least 50 mm.

Clause 117. The upper according to any one of Clauses 82 to 113 or Clause 116, wherein the second continuous path of the second filament includes at least 5 second non-intersecting path segments, wherein each non-intersecting path segment of the at least 5 second non-intersecting path segments is spaced apart from each directly adjacent second non-intersecting path segments by less than 5 mm over a length dimension of at least 50 mm.

Clause 118. The upper blank according to any one of Clauses 46 to 117, wherein the first filament comprises a first material, wherein the second filament comprises a second material, and wherein the first material is the same as the second material.

Clause 119. The upper blank according to any one of Clauses 46 to 117, wherein the first filament comprises a first material, wherein the second filament comprises a second material, and wherein the first material is different from the second material.

Clause 120. The upper blank according to Clause 118 or 119, wherein the first filament is a first color and the second filament is a second color that differs from the first color.

Clause 121. The upper blank according to Clause 118 or 119, wherein at least one of the first filament or the second filament comprises a translucent or transparent material.

Clause 122. The upper blank according to any one of Clauses 46 to 117, wherein the filament in each layer of the upper blank is made from the same material.

Clause 123. The upper blank according to any one of Clauses 46 to 117, wherein the filament in each layer of the

63

upper blank comprises a thermoplastic polyurethane material or other thermoplastic material.

Clause 124. The upper blank according to any one of Clauses 46 to 117, wherein the filament in each layer of the upper blank comprises a material that does not substantially absorb water.

Clause 125. The upper blank according to any one of Clauses 46 to 117, wherein the filament in each layer of the upper blank comprises a hydrophobic material.

Clause 126. The upper blank according to any one of Clauses 46 to 125, wherein two or more layers of the upper blank collectively produce a moiré effect in a least a portion of the upper blank.

Clause 127. An article of footwear, comprising:

an upper formed from an upper blank according to any one of Clauses 46 to 126; and
a sole structure engaged with the upper.

Clause 128. A method of forming an upper for an article of footwear, comprising:

extruding a first material to form a first layer comprising a first extruded filament including first plural, non-intersecting, spaced apart path segments, wherein the first extruded filament has a width dimension of less than 1 mm wide; and

extruding a second material to form a second layer comprising a second extruded filament including second plural, non-intersecting, spaced apart path segments, wherein the second extruded filament has a width dimension of less than 1 mm wide, and wherein the step of extruding the second material includes fusing the second layer to the first layer at locations where the second layer contacts the first layer.

Clause 129. The method according to Clause 128, wherein the step of extruding the first material includes forming the first plural, non-intersecting, spaced apart path segments in a serpentine configuration including at least two peaks and at least two valleys, and/or wherein the step of extruding the second material includes forming the second plural, non-intersecting, spaced apart path segments in a serpentine configuration including at least two peaks and at least two valleys.

Clause 130. The method according to Clause 128 or 129, wherein the step of extruding the first material includes forming the first plural, non-intersecting, spaced apart path segments to include at least 5 first non-intersecting path segments, wherein each non-intersecting path segment of the at least 5 first non-intersecting path segments is spaced apart from each directly adjacent first non-intersecting path segments by less than 5 mm over a length dimension of at least 25 mm.

Clause 131. The method according to any one of Clauses 128 to 130, wherein the step of extruding the second material includes forming the second plural, non-intersecting, spaced apart path segments to include at least 5 second non-intersecting path segments, wherein each non-intersecting path segment of the at least 5 second non-intersecting path segments is spaced apart from each directly adjacent second non-intersecting path segments by less than 5 mm over a length dimension of at least 25 mm.

Clause 132. The method according to any one of Clauses 128 to 131, further comprising:

extruding a third material to form a third layer comprising a third extruded filament including third plural, non-intersecting, spaced apart path segments, wherein the third extruded filament has a width dimension of less than 1 mm wide, and wherein the step of extruding the third material includes fusing the third layer to either or

64

both of the first layer and to the second layer at locations where the third layer contacts either or both of the first layer and the second layer, respectively.

Clause 133. The method according to Clause 132, further comprising:

extruding a fourth material to form a fourth layer comprising a fourth extruded filament including fourth plural, non-intersecting, spaced apart path segments, wherein the fourth extruded filament has a width dimension of less than 1 mm wide, and wherein the step of extruding the fourth material includes fusing the fourth layer to any one of or any combination of the first layer, the second layer, and the third layer at locations where the fourth layer contacts any one of or any combination of the first layer, the second layer, and the third layer, respectively.

Clause 134. The method according to Clause 133, further comprising:

extruding a fifth material to form a fifth layer comprising a fifth extruded filament including fifth plural, non-intersecting, spaced apart path segments, wherein the fifth extruded filament has a width dimension of less than 1 mm wide, and wherein the step of extruding the fifth material includes fusing the fifth layer to any one of or any combination of the first layer, the second layer, the third layer, and the fourth layer at locations where the fifth layer contacts any one of or any combination of the first layer, the second layer, the third layer, and the fourth layer, respectively.

Clause 135. The method according to Clause 134, further comprising:

extruding a sixth material to form a sixth layer comprising a sixth extruded filament including sixth plural, non-intersecting, spaced apart path segments, wherein the sixth extruded filament has a width dimension of less than 1 mm wide, and wherein the step of extruding the sixth material includes fusing the sixth layer to any one of or any combination of the first layer, the second layer, the third layer, the fourth layer, and the fifth layer at locations where the sixth layer contacts any one of or any combination of the first layer, the second layer, the third layer, the fourth layer, and the fifth layer, respectively.

Clause 136. The method according to Clause 135, further comprising:

extruding a seventh material to form a seventh layer comprising a seventh extruded filament including seventh plural, non-intersecting, spaced apart path segments, wherein the seventh extruded filament has a width dimension of less than 1 mm wide, and wherein the step of extruding the seventh material includes fusing the seventh layer to any one of or any combination of the first layer, the second layer, the third layer, the fourth layer, the fifth layer, and the sixth layer at locations where the seventh layer contacts any one of or any combination of the first layer, the second layer, the third layer, the fourth layer, the fifth layer, and the sixth layer, respectively.

Clause 137. The method according to Clause 136, further comprising:

extruding an eighth material to form an eighth layer comprising an eighth extruded filament including eighth plural, non-intersecting, spaced apart path segments, wherein the eighth extruded filament has a width dimension of less than 1 mm wide, and wherein the step of extruding the eighth material includes fusing the eighth layer to any one of or any combination of the

65

first layer, the second layer, the third layer, the fourth layer, the fifth layer, the sixth layer, and the seventh layer at locations where the eighth layer contacts any one of or any combination of the first layer, the second layer, the third layer, the fourth layer, the fifth layer, the sixth layer, and the seventh layer, respectively.

Clause 138. The method according to any one of Clauses 128 to 137, wherein the extruded and fused layers form a fused upper component, wherein at least one of the layers is extruded onto a base surface, and wherein the method further comprises:

removing the fused upper component from the base substrate; and

engaging the fused upper component with another upper component.

Clause 139. The method according to any one of Clauses 128 to 138, wherein the materials that are extruded and fused in the extruding steps are same material.

Clause 140. The method according to any one of Clauses 128 to 138, wherein at least one of the materials that is extruded and fused in the extruding steps differs from the material extruded in at least one of the other extruding steps.

Clause 141. The method according to Clause 139 or 140, wherein at least one of the materials that is extruded and fused in the extruding steps differs in color from a color of the material extruded in at least one of the other extruding steps.

Clause 142. The method according to Clause 139 or 140, wherein at least one of the materials that is extruded and fused in the extruding steps comprises a translucent or transparent material.

Clause 143. The method according to any one of Clauses 128 to 142, wherein the material that is extruded and fused in each of the extruding steps comprises a thermoplastic polyurethane material or other thermoplastic material.

Clause 144. The method according to any one of Clauses 128 to 142, wherein the material that is extruded and fused in each of the extruding steps comprises a material that does not substantially absorb water.

Clause 145. The method according to any one of Clauses 128 to 142, wherein the material that is extruded and fused in each of the extruding steps comprises a hydrophobic material.

Clause 146. The method according to any one of Clauses 128 to 145, wherein two or more layers of the upper are extruded and fused into relative positions to collectively produce a moiré effect in at least a portion of the upper.

Clause 147. The method according to any one of Clauses 128 to 146, wherein a plurality of the second plural, non-intersecting, spaced apart path segments of the second extruded filament extend parallel to and partially overlap with a plurality of the first plural, non-intersecting, spaced apart path segments of the first extruded filament over a path segment length of at least 25 mm.

Clause 148. The method according to any one of Clauses 128 to 146, wherein a plurality of the second plural, non-intersecting, spaced apart path segments of the second extruded filament intersect and form an angle with a plurality of the first plural, non-intersecting, spaced apart path segments of the first extruded filament.

Clause 149. The method according to Clause 148, wherein the angle is within a range of 65° to 90°.

Clause 150. The method according to Clause 148 or 149, wherein the plurality of the second plural, non-intersecting, spaced apart path segments of the second extruded filament and the first plural, non-intersecting, spaced apart path

66

segments of the first extruded filament cooperate to form one or more diamond shapes in a forefoot region of the upper.

Clause 151. The method according to Clause 150, wherein a long axis of each of the one or more diamond shapes extends in substantially an anterior-posterior direction of the upper.

Clause 152. The method according to any one of Clauses 128 to 151, wherein the first extruded filament has a first thickness at a first region of the upper, wherein the first extruded filament has a second thickness at a second region of the upper, and wherein the first thickness differs from the second thickness.

Clause 153. The method according to any one of Clauses 128 to 151, wherein the first extruded filament has a first diameter at a first region of the upper, wherein the first extruded filament has a second diameter at a second region of the upper, and wherein the first diameter differs from the second diameter.

Clause 154. The method according to any one of Clauses 128 to 153, wherein the first extruded filament has a first thickness at a first region of the upper, wherein the second extruded filament has a second thickness at a second region of the upper, and wherein the first thickness differs from the second thickness.

Clause 155. The method according to any one of Clauses 128 to 153, wherein the first extruded filament has a first diameter at a first region of the upper, wherein the second extruded filament has a second diameter at a second region of the upper, and wherein the first diameter differs from the second diameter.

Clause 156. The method according to any one of Clauses 128 to 155, wherein the step of extruding the first material includes extruding the first layer to form a perimeter of the upper.

Clause 157. A method of making an article of footwear, comprising:

forming an upper according to any one of Clauses 128 to 156; and

engaging the upper with a sole structure.

Clause 158. A method of forming an upper for an article of footwear, comprising:

extruding a first material to form a first layer comprising a first extruded filament as a first path, wherein the first path of the first extruded filament forms a first lateral rear heel portion, a first lateral midfoot portion, a first forefoot portion, a first medial midfoot portion, and a first medial rear heel portion, and wherein the first extruded filament has a width dimension of less than 1 mm wide; and

extruding a second material to form a second layer comprising a second extruded filament as a second path, wherein the second path of the second extruded filament forms a second lateral rear heel portion, a second lateral midfoot portion, a second forefoot portion, a second medial midfoot portion, and a second medial rear heel portion, wherein the second extruded filament has a width dimension of less than 1 mm wide, and wherein the step of extruding the second material includes fusing the second layer to the first layer at locations where the second layer contacts the first layer.

Clause 159. The method according to Clause 158, wherein the step of extruding the second material includes forming the second path to overlap the first path over at least 90% of an overall length of the second path and/or forming the second path to overlap the first path over at least 90% of an overall length of the first path.

67

Clause 160. The method according to Clause 158 or 159, further comprising:

extruding a third material to form a third layer comprising a third extruded filament as a third path, wherein the third path of the third extruded filament forms a third lateral rear heel portion, a third lateral midfoot portion, a third forefoot portion, a third medial midfoot portion, and a third medial rear heel portion, wherein the third extruded filament has a width dimension of less than 1 mm wide, and wherein the step of extruding the third material includes fusing the third layer to either or both of the first layer and to the second layer at locations where the third layer contacts either or both of the first layer and the second layer, respectively.

Clause 161. The method according to Clause 160, further comprising:

extruding a fourth material to form a fourth layer comprising a fourth extruded filament formed as a fourth path, wherein the fourth path of the fourth extruded filament forms a fourth lateral rear heel portion, a fourth lateral midfoot portion, a fourth forefoot portion, a fourth medial midfoot portion, and a fourth medial rear heel portion, wherein the fourth extruded filament has a width dimension of less than 1 mm wide, and wherein the step of extruding the fourth material includes fusing the fourth layer to any one of or any combination of the first layer, the second layer, and the third layer at locations where the fourth layer contacts any one of or any combination of the first layer, the second layer, and the third layer, respectively.

Clause 162. The method according to Clause 161, further comprising:

extruding a fifth material to form a fifth layer comprising a fifth extruded filament as a fifth path, wherein the fifth path of the fifth extruded filament forms a fifth lateral rear heel portion, a fifth lateral midfoot portion, a fifth forefoot portion, a fifth medial midfoot portion, and a fifth medial rear heel portion, wherein the fifth extruded filament has a width dimension of less than 1 mm wide, and wherein the step of extruding the fifth material includes fusing the fifth layer to any one of or any combination of the first layer, the second layer, the third layer, and the fourth layer at locations where the fifth layer contacts any one of or any combination of the first layer, the second layer, the third layer, and the fourth layer, respectively.

Clause 163. The method according to Clause 162, further comprising:

extruding a sixth material to form a sixth layer comprising a sixth extruded filament as a sixth path, wherein the sixth path of the sixth extruded filament forms a sixth lateral rear heel portion, a sixth lateral midfoot portion, a sixth forefoot portion, a sixth medial midfoot portion, and a sixth medial rear heel portion, wherein the sixth extruded filament has a width dimension of less than 1 mm wide, and wherein the step of extruding the sixth material includes fusing the sixth layer to any one of or any combination of the first layer, the second layer, the third layer, the fourth layer, and the fifth layer at locations where the sixth layer contacts any one of or any combination of the first layer, the second layer, the third layer, the fourth layer, and the fifth layer, respectively.

Clause 164. The method according to Clause 163, further comprising:

extruding a seventh material to form a seventh layer comprising a seventh extruded filament as a seventh

68

path, wherein the seventh path of the seventh extruded filament forms a seventh lateral rear heel portion, a seventh lateral midfoot portion, a seventh forefoot portion, a seventh medial midfoot portion, and a seventh medial rear heel portion, wherein the seventh extruded filament has a width dimension of less than 1 mm wide, and wherein the step of extruding the seventh material includes fusing the seventh layer to any one of or any combination of the first layer, the second layer, the third layer, the fourth layer, the fifth layer, and the sixth layer at locations where the seventh layer contacts any one of or any combination of the first layer, the second layer, the third layer, the fourth layer, the fifth layer, and the sixth layer, respectively.

Clause 165. The method according to Clause 164, further comprising:

extruding an eighth material to form an eighth layer comprising an eighth extruded filament as an eighth path, wherein the eighth path of the eighth extruded filament forms an eighth lateral rear heel portion, an eighth lateral midfoot portion, an eighth forefoot portion, an eighth medial midfoot portion, and an eighth medial rear heel portion, wherein the eighth extruded filament has a width dimension of less than 1 mm wide, and wherein the step of extruding the eighth material includes fusing the eighth layer to any one of or any combination of the first layer, the second layer, the third layer, the fourth layer, the fifth layer, the sixth layer, and the seventh layer at locations where the eighth layer contacts any one of or any combination of the first layer, the second layer, the third layer, the fourth layer, the fifth layer, the sixth layer, and the seventh layer, respectively.

Clause 166. A method of forming an upper for an article of footwear, comprising:

extruding a first material to form a first layer comprising a first extruded filament as a first continuous path, wherein the first continuous path of the first extruded filament forms a first lateral rear heel portion, a first lateral midfoot portion, a first forefoot portion, a first medial midfoot portion, and a first medial rear heel portion, and wherein the first extruded filament has a width dimension of less than 1 mm wide; and

extruding a second material to form a second layer comprising a second extruded filament as a second continuous path, wherein the second continuous path of the second extruded filament forms a second lateral rear heel portion, a second lateral midfoot portion, a second forefoot portion, a second medial midfoot portion, and a second medial rear heel portion, wherein the second extruded filament has a width dimension of less than 1 mm wide, and wherein the step of extruding the second material includes fusing the second layer to the first layer at locations where the second layer contacts the first layer.

Clause 167. The method according to Clause 166, wherein the step of extruding the second material includes forming the second continuous path to overlap the first continuous path over at least 90% of an overall length of the second continuous path and/or forming the second continuous path to overlap the first continuous path over at least 90% of an overall length of the first continuous path.

Clause 168. The method according to Clause 166 or 167, further comprising:

extruding a third material to form a third layer comprising a third extruded filament as a third continuous path, wherein the third continuous path of the third extruded

69

filament forms a third lateral rear heel portion, a third lateral midfoot portion, a third forefoot portion, a third medial midfoot portion, and a third medial rear heel portion, wherein the third extruded filament has a width dimension of less than 1 mm wide, and wherein the step of extruding the third material includes fusing the third layer to either or both of the first layer and to the second layer at locations where the third layer contacts either or both of the first layer and the second layer, respectively.

Clause 169. The method according to Clause 168, further comprising:

extruding a fourth material to form a fourth layer comprising a fourth extruded filament formed as a fourth continuous path, wherein the fourth continuous path of the fourth extruded filament forms a fourth lateral rear heel portion, a fourth lateral midfoot portion, a fourth forefoot portion, a fourth medial midfoot portion, and a fourth medial rear heel portion, wherein the fourth extruded filament has a width dimension of less than 1 mm wide, and wherein the step of extruding the fourth material includes fusing the fourth layer to any one of or any combination of the first layer, the second layer, and the third layer at locations where the fourth layer contacts any one of or any combination of the first layer, the second layer, and the third layer, respectively.

Clause 170. The method according to Clause 169, further comprising:

extruding a fifth material to form a fifth layer comprising a fifth extruded filament as a fifth continuous path, wherein the fifth continuous path of the fifth extruded filament forms a fifth lateral rear heel portion, a fifth lateral midfoot portion, a fifth forefoot portion, a fifth medial midfoot portion, and a fifth medial rear heel portion, wherein the fifth extruded filament has a width dimension of less than 1 mm wide, and wherein the step of extruding the fifth material includes fusing the fifth layer to any one of or any combination of the first layer, the second layer, the third layer, and the fourth layer at locations where the fifth layer contacts any one of or any combination of the first layer, the second layer, the third layer, and the fourth layer, respectively.

Clause 171. The method according to Clause 170, further comprising:

extruding a sixth material to form a sixth layer comprising a sixth extruded filament as a sixth continuous path, wherein the sixth continuous path of the sixth extruded filament forms a sixth lateral rear heel portion, a sixth lateral midfoot portion, a sixth forefoot portion, a sixth medial midfoot portion, and a sixth medial rear heel portion, wherein the sixth extruded filament has a width dimension of less than 1 mm wide, and wherein the step of extruding the sixth material includes fusing the sixth layer to any one of or any combination of the first layer, the second layer, the third layer, the fourth layer, and the fifth layer at locations where the sixth layer contacts any one of or any combination of the first layer, the second layer, the third layer, the fourth layer, and the fifth layer, respectively.

Clause 172. The method according to Clause 171, further comprising:

extruding a seventh material to form a seventh layer comprising a seventh extruded filament as a seventh continuous path, wherein the seventh continuous path of the seventh extruded filament forms a seventh lateral rear heel portion, a seventh lateral midfoot portion, a seventh forefoot portion, a seventh medial midfoot

70

portion, and a seventh medial rear heel portion, wherein the seventh extruded filament has a width dimension of less than 1 mm wide, and wherein the step of extruding the seventh material includes fusing the seventh layer to any one of or any combination of the first layer, the second layer, the third layer, the fourth layer, the fifth layer, and the sixth layer at locations where the seventh layer contacts any one of or any combination of the first layer, the second layer, the third layer, the fourth layer, the fifth layer, and the sixth layer, respectively.

Clause 173. The method according to Clause 172, further comprising:

extruding an eighth material to form an eighth layer comprising an eighth extruded filament as an eighth continuous path, wherein the eighth continuous path of the eighth extruded filament forms an eighth lateral rear heel portion, an eighth lateral midfoot portion, an eighth forefoot portion, an eighth medial midfoot portion, and an eighth medial rear heel portion, wherein the eighth extruded filament has a width dimension of less than 1 mm wide, and wherein the step of extruding the eighth material includes fusing the eighth layer to any one of or any combination of the first layer, the second layer, the third layer, the fourth layer, the fifth layer, the sixth layer, and the seventh layer at locations where the eighth layer contacts any one of or any combination of the first layer, the second layer, the third layer, the fourth layer, the fifth layer, the sixth layer, and the seventh layer, respectively.

Clause 174. The method according to any one of Clauses 158 to 173, wherein the materials that are extruded and fused in the extruding steps are same material.

Clause 175. The method according to any one of Clauses 158 to 173, wherein at least one of the materials that is extruded and fused in the extruding steps differs from the material extruded in at least one of the other extruding steps.

Clause 176. The method according to Clause 174 or 175, wherein at least one of the materials that is extruded and fused in the extruding steps differs in color from a color of the material extruded in at least one of the other extruding steps.

Clause 177. The method according to Clause 174 or 175, wherein at least one of the materials that is extruded and fused in the extruding steps comprises a translucent or transparent material.

Clause 178. The method according to any one of Clauses 158 to 177, wherein the material that is extruded and fused in each of the extruding steps comprises a thermoplastic polyurethane material or other thermoplastic material.

Clause 179. The method according to any one of Clauses 158 to 177, wherein the material that is extruded and fused in each of the extruding steps comprises a material that does not substantially absorb water.

Clause 180. The method according to any one of Clauses 158 to 177, wherein the material that is extruded and fused in each of the extruding steps comprises a hydrophobic material.

Clause 181. The method according to any one of Clauses 158 to 180, wherein two or more layers of the upper are extruded and fused into relative positions to cooperatively produce a moiré effect in at least a portion of the upper.

Clause 182. The method according to any one of Clauses 158 to 181, wherein the extruding steps form an upper blank, and wherein the method further comprises:

trimming the upper blank to form an upper component, and optionally, engaging the upper component with another upper component.

71

Clause 183. The method according to any one of Claims 158 to 181, wherein the extruding steps form an upper component, and wherein the method further comprises: engaging the upper component with another upper component.

Clause 184. The method according to Claim 182 or 183, wherein the engaging step includes bonding the upper component with said another upper component by an adhesive or cement.

Clause 185. A method of making an article of footwear, comprising:

forming an upper according to the method in any one of Claims 158 to 184; and
engaging the upper with a sole structure.

Clause 186. A method of making an article of footwear, comprising:

forming an upper according to any one of Claims 1 to 44; and
engaging the upper with a sole structure.

Clause 187. A method of making an article of footwear, comprising:

forming an upper blank according to any one of Claims 46 to 126; and
engaging the upper blank with a sole structure.

Clause 188. An upper formed by the method according to any one of Claims 128 to 156 or 158 to 184.

What is claimed is:

1. An upper for an article of footwear, comprising:
an upper component that includes:

(a) a first layer comprising a first filament formed as a first continuous path including first plural, non-intersecting, spaced apart path segments that form a first heel portion, a first midfoot portion, and a first forefoot portion; and

(b) a second layer comprising a second filament formed as a second continuous path including second plural, non-intersecting, spaced apart path segments that form a second heel portion, a second midfoot portion, and a second forefoot portion, and wherein the second layer is fused to the first layer at locations where the second layer contacts the first layer,

wherein a plurality of the second plural, non-intersecting, spaced apart path segments of the second filament intersect and form an angle with a plurality of the first plural, non-intersecting, spaced apart path segments of the first filament.

2. The upper according to claim 1, wherein the first plural, non-intersecting, spaced apart path segments extend in a substantially medio-lateral direction of the upper, and wherein the second plural, non-intersecting, spaced apart path segments extend in a substantially medio-lateral direction of the upper.

3. The upper according to claim 1, wherein the first plural, non-intersecting, spaced apart path segments extend in a substantially medio-lateral direction of the upper, and wherein the second plural, non-intersecting, spaced apart path segments extend in a substantially anterior-posterior direction of the upper.

4. The upper according to claim 1, wherein the first plural, non-intersecting, spaced apart path segments extend in a substantially anterior-posterior direction of the upper and include a serpentine configuration having at least two peaks and at least two valleys.

5. The upper according to claim 4, wherein the serpentine configuration is formed at least at a medial midfoot portion of the upper component, or

72

wherein the serpentine configuration is formed at least at a lateral midfoot portion of the upper component.

6. The upper according to claim 1, wherein the upper component includes a unitary construction in which the first layer and the second layer are fixed together only in a non-adhesive fused manner, and wherein the unitary construction of the upper component includes a medial heel portion, a medial midfoot portion, a lateral heel portion, and lateral midfoot portion, and a forefoot portion extending between the medial midfoot portion and the lateral midfoot portion.

7. The upper according to claim 1, wherein the first plural, non-intersecting, spaced apart path segments include at least 5 first non-intersecting path segments, wherein each non-intersecting path segment of the at least 5 first non-intersecting path segments is spaced apart from each directly adjacent first non-intersecting path segments by less than 5 mm over a length dimension of at least 50 mm, and

wherein the second plural, non-intersecting, spaced apart path segments include at least 5 second non-intersecting path segments, wherein each non-intersecting path segment of the at least 5 second non-intersecting path segments is spaced apart from each directly adjacent second non-intersecting path segments by less than 5 mm over a length dimension of at least 50 mm.

8. The upper according to claim 1, wherein the upper component further includes a third layer comprising a third filament including third plural, non-intersecting, spaced apart path segments, wherein the third filament has a width dimension of less than 2 mm wide, and wherein the third layer is fused to either or both of the first layer and to the second layer at locations where the third layer contacts either or both of the first layer and the second layer, respectively.

9. The upper according to claim 8, wherein the upper component further includes a fourth layer comprising a fourth filament including fourth plural, non-intersecting, spaced apart path segments, wherein the fourth filament has a width dimension of less than 2 mm wide, and wherein the fourth layer is fused to any one of or any combination of the first layer, the second layer, and the third layer at locations where the fourth layer contacts any one of or any combination of the first layer, the second layer, and the third layer, respectively;

a fifth layer comprising a fifth filament including fifth plural, non-intersecting, spaced apart path segments, wherein the fifth filament has a width dimension of less than 2 mm wide, and wherein the fifth layer is fused to any one of or any combination of the first layer, the second layer, the third layer, and the fourth layer at locations where the fifth layer contacts any one of or any combination of the first layer, the second layer, the third layer, and the fourth layer, respectively;

a sixth layer comprising a sixth filament including sixth plural, non-intersecting, spaced apart path segments, wherein the sixth filament has a width dimension of less than 2 mm wide, and wherein the sixth layer is fused to any one of or any combination of the first layer, the second layer, the third layer, the fourth layer, and the fifth layer at locations where the sixth layer contacts any one of or any combination of the first layer, the second layer, the third layer, the fourth layer, and the fifth layer, respectively;

a seventh layer comprising a seventh filament including seventh plural, non-intersecting, spaced apart path segments, wherein the seventh filament has a width dimension of less than 2 mm wide, and wherein the seventh

73

layer is fused to any one of or any combination of the first layer, the second layer, the third layer, the fourth layer, the fifth layer, and the sixth layer at locations where the seventh layer contacts any one of or any combination of the first layer, the second layer, the third layer, the fourth layer, the fifth layer, and the sixth layer, respectively; and

an eighth layer comprising an eighth filament including eighth plural, non-intersecting, spaced apart path segments, wherein the eighth filament has a width dimension of less than 2 mm wide, and wherein the eighth layer is fused to any one of or any combination of the first layer, the second layer, the third layer, the fourth layer, the fifth layer, the sixth layer, and the seventh layer at locations where the eighth layer contacts any one of or any combination of the first layer, the second layer, the third layer, the fourth layer, the fifth layer, the sixth layer, and the seventh layer, respectively.

10. The upper according to claim 1, wherein the first layer and the second layer of the upper component collectively produce a moiré effect in at least a portion of the upper.

11. An upper for an article of footwear, comprising an upper component that includes:

a first layer comprising a first filament formed as a first continuous path including first plural, non-intersecting, spaced apart path segments that form a first heel portion, a first midfoot portion, and a first forefoot portion; and

a second layer comprising a second filament formed as a second continuous path including second plural, non-intersecting, spaced apart path segments that form a second heel portion, a second midfoot portion, and a second forefoot portion, and wherein the second layer is fused to the first layer at locations where the second layer contacts the first layer.

12. The upper according to claim 11, wherein at least one of the first plural, non-intersecting, spaced apart path segments and the second plural, non-intersecting, spaced apart path segments is located closer together in a forefoot region of the upper as compared to in a midfoot region and/or a heel region of the upper.

13. The upper according to claim 11, wherein the first plural, non-intersecting, spaced apart path segments extend in a substantially medio-lateral direction of the upper, and wherein the second plural, non-intersecting, spaced apart path segments extend in a substantially anterior-posterior direction of the upper.

14. The upper according to claim 11, wherein a plurality of the second plural, non-intersecting, spaced apart path segments of the second filament extend parallel to and partially overlap with a plurality of the first plural, non-

74

intersecting, spaced apart path segments of the first filament over a path segment length of at least 25 mm.

15. The upper according to claim 11, wherein the first plural, non-intersecting, spaced apart path segments extend in a serpentine configuration including at least two peaks and at least two valleys, and/or wherein the second plural, non-intersecting, spaced apart path segments extend in a serpentine configuration including at least two peaks and at least two valleys.

16. An upper for an article of footwear, comprising:

a first layer comprising a first filament formed as a first continuous path including first plural, non-intersecting, spaced apart path segments, wherein the first filament forms a first lateral rear heel portion, a first lateral midfoot portion, a first forefoot portion, a first medial midfoot portion, and a first medial rear heel portion; and

a second layer comprising a second filament formed as a second continuous path including second plural, non-intersecting, spaced apart path segments, wherein the second filament forms a second lateral rear heel portion, a second lateral midfoot portion, a second forefoot portion, a second medial midfoot portion, and a second medial rear heel portion, and wherein the second layer is fused to the first layer at locations where the second layer contacts the first layer,

wherein a plurality of the second plural, non-intersecting, spaced apart path segments of the second filament intersect and form an angle with a plurality of the first plural, non-intersecting, spaced apart path segments of the first filament.

17. The upper according to claim 16, wherein the first filament has a width dimension of less than 3 mm wide, and the second filament has a width dimension of less than 3 mm wide.

18. The upper according to claim 16, wherein:

the first plural, non-intersecting, spaced apart path segments include at least 5 first non-intersecting path segments, wherein each non-intersecting path segment of the at least 5 first non-intersecting path segments is spaced apart from each directly adjacent first non-intersecting path segments by less than 5 mm over a length dimension of at least 25 mm; and

the second plural, non-intersecting, spaced apart path segments include at least 5 second non-intersecting path segments, wherein each non-intersecting path segment of the at least 5 second non-intersecting path segments is spaced apart from each directly adjacent second non-intersecting path segments by less than 5 mm over a length dimension of at least 25 mm.

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